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(54) REFRIGERATOR

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(21) Appl. No.: **19/191,711**

(22) Filed: **Apr. 28, 2025**

Related U.S. Application Data

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(52) **U.S. Cl.**
CPC ***F25C 1/18*** (2013.01); ***F25C 1/10*** (2013.01); ***F25C 1/25*** (2018.01); ***F25C 5/02*** (2013.01); *F25C 2305/0221* (2021.08); *F25C 2400/10* (2013.01); *F25C 2400/14* (2013.01); *F25C 2700/14* (2013.01)

(57) **ABSTRACT**

Provided is a refrigerator. The refrigerator may include a first tray assembly defining a portion of an ice making cell and a second tray assembly defining another portion of the ice making cell. The one tray assembly of the first and second tray assemblies may include a first portion that defines at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion.

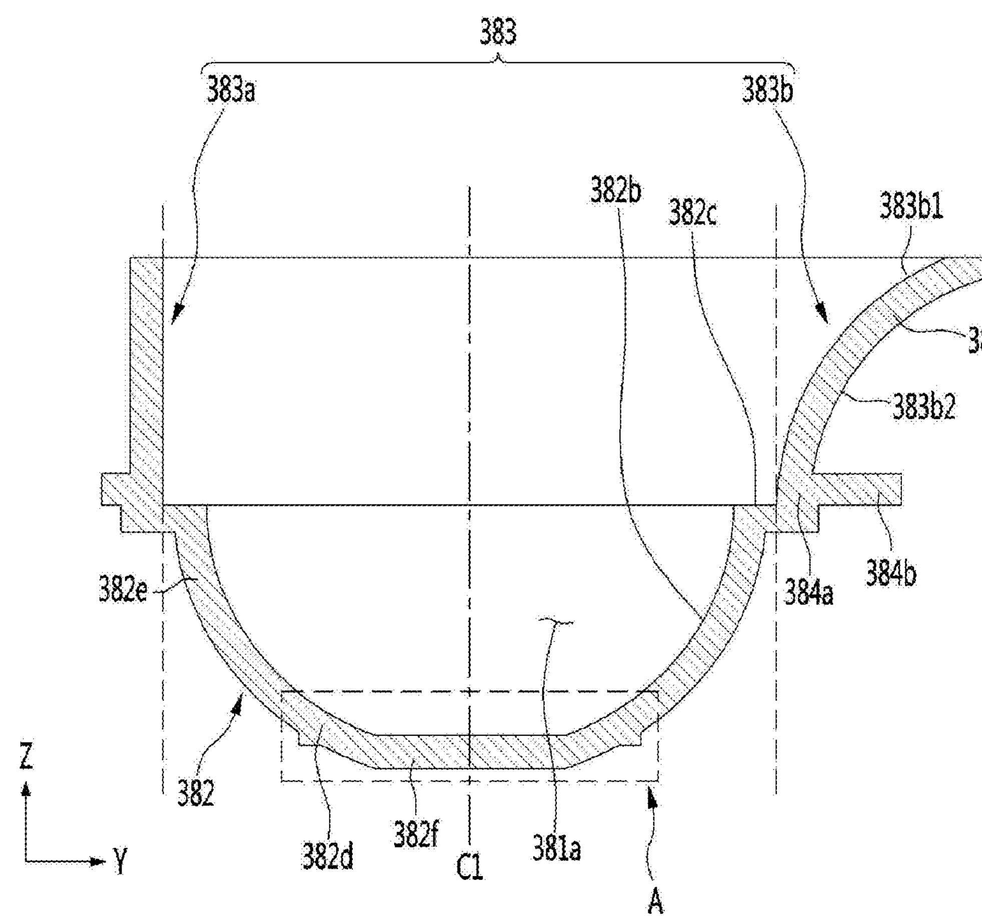


FIG. 1

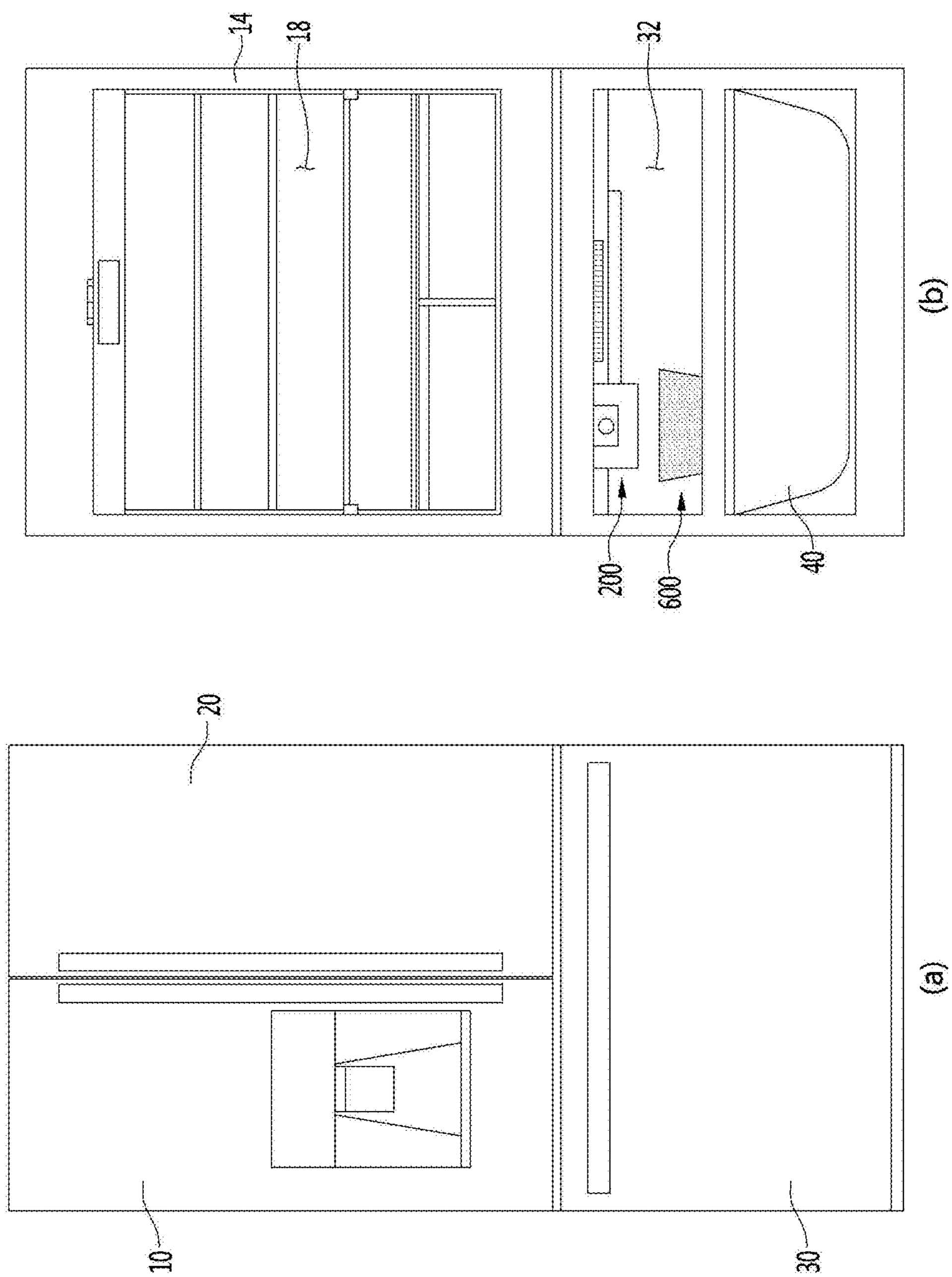


FIG. 2

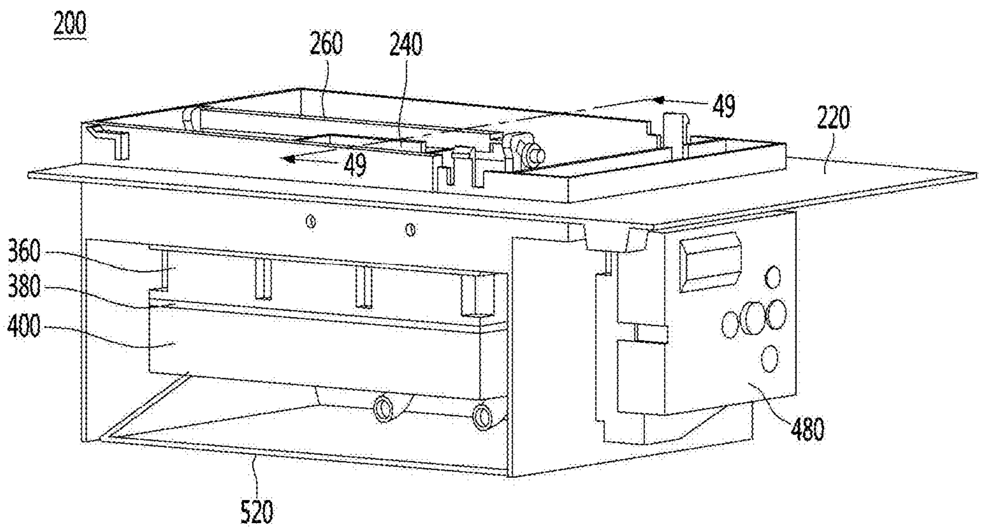

200
260
240
49
220
49
360
380
400
480
520

FIG. 3

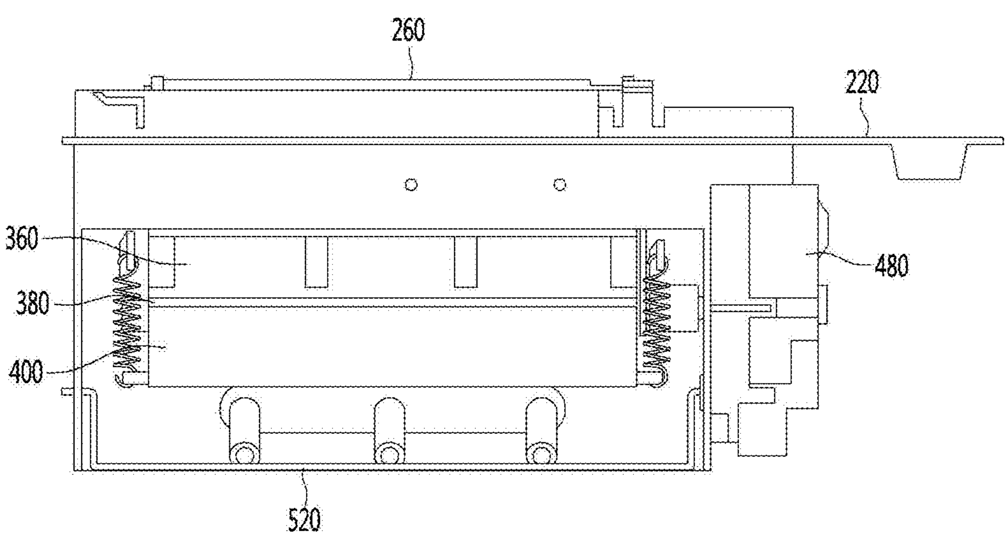

260
220
360
480
380
400
520

FIG. 4

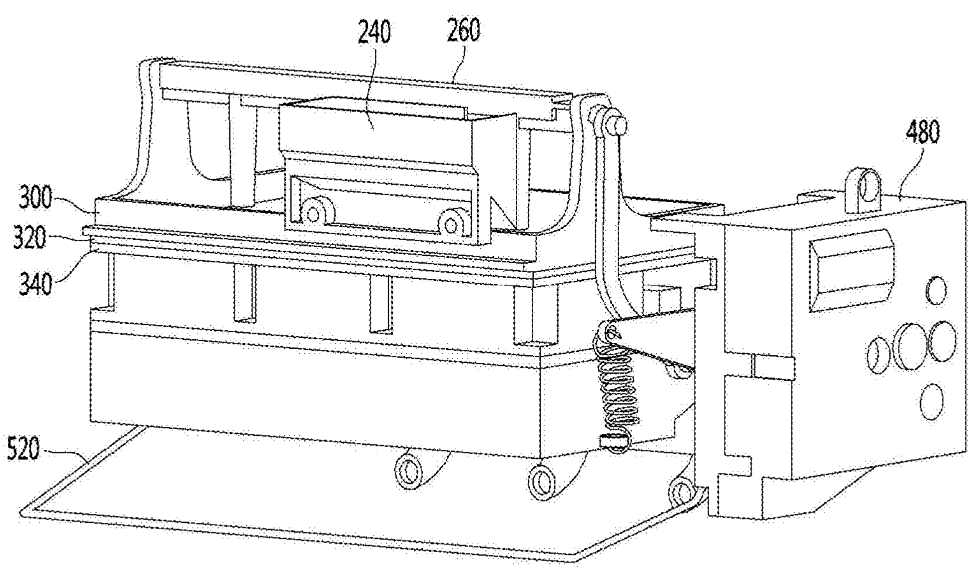
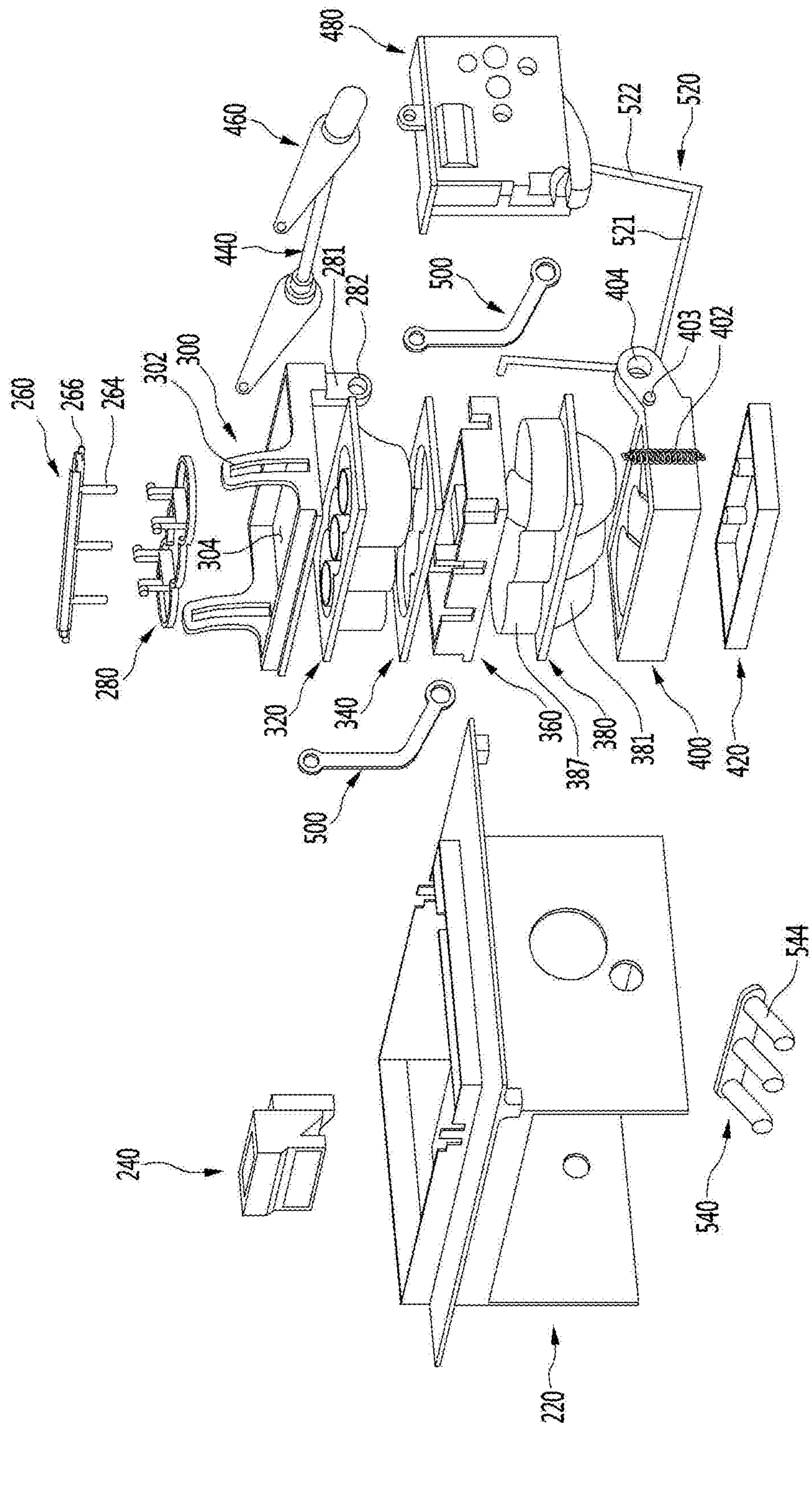


FIG. 5



480
460
440
281
282
522
520
521
500
404
403
402
260
266
264
302
300
304
280
320
340
360
387
380
381
400
420
500
240
220
540
544

FIG. 6

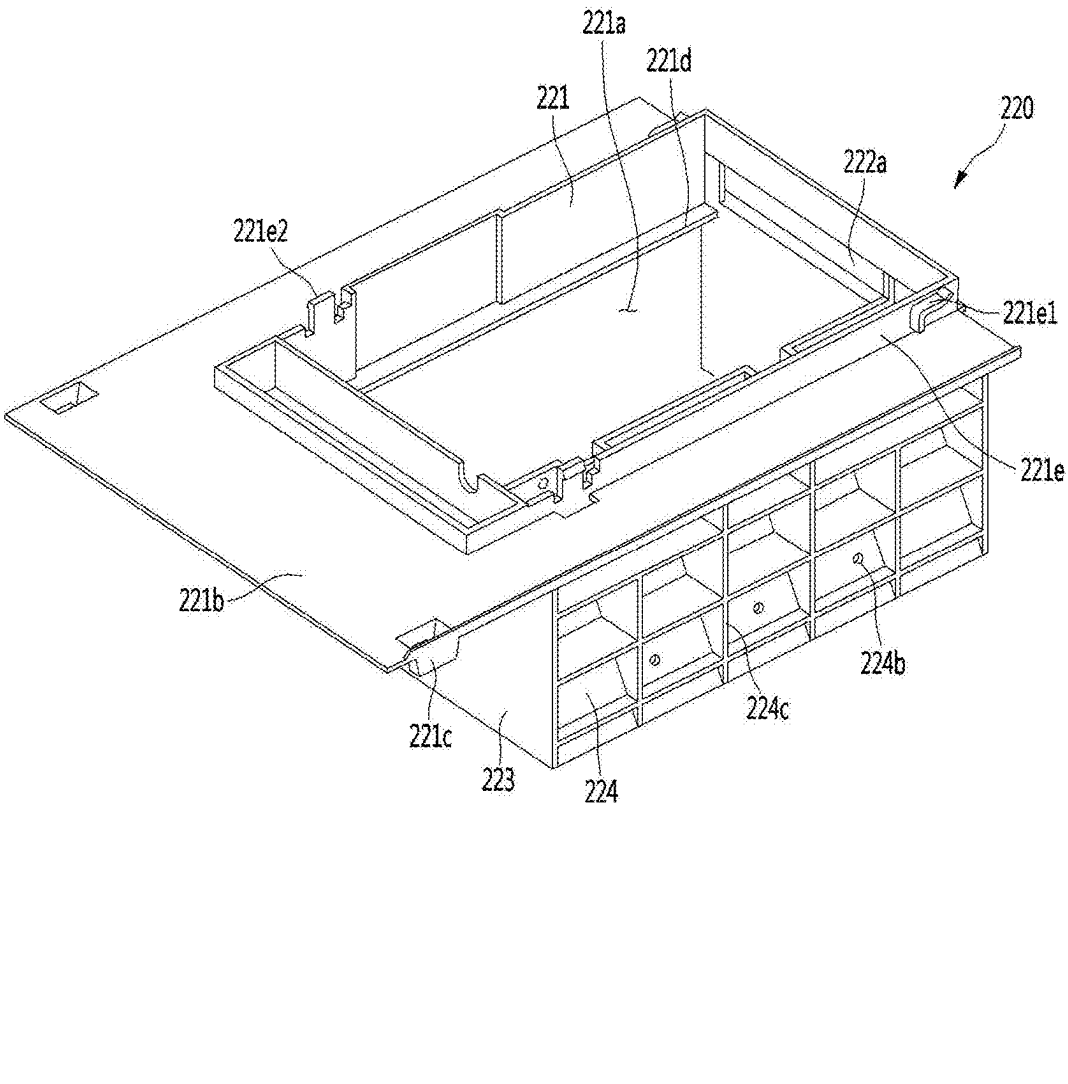

221a
221d
221
220
222a
221e2
221e1
221e
221b
224b
224c
221c
223
224

FIG. 7

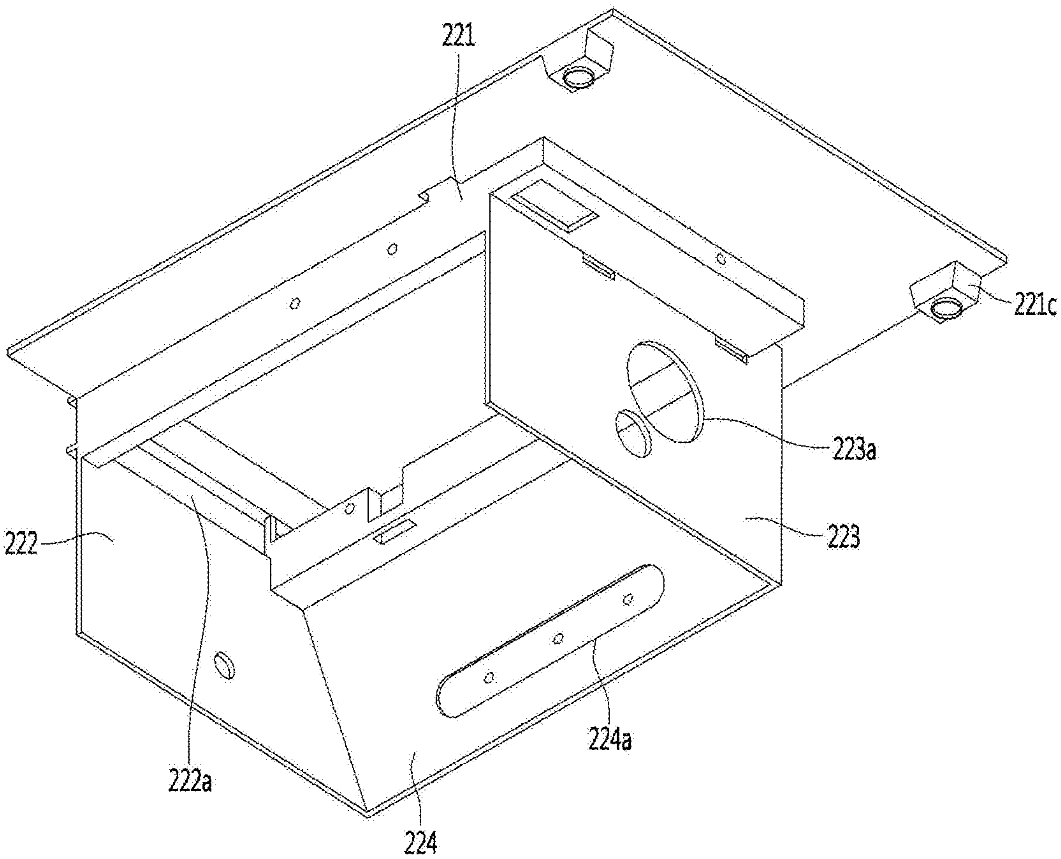
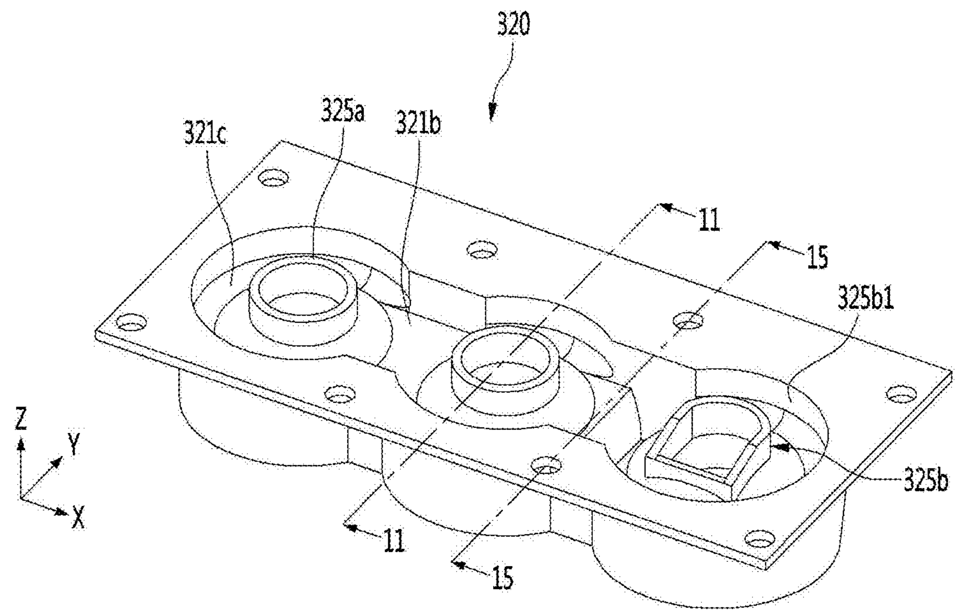
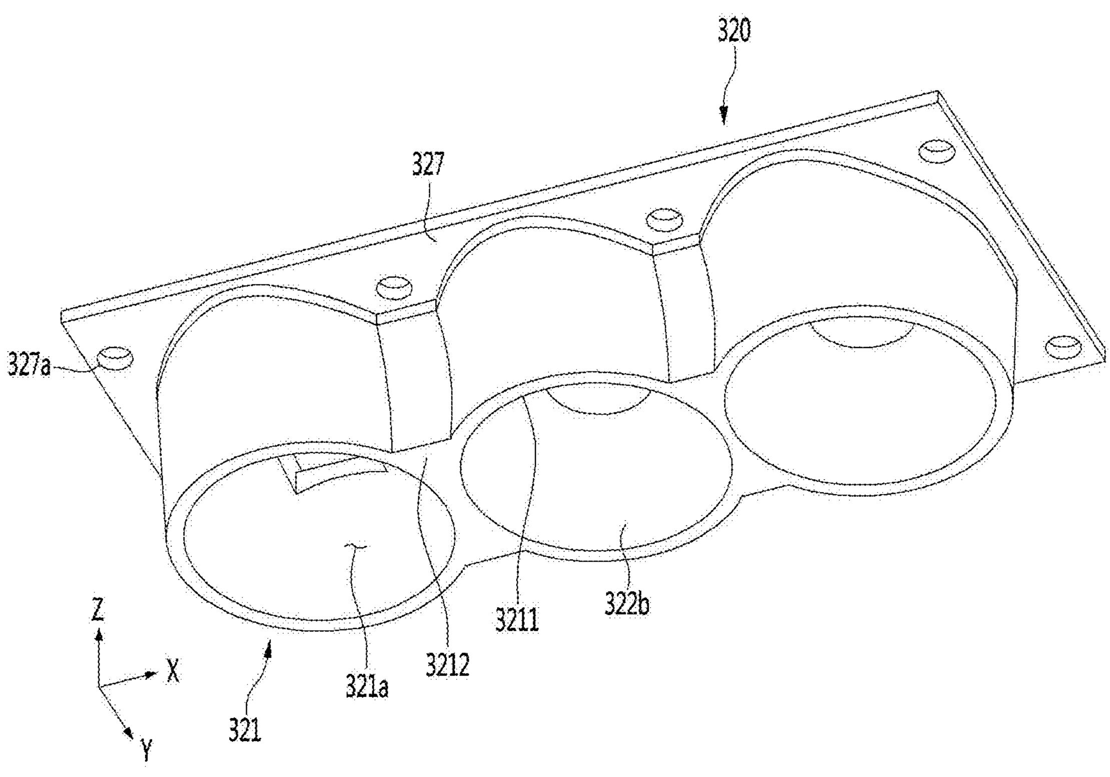


FIG. 8



320
321c
325a
321b
11
15
325b1
325b
Z
Y
X
11
15

FIG. 9



320
327
327a
Z
X
Y
321
321a
3212
3211
322b

FIG. 10

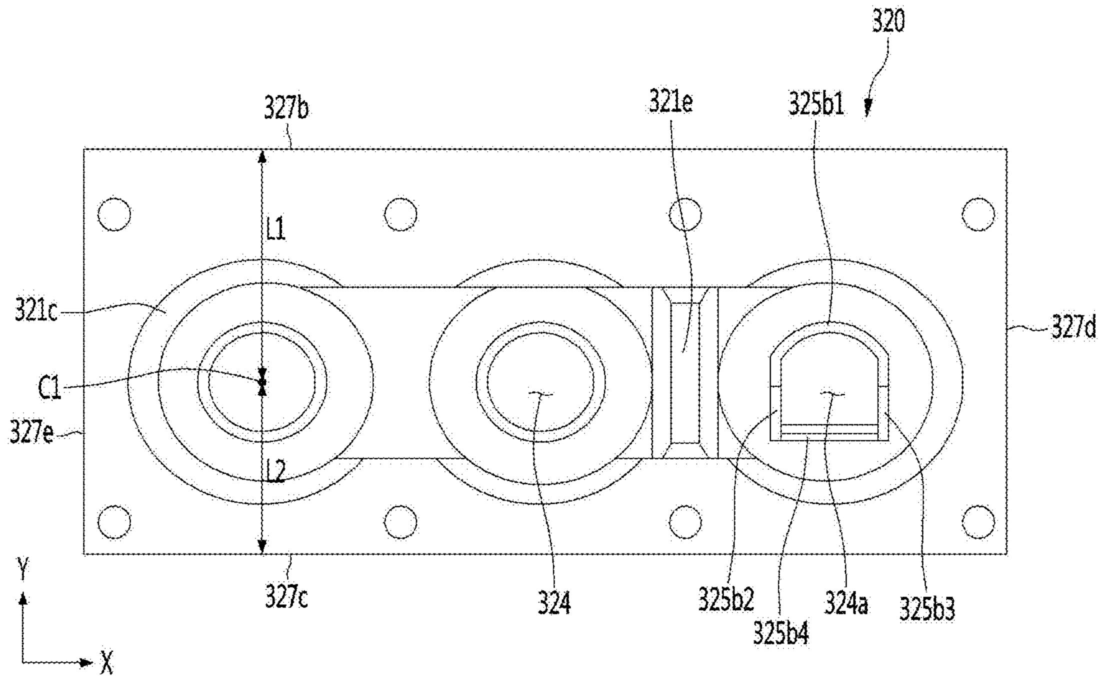

320
327b
321e
325b1
L1
321c
327d
C1
327e
L2
Y
327c
324
325b2
325b4
324a
325b3
X

FIG. 11

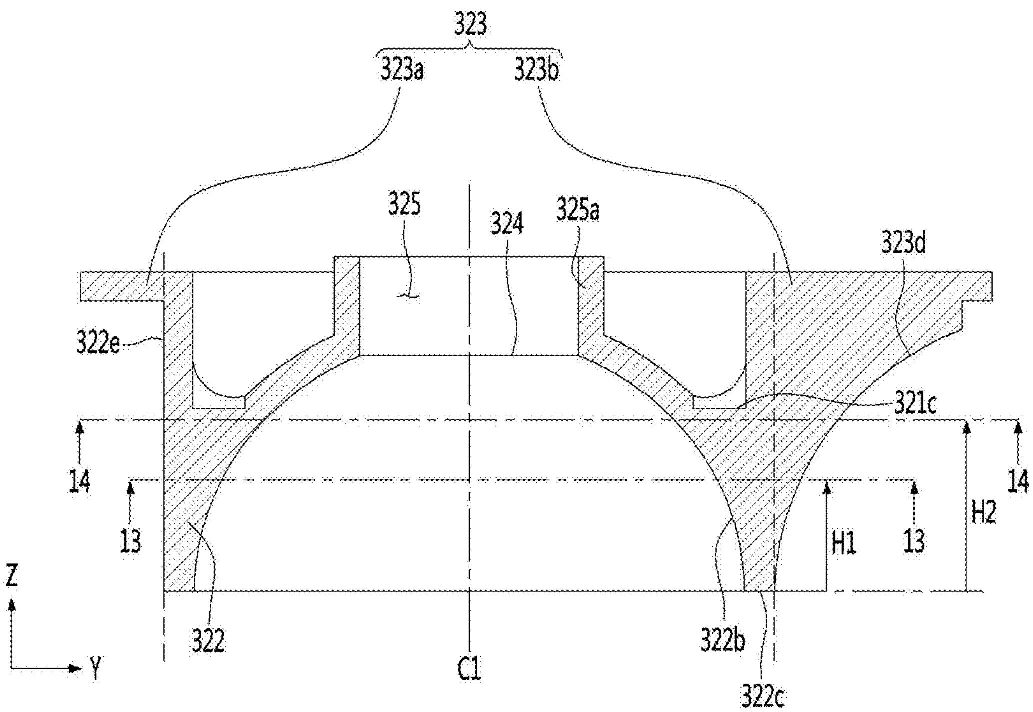

323
323a
323b
325
324
325a
323d
322e
321c
14
14
13
13
H1
H2
Z
Y
322
C1
322b
322c

FIG. 12

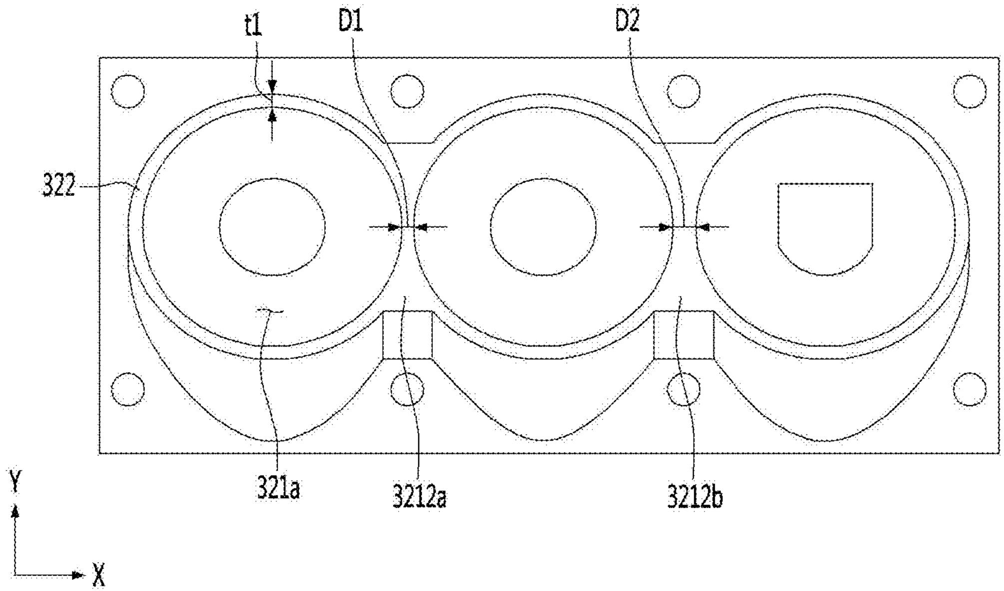


FIG. 13

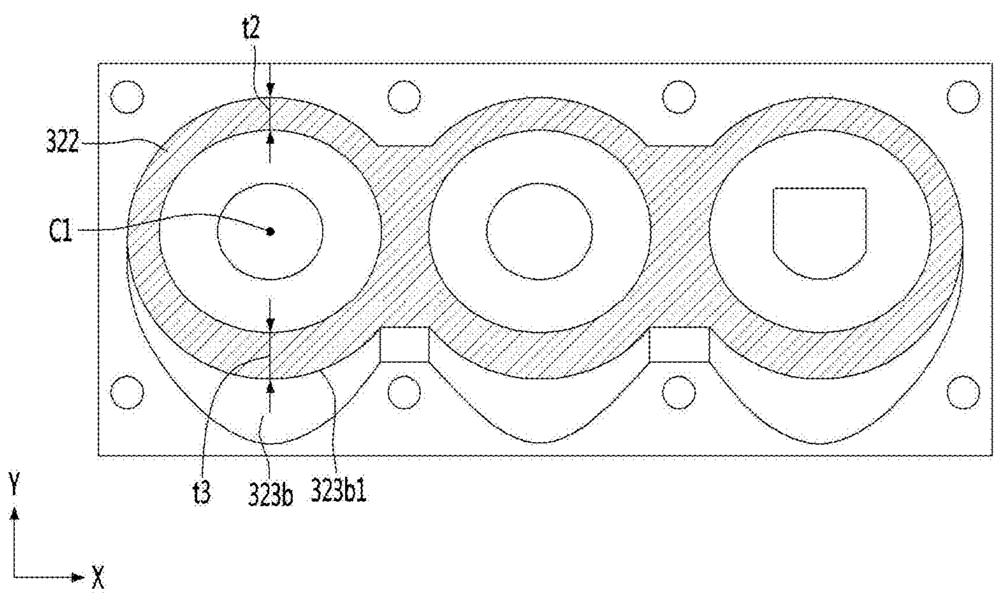


FIG. 14

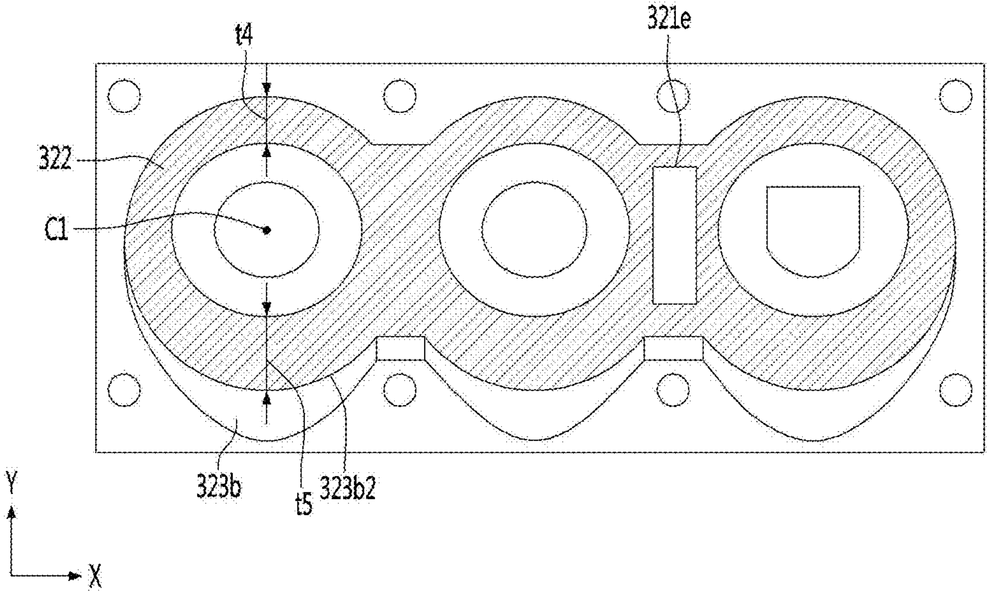


FIG. 15

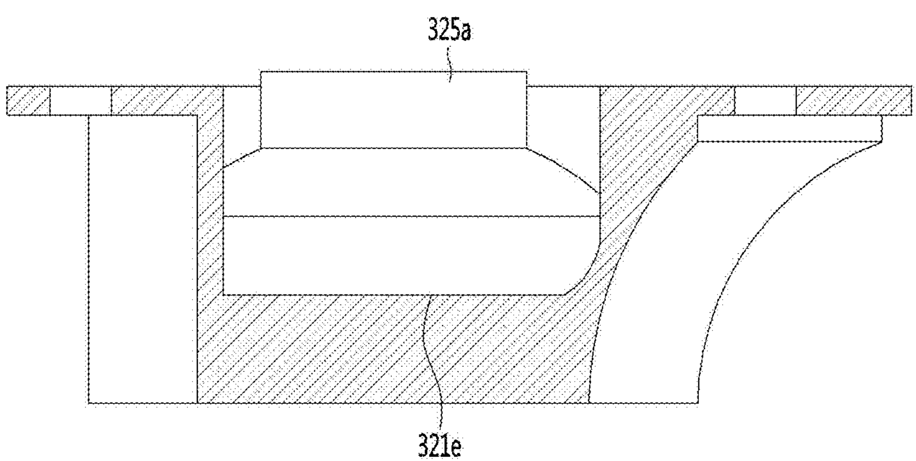


FIG. 16

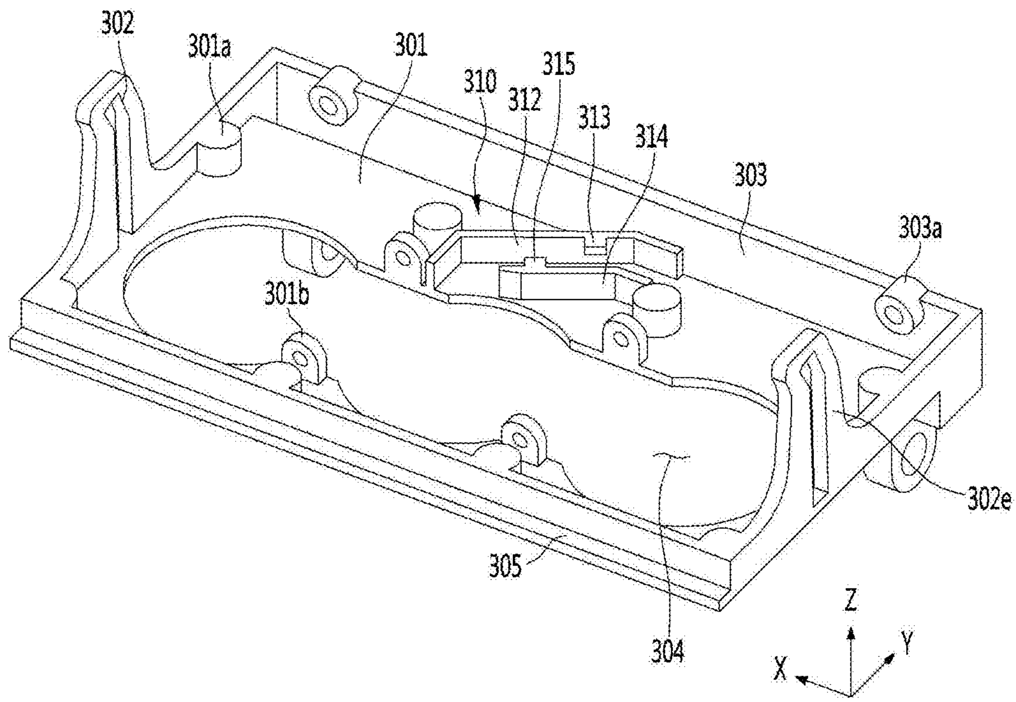

302
301a
301
310
312
315
313
314
303
303a
301b
302e
305
304
Z
Y
X

FIG. 17

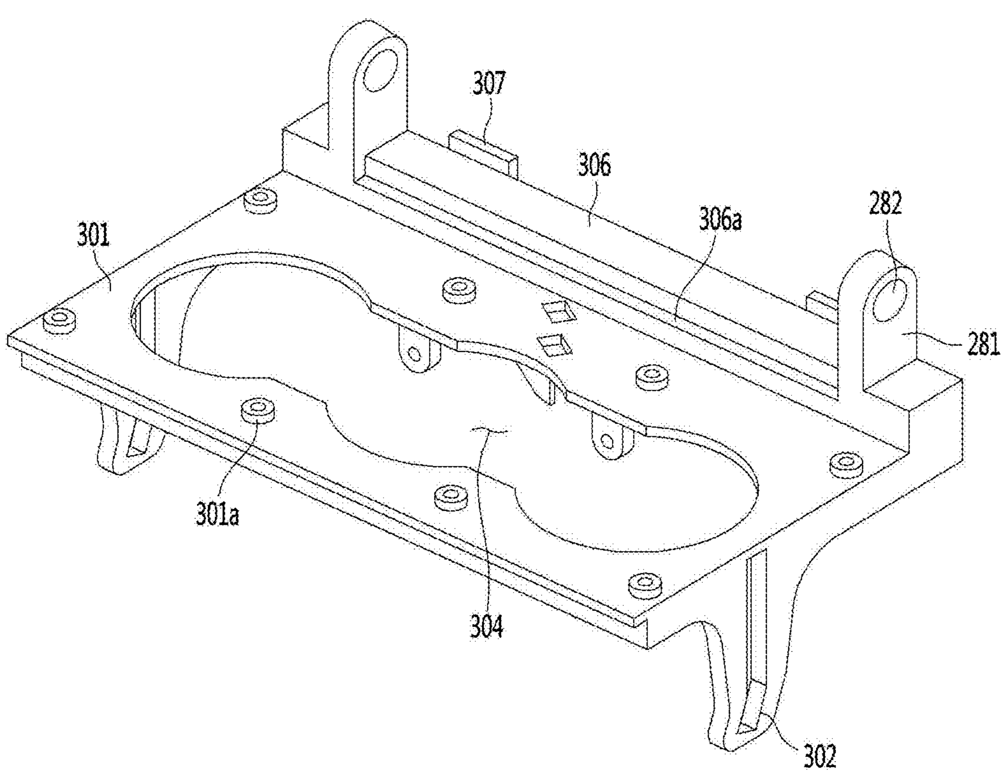

307
306
306a
282
301
281
301a
304
302

FIG. 18

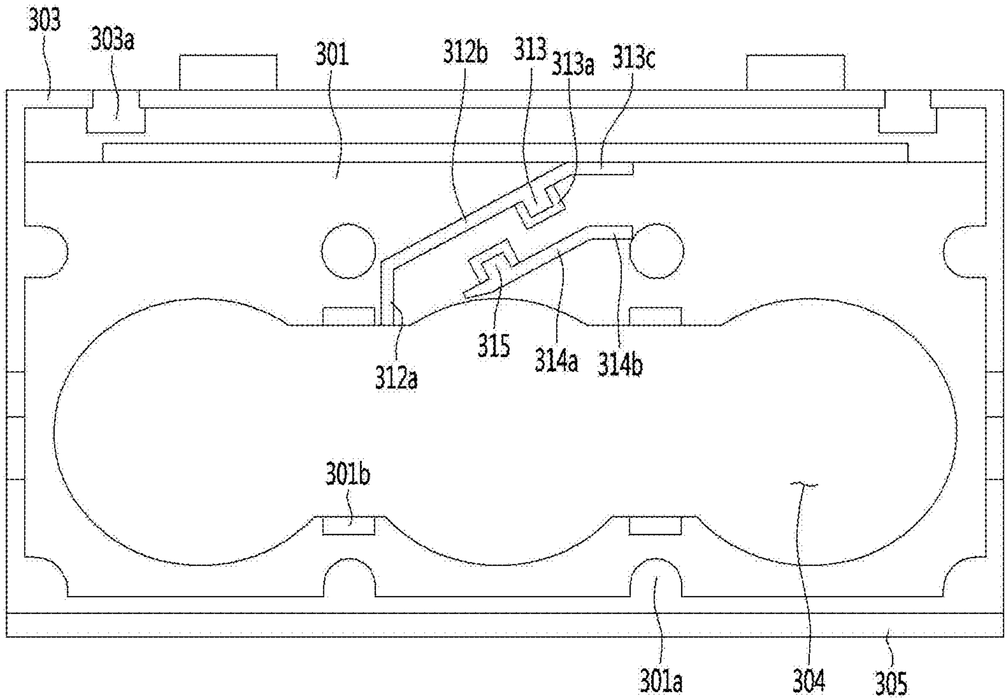

303
303a
301
312b
313
313a
313c
312a
315
314a
314b
301b
301a
304
305

FIG. 19

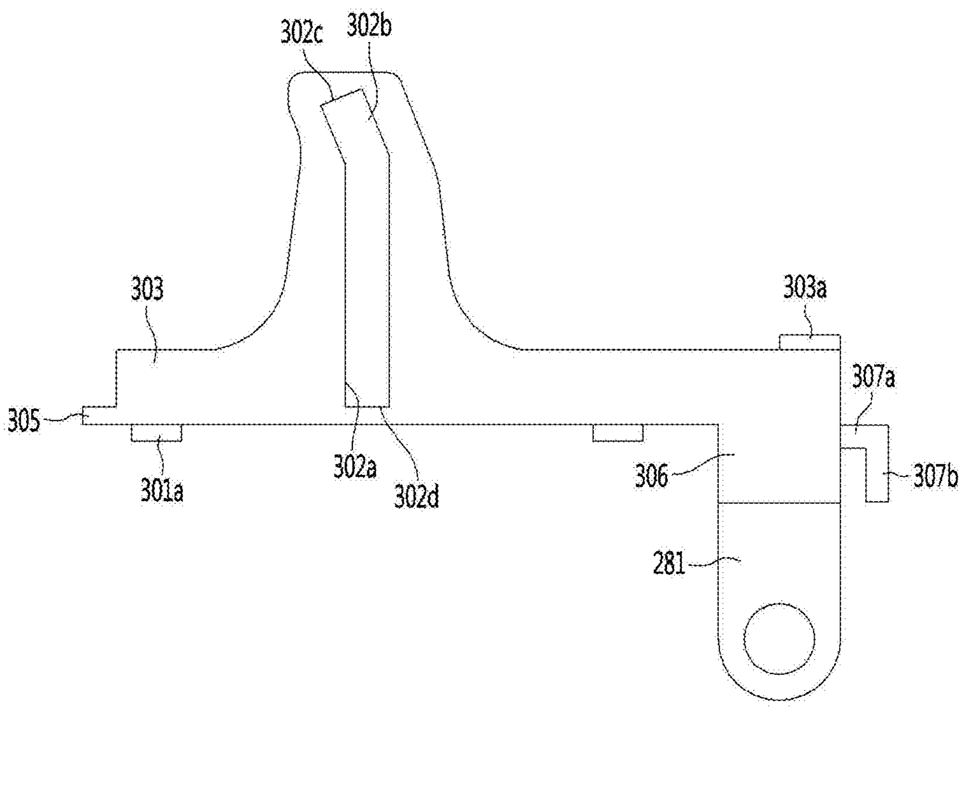

302c
302b
303
303a
307a
305
301a
302a
302d
306
307b
281

FIG. 20

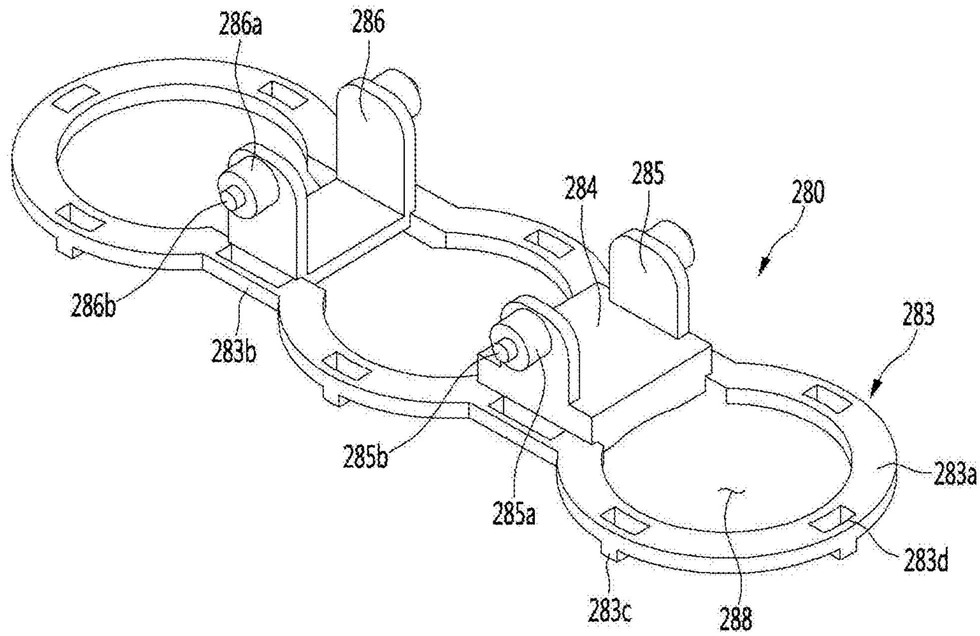

286a
286
285
284
280
286b
283b
283
285b
283a
285a
283d
283c
288

FIG. 21

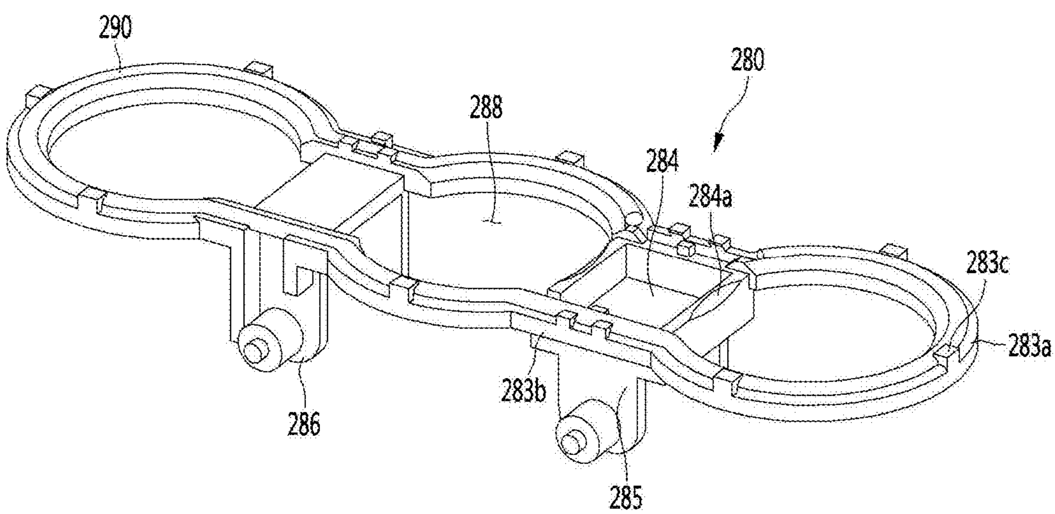

290
280
288
284
284a
283c
283a
283b
286
285

FIG. 22

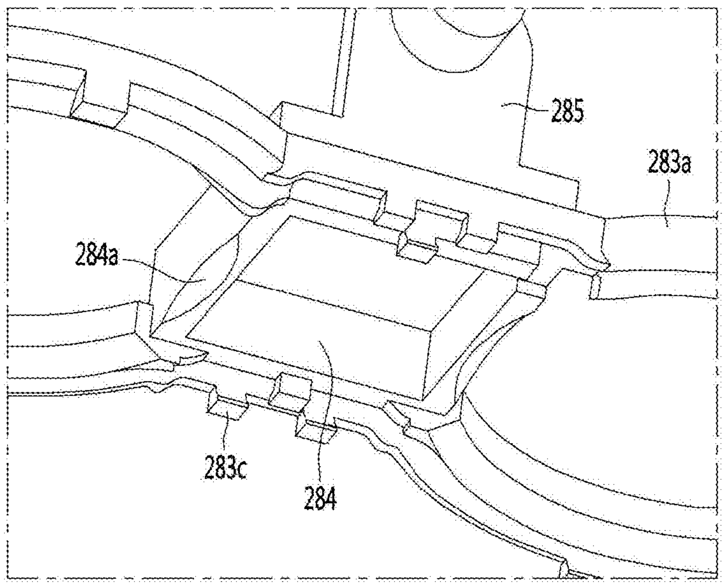


FIG. 23

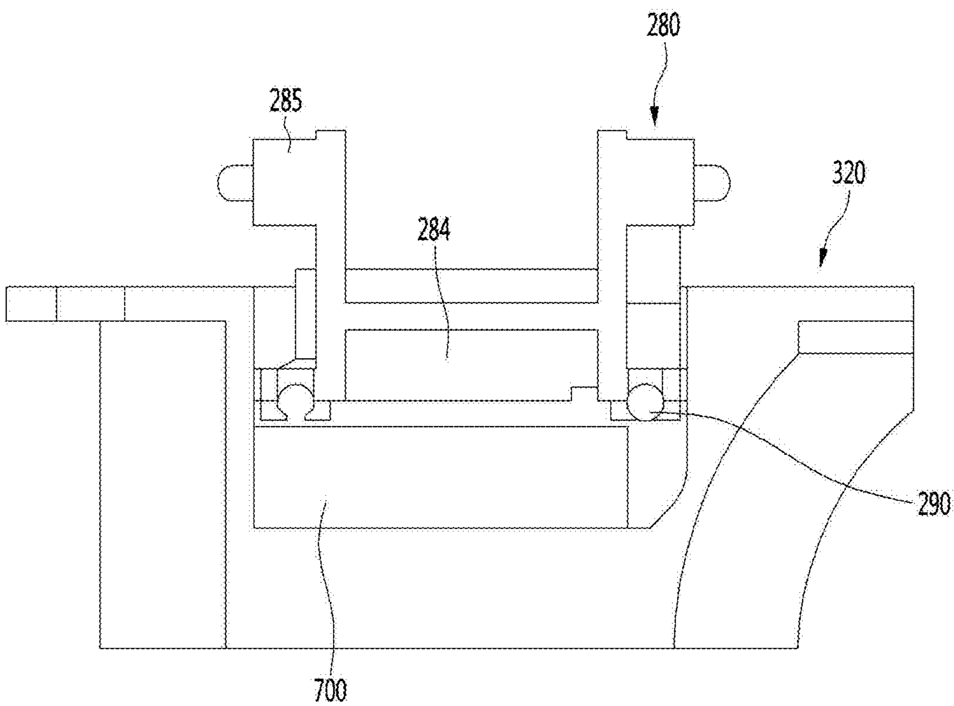


FIG. 24

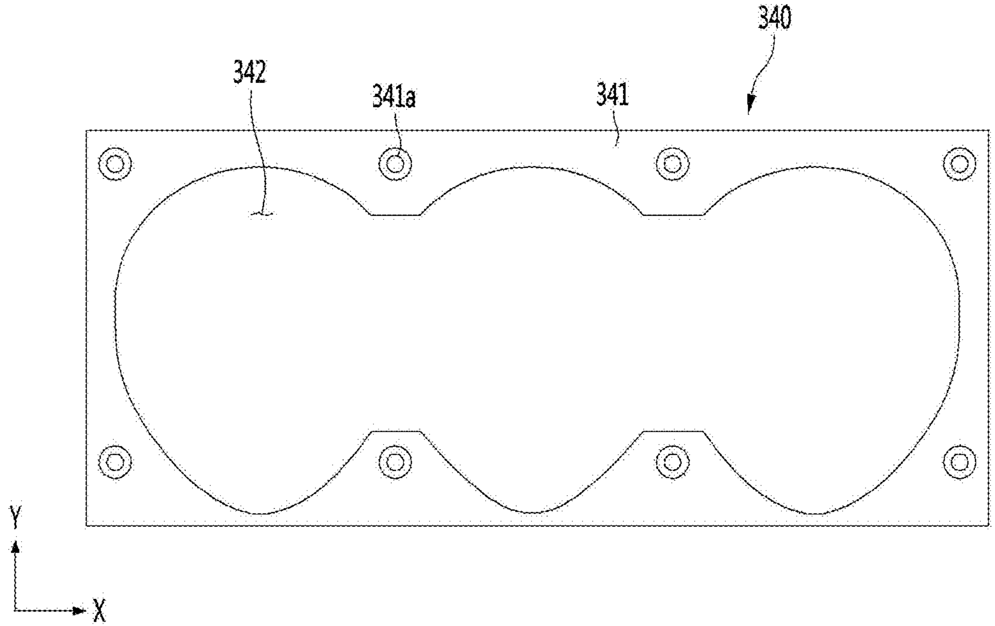

340
342
341a
341
Y
X

FIG. 25

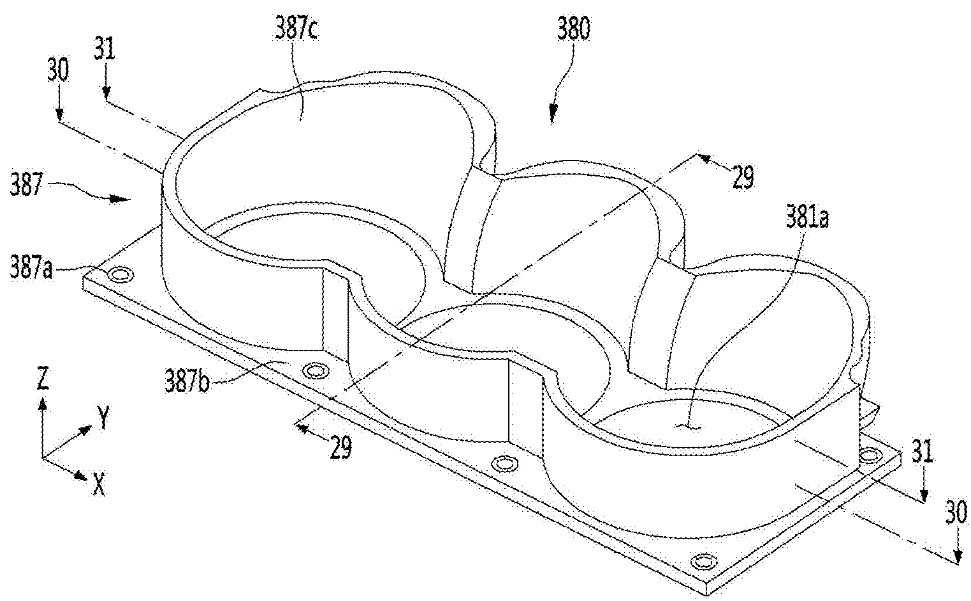

387c
380
31
30
387
29
381a
387a
387b
Z
Y
X
29
31
30

FIG. 26

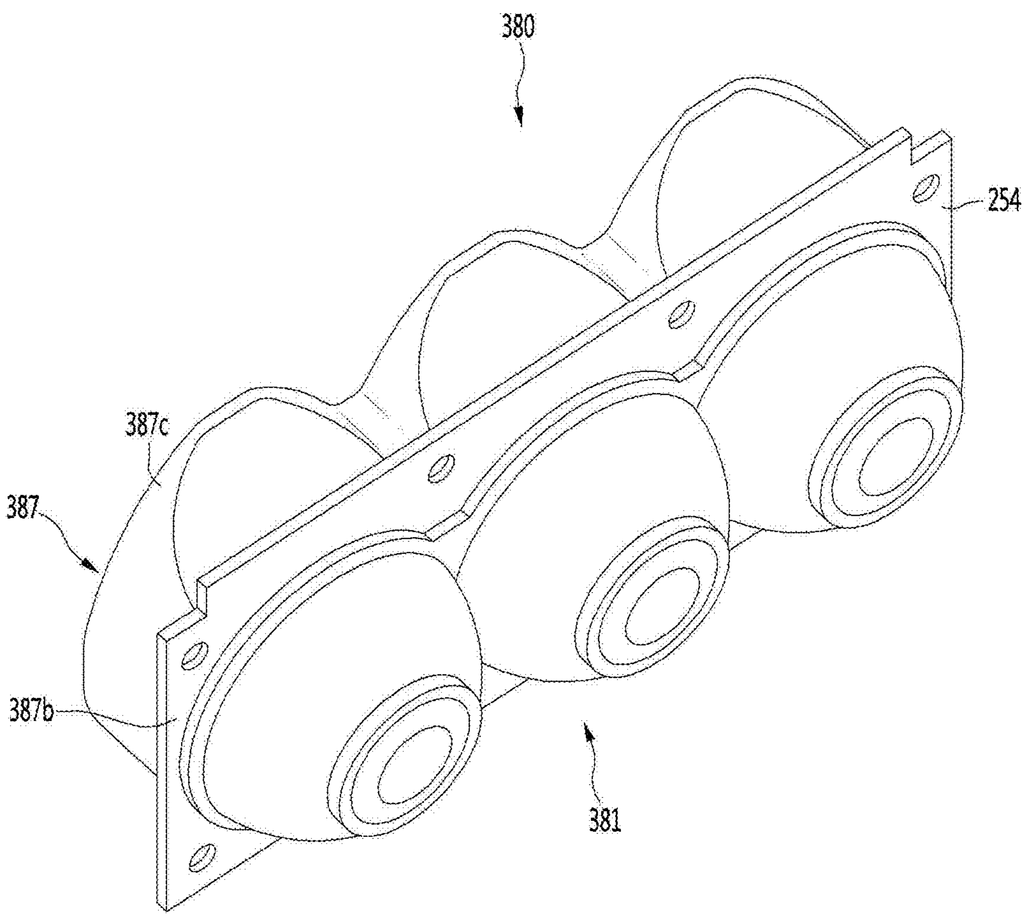


FIG. 27

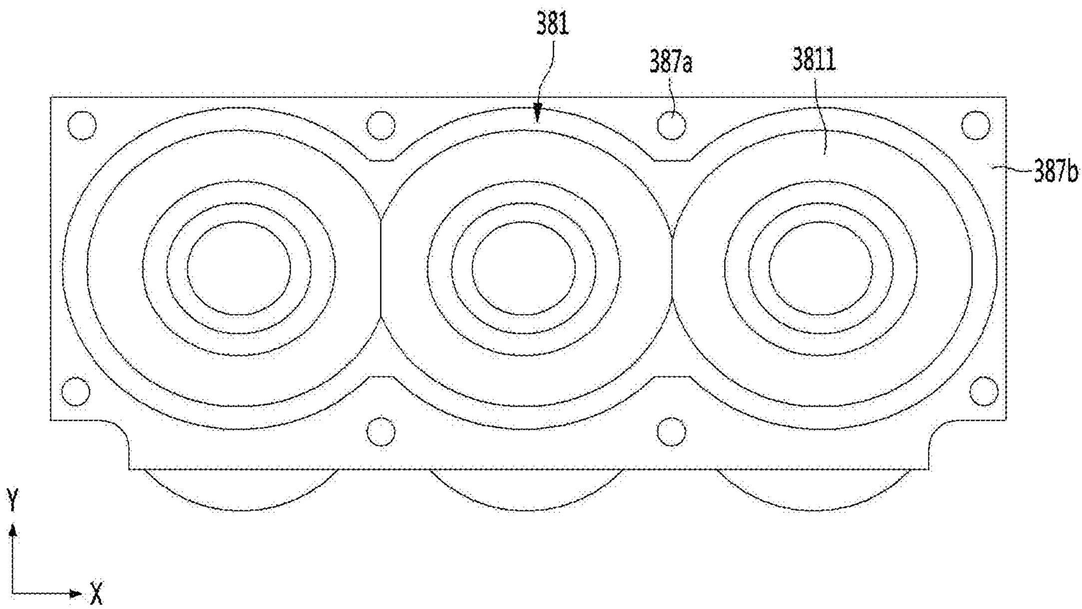


FIG. 28

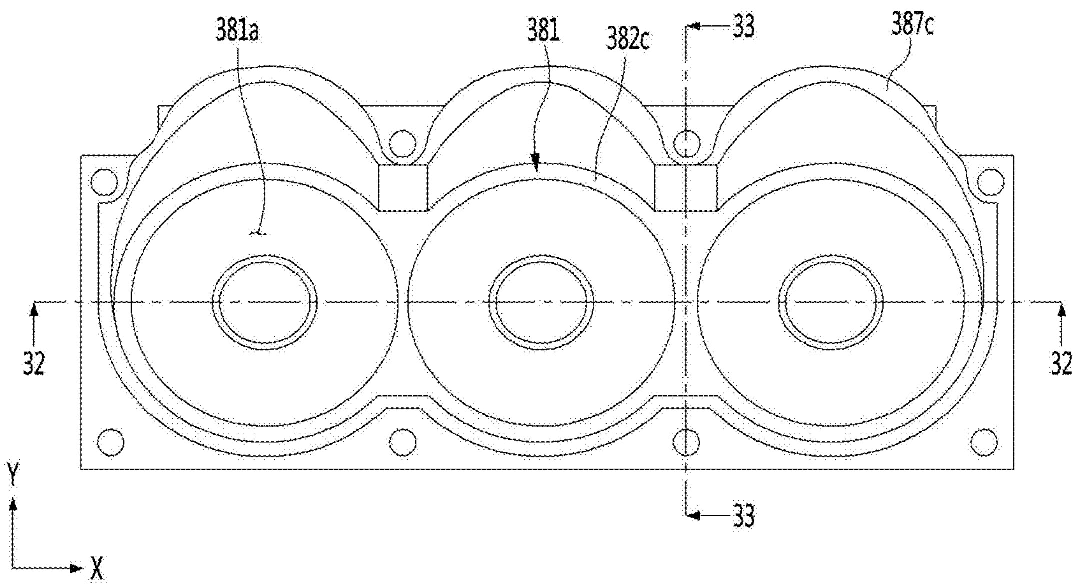


FIG. 29

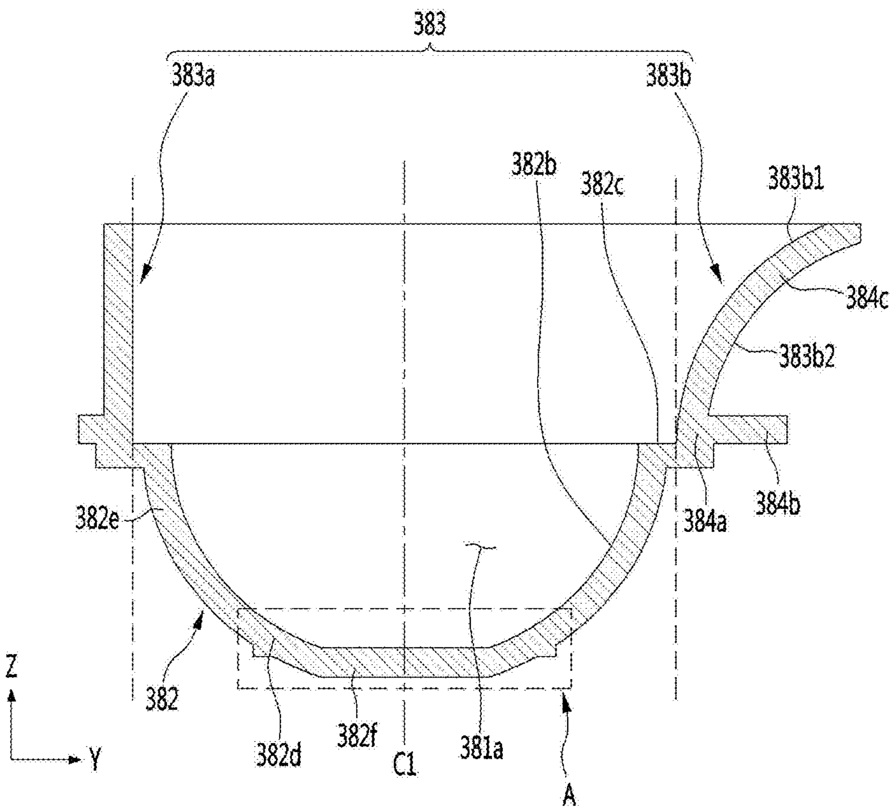

383
383a
383b
382b
382c
383b1
384c
383b2
382e
384a
384b
Z
Y
382
382d
382f
C1
381a
A

FIG. 30

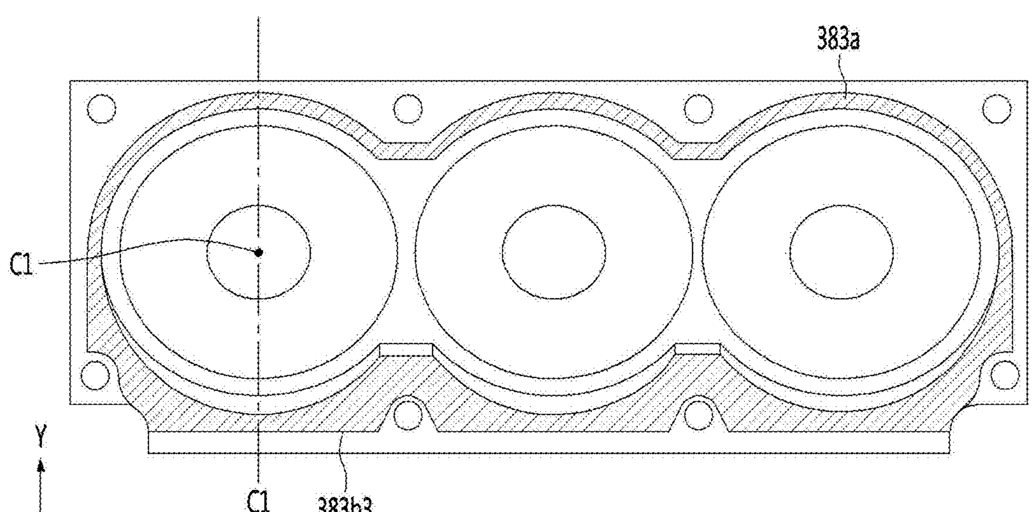

383a
C1
Y
X
C1
383b3

FIG. 31

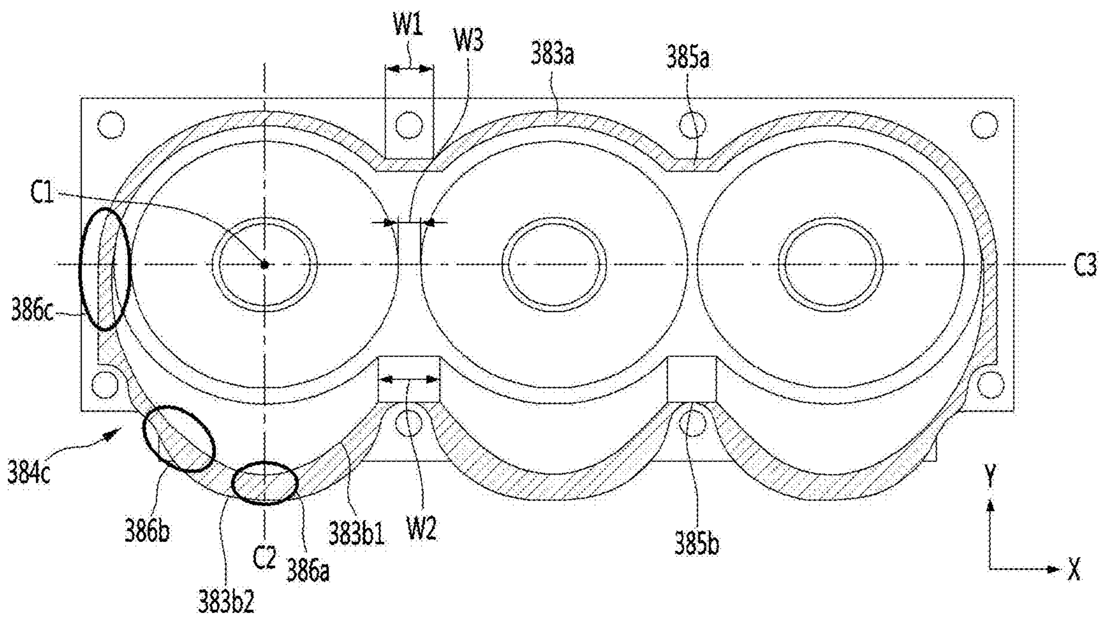

W1
W3
383a
385a
C1
C3
386c
384c
386b
383b2
C2
386a
383b1
W2
385b
Y
X

FIG. 32

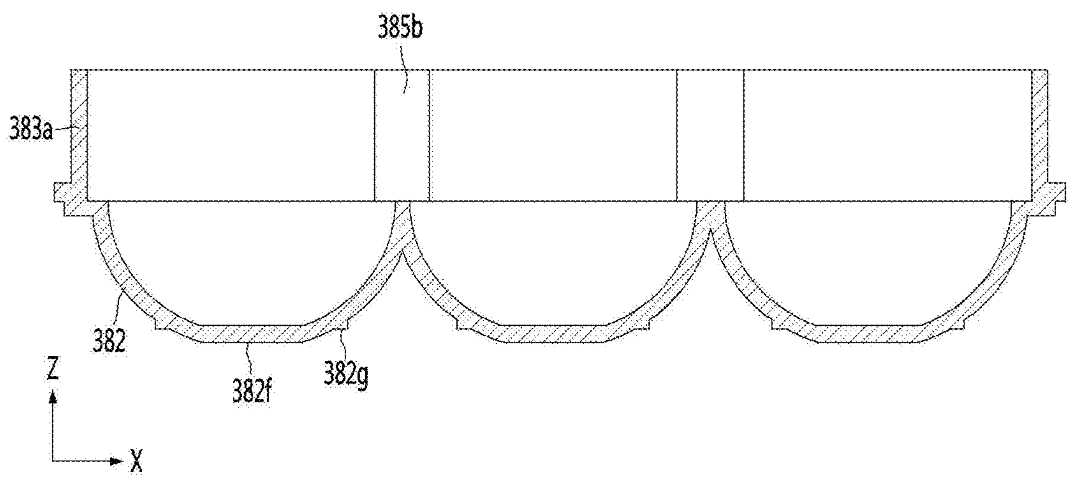

385b
383a
382
382f
382g
Z
X

FIG. 33

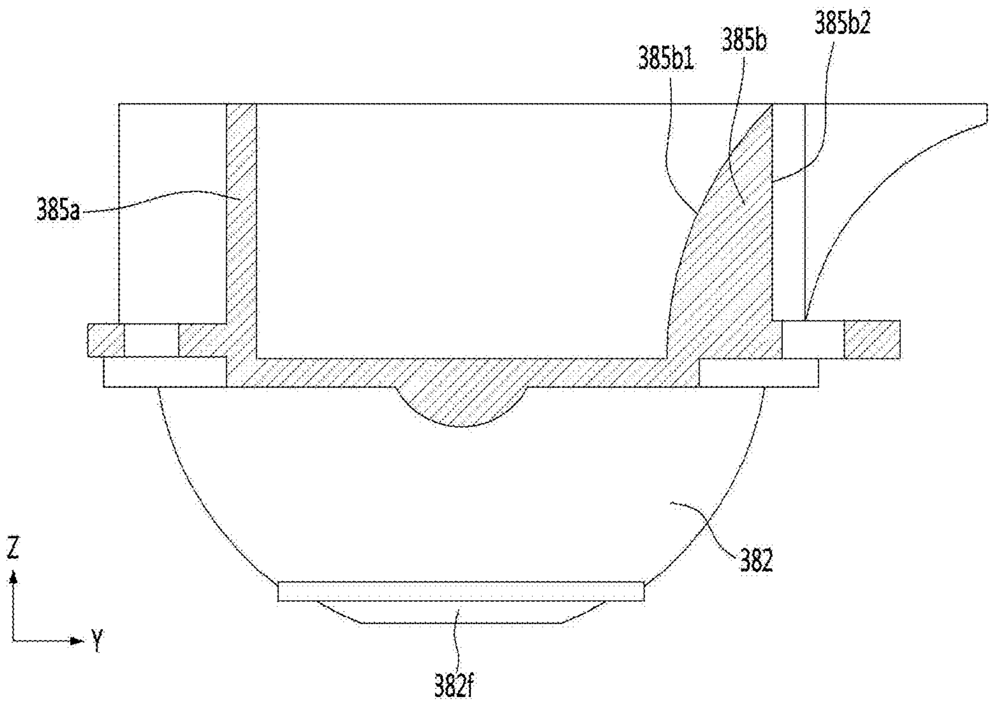

385b1
385b
385b2
385a
Z
Y
382
382f

FIG. 34

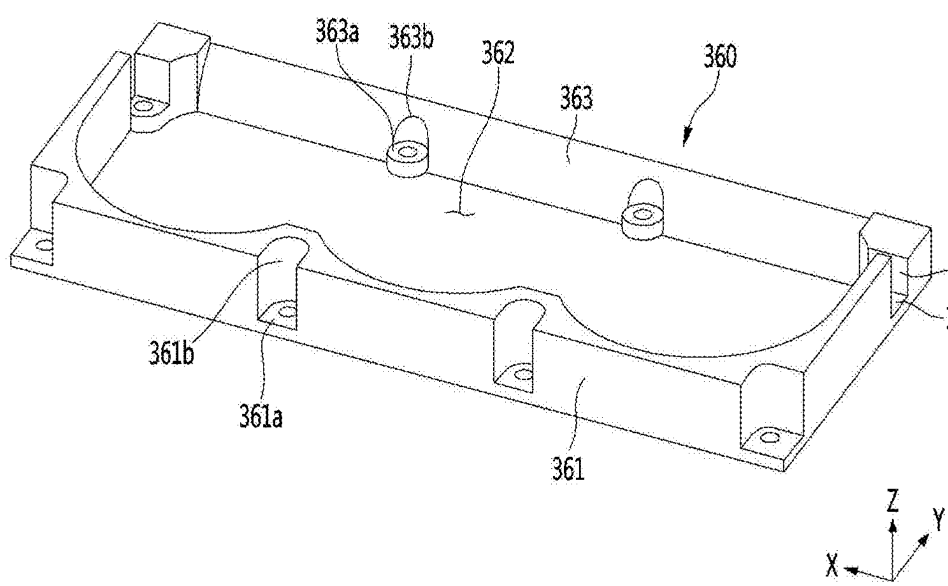

363a
363b
362
360
363
361d
361c
361b
361a
361
Z
Y
X

FIG. 35

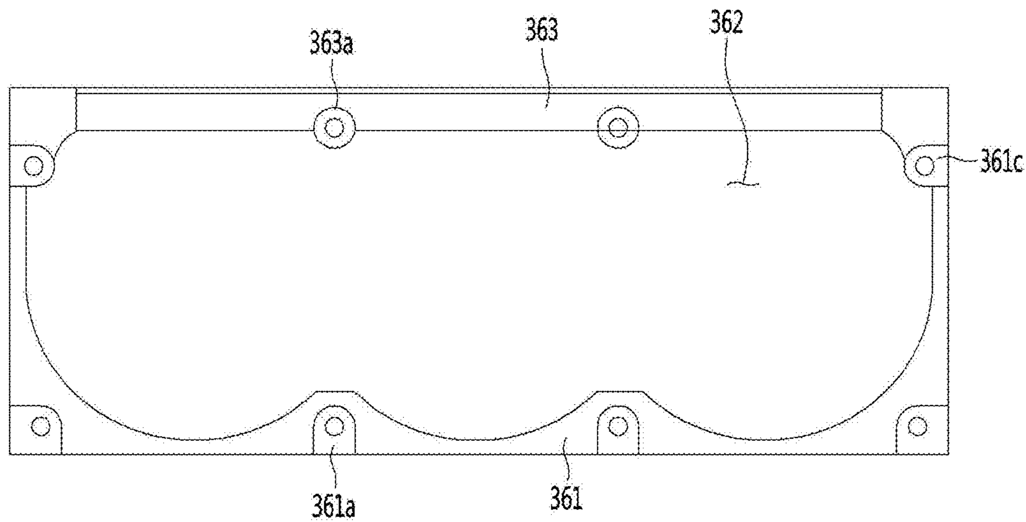

363a
363
362
361c
361a
361

FIG. 36

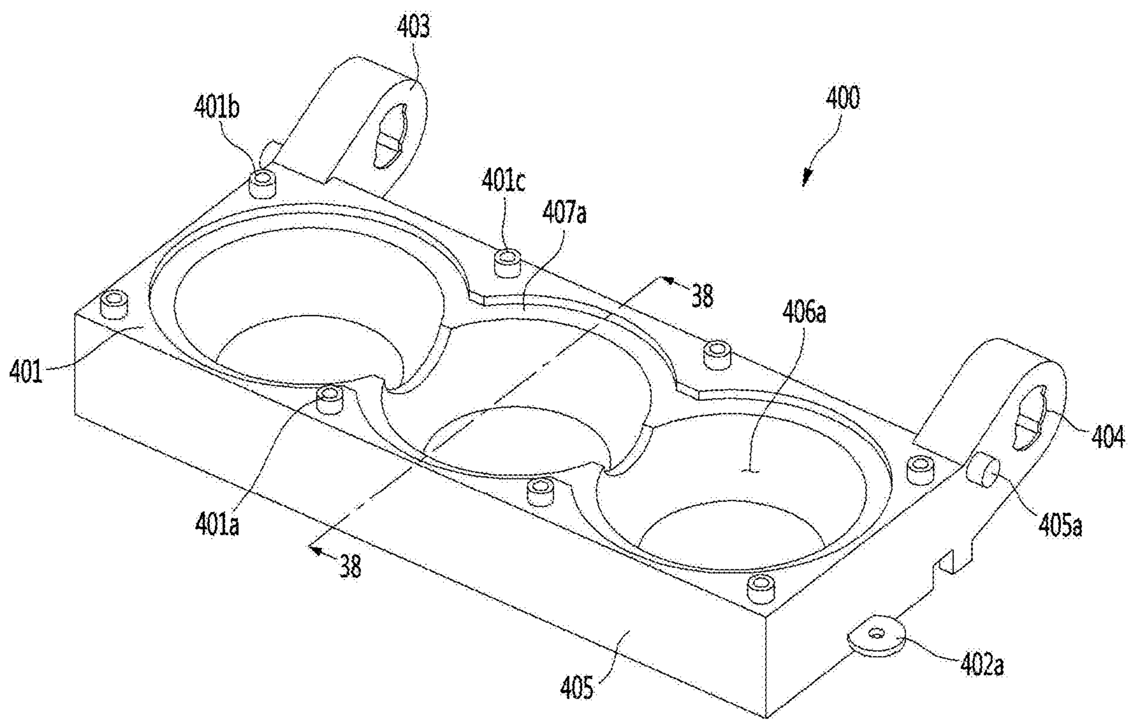

403
401b
400
401c
407a
38
406a
401
404
405a
401a
38
402a
405

FIG. 37

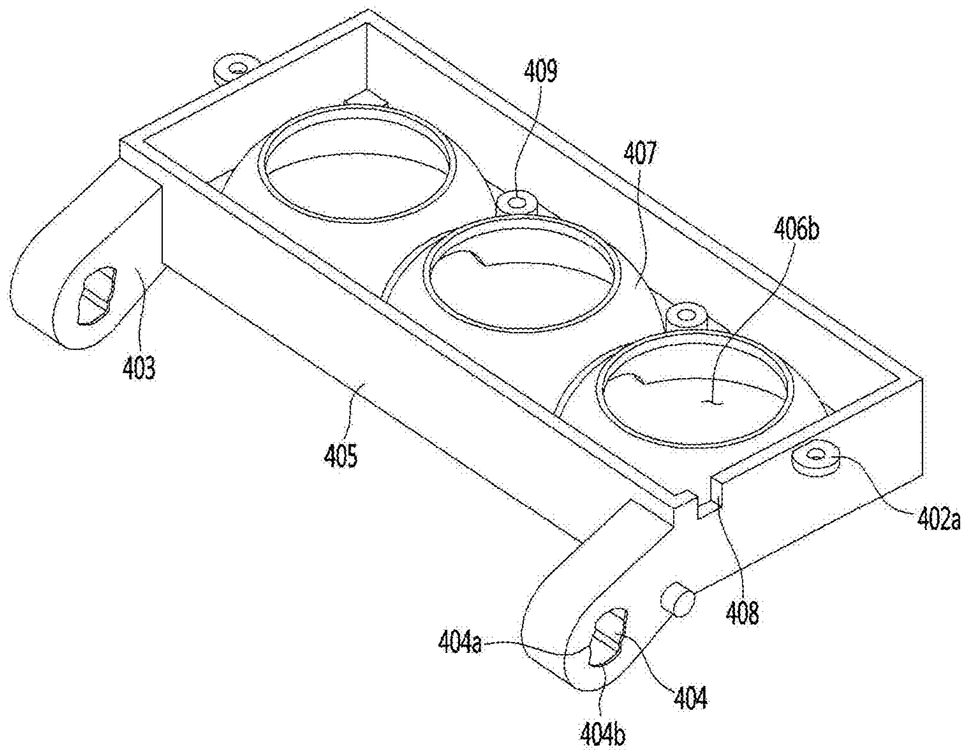

409
407
406b
403
405
402a
408
404a
404
404b

FIG. 38

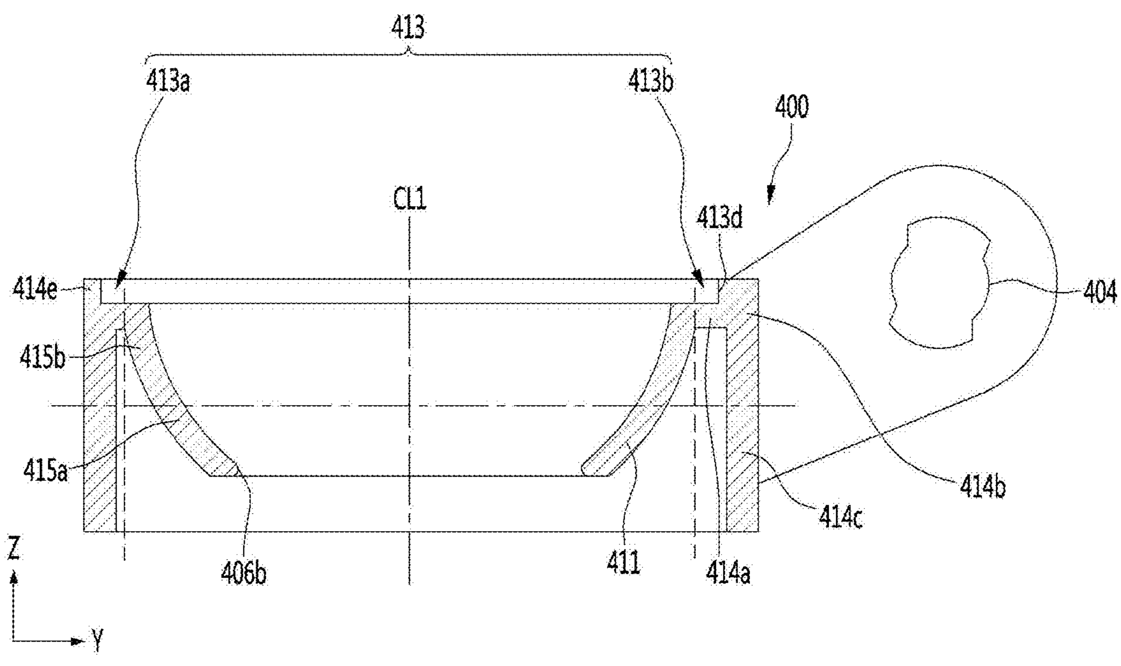

413
413a
413b
400
CL1
413d
414e
404
415b
415a
414b
414c
406b
411
414a
Z
Y

FIG. 39

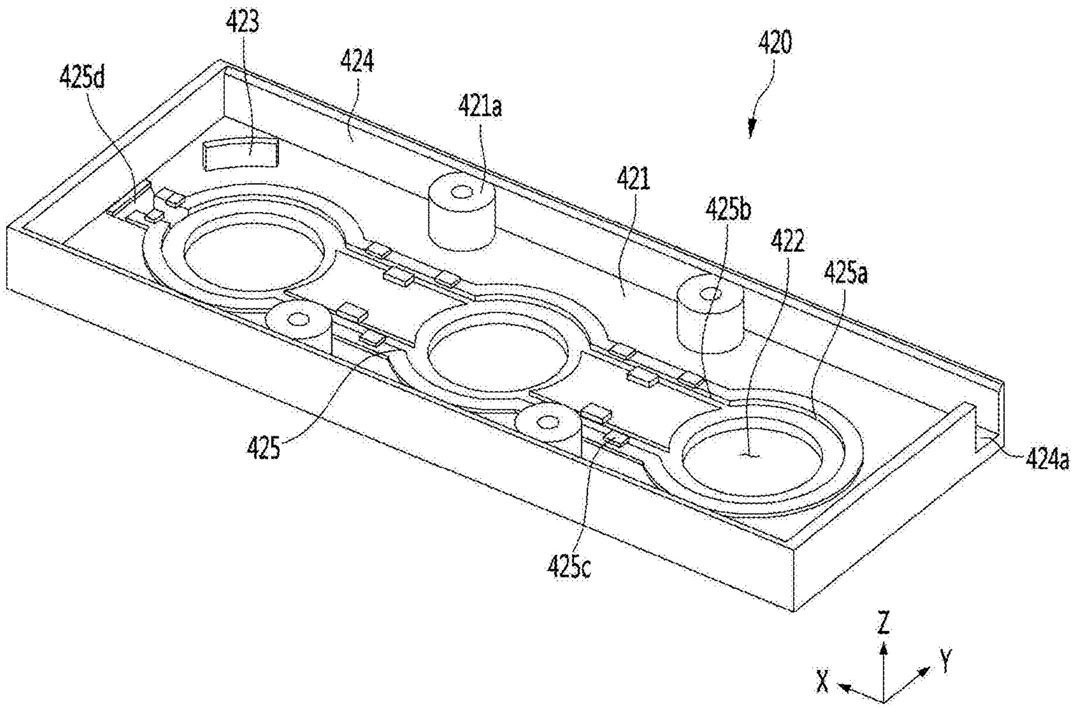

423
424
425d
420
421a
421
425b
422
425a
425
424a
425c
Z
Y
X

FIG. 40

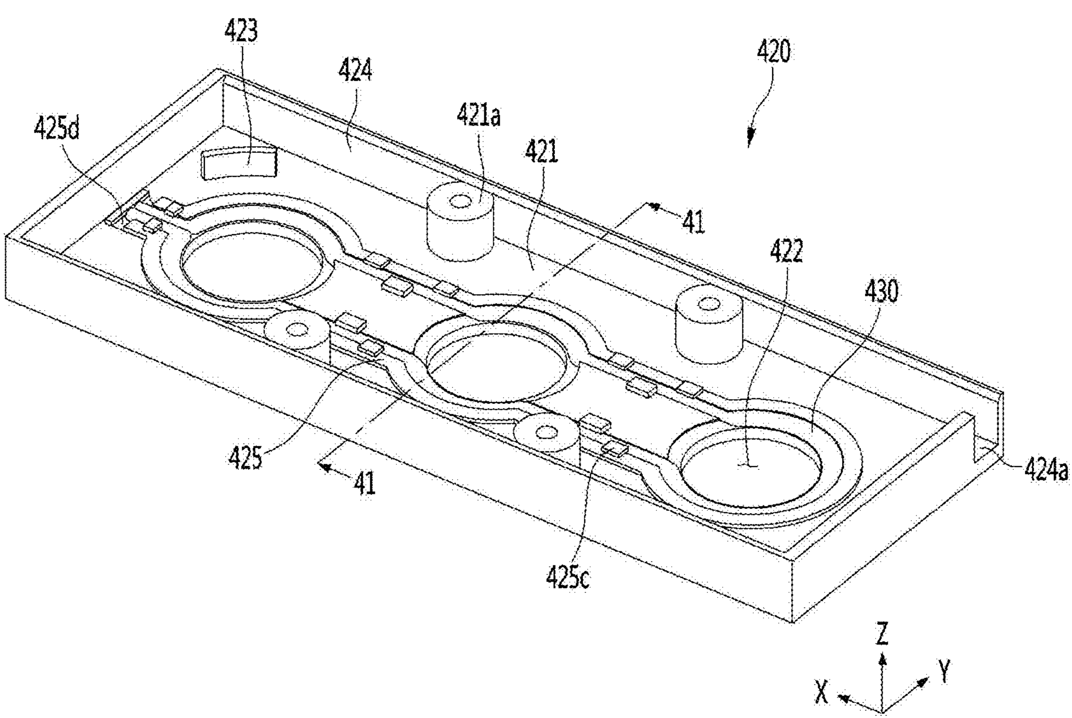

423
424
420
425d
421a
421
41
422
430
425
424a
41
425c
Z
Y
X

FIG. 41

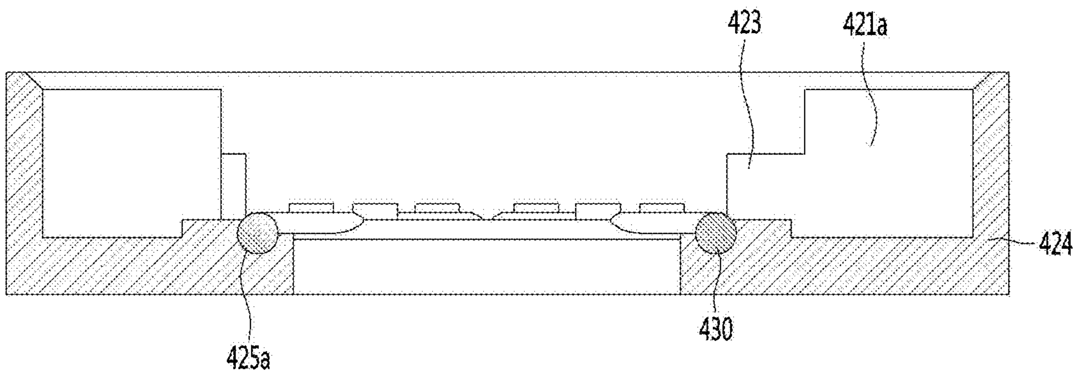


FIG. 42

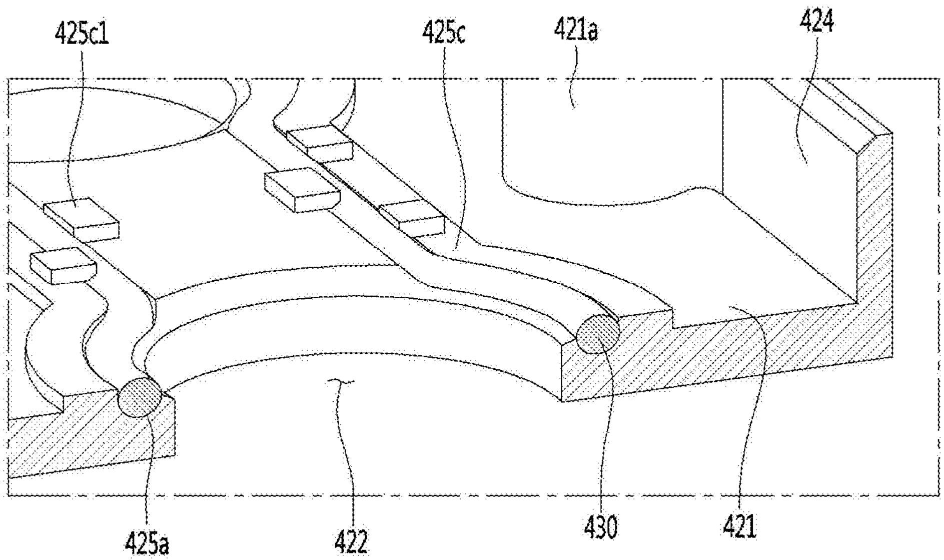
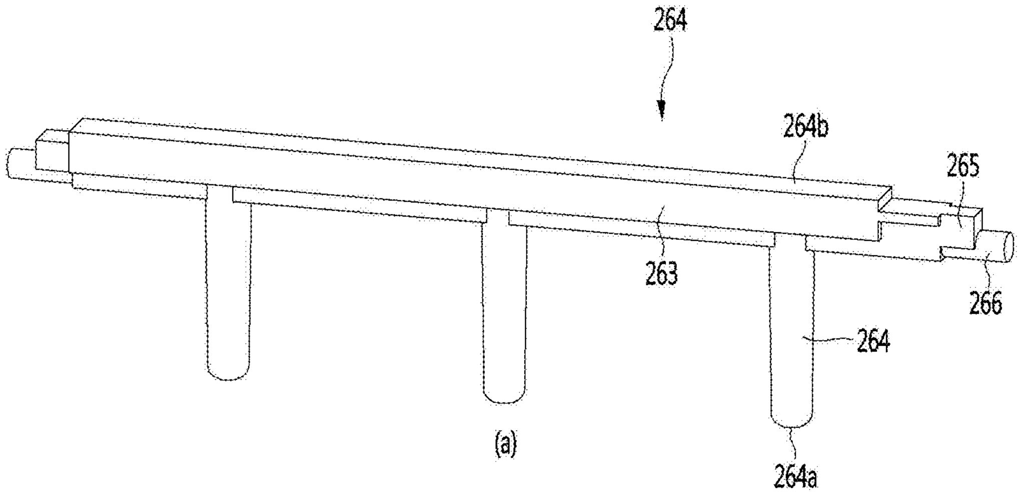
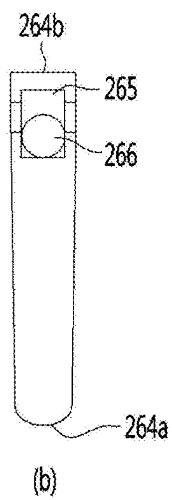


FIG. 43



(a)



(b)

FIG. 44

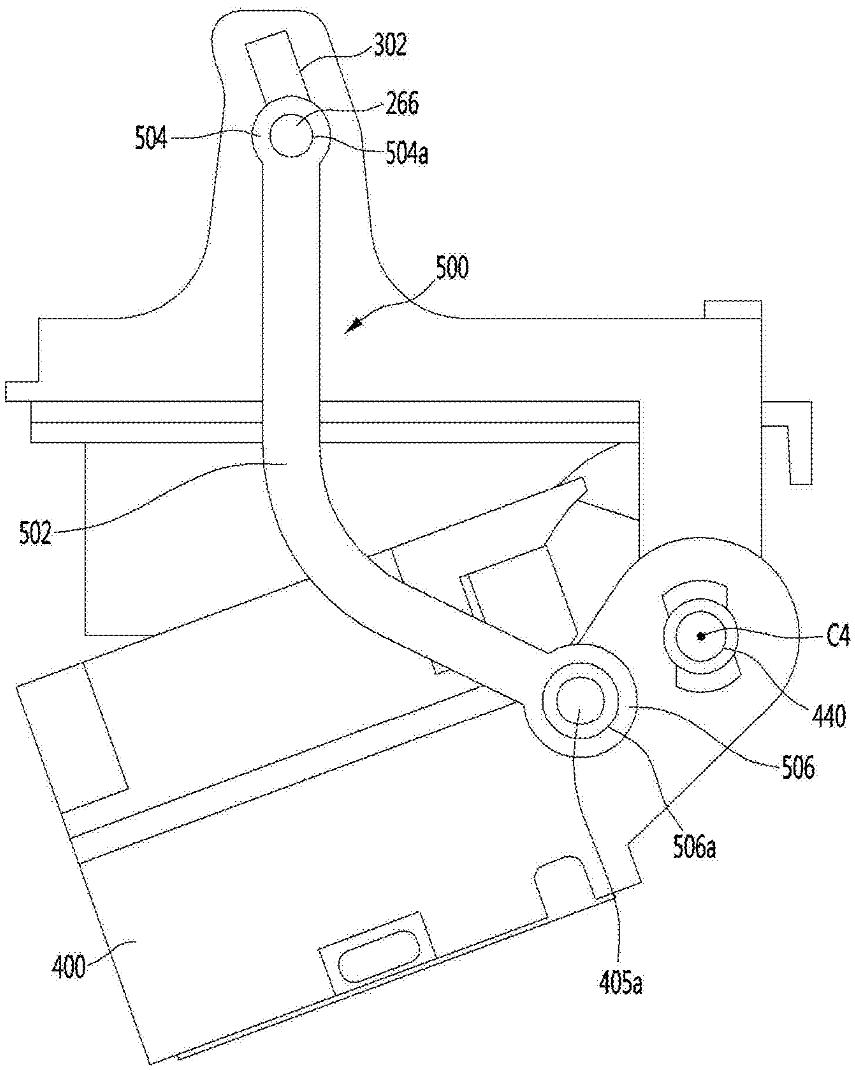


FIG. 45

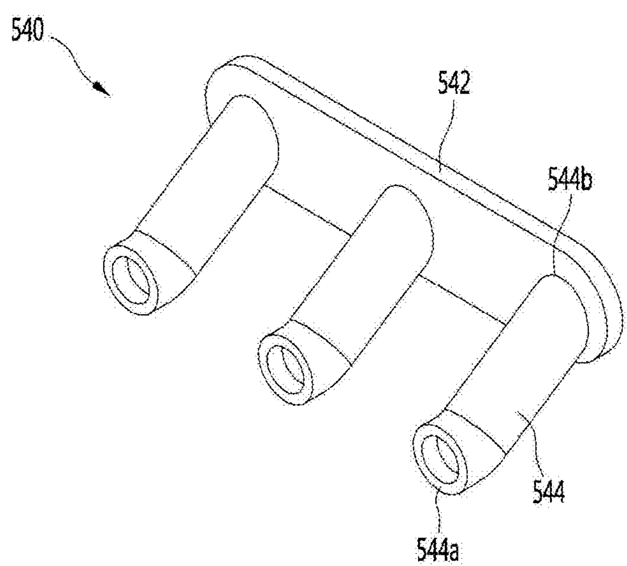


FIG. 46

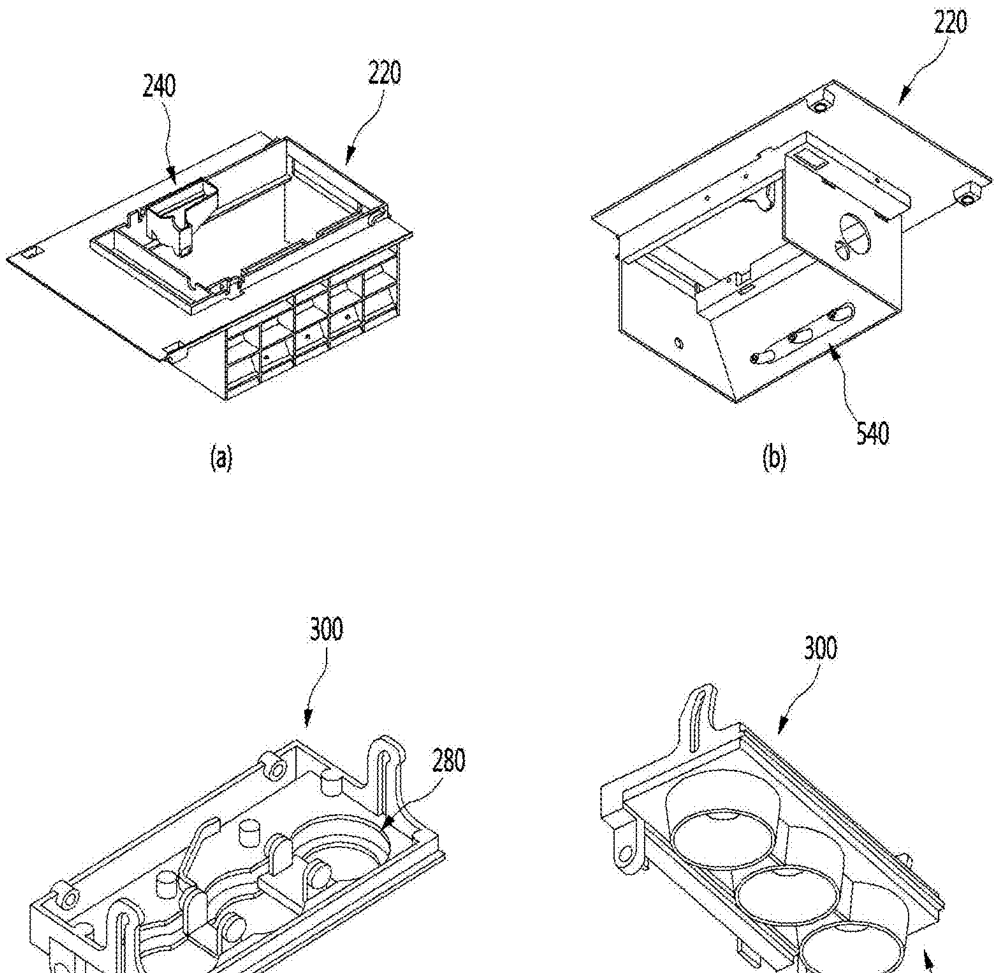


FIG. 47

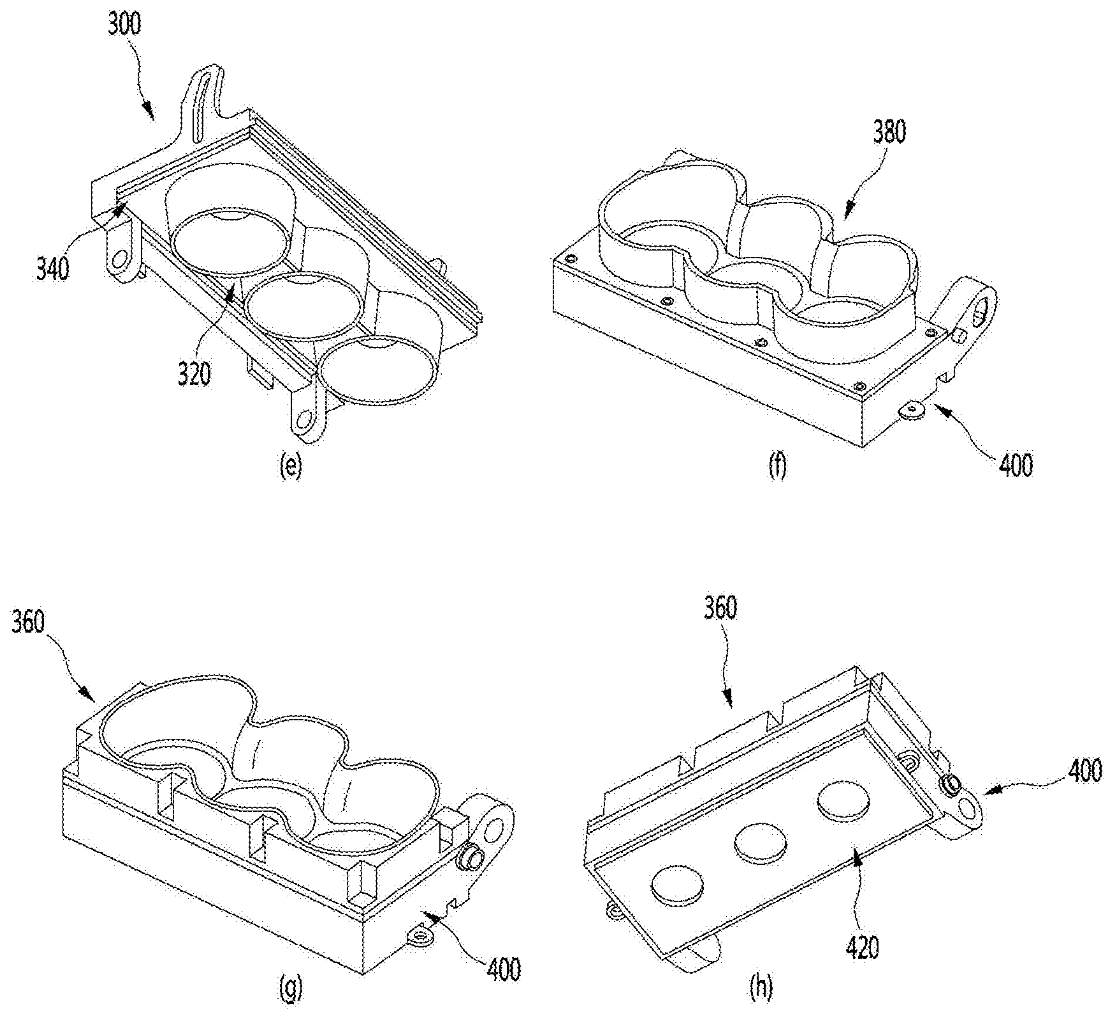


FIG. 48

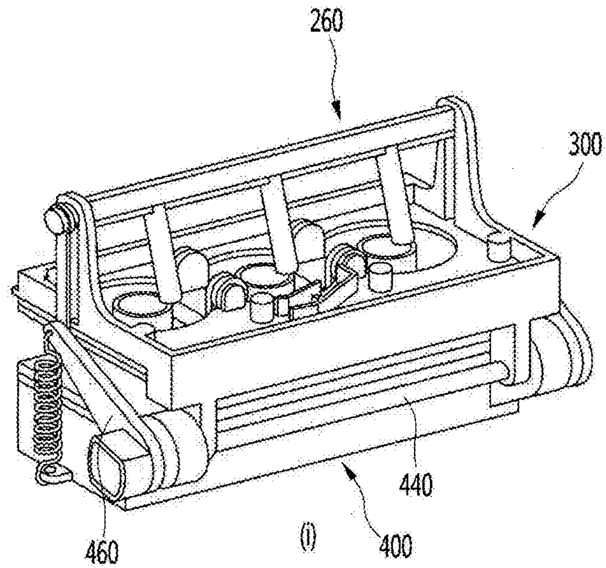

260
300
440
460
(i)
400

FIG. 49

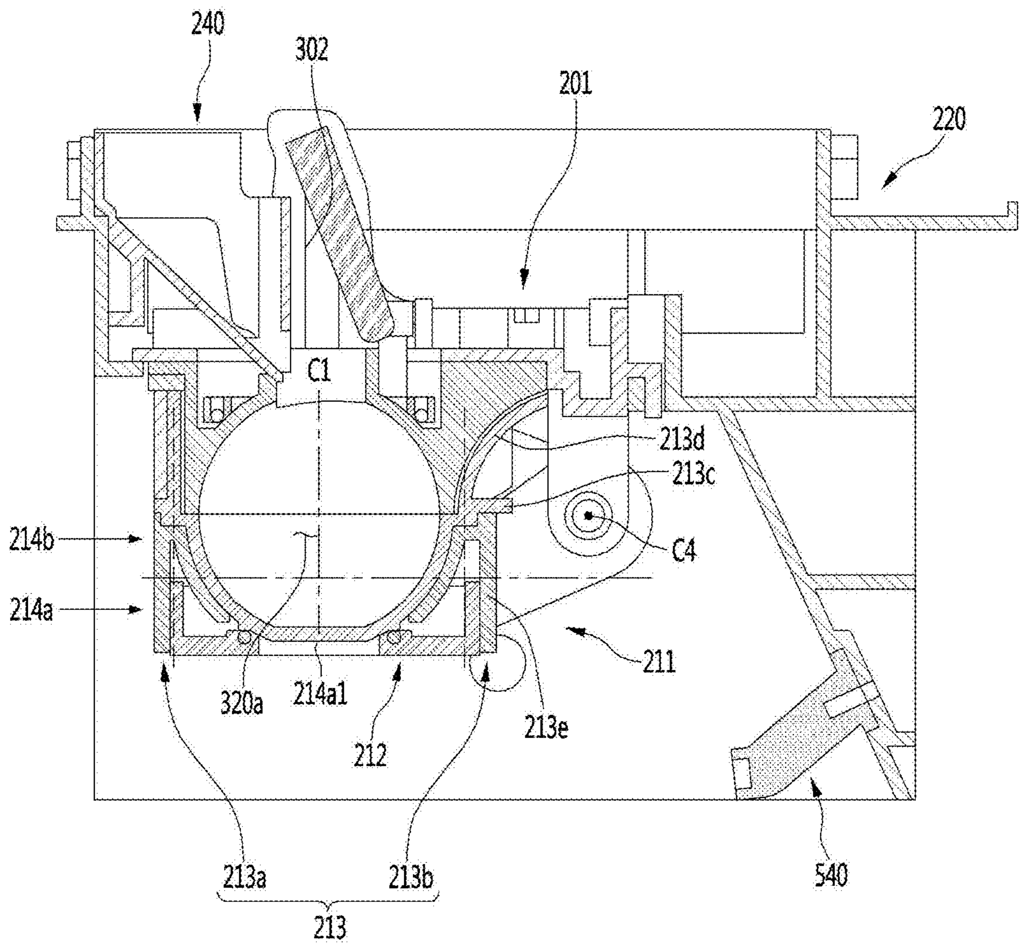

240
302
201
220
C1
213d
213c
C4
214b
214a
211
320a
214a1
212
213e
540
213a
213b
213

FIG. 50

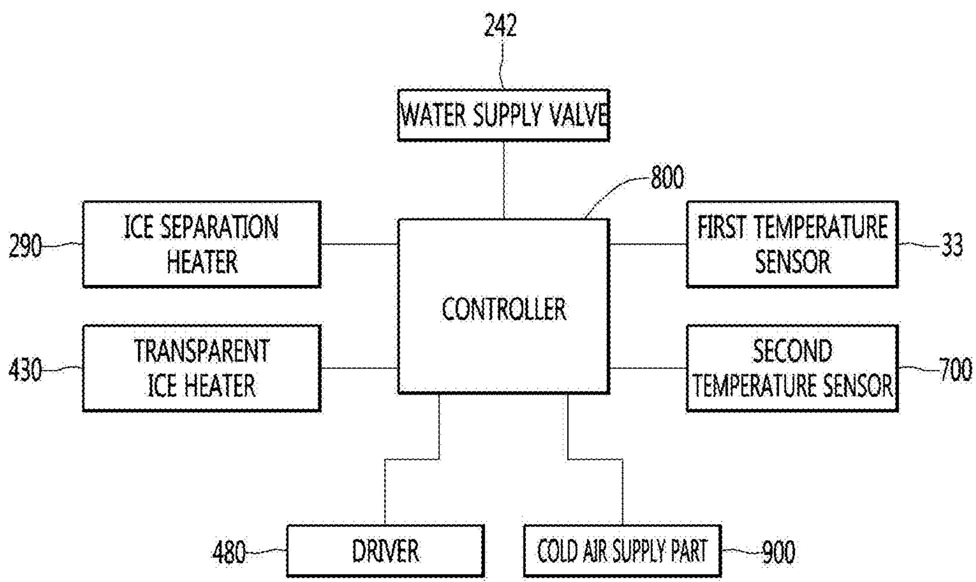

242
WATER SUPPLY VALVE
800
290
ICE SEPARATION HEATER
CONTROLLER
FIRST TEMPERATURE SENSOR
33
430
TRANSPARENT ICE HEATER
SECOND TEMPERATURE SENSOR
700
480
DRIVER
COLD AIR SUPPLY PART
900

FIG. 51

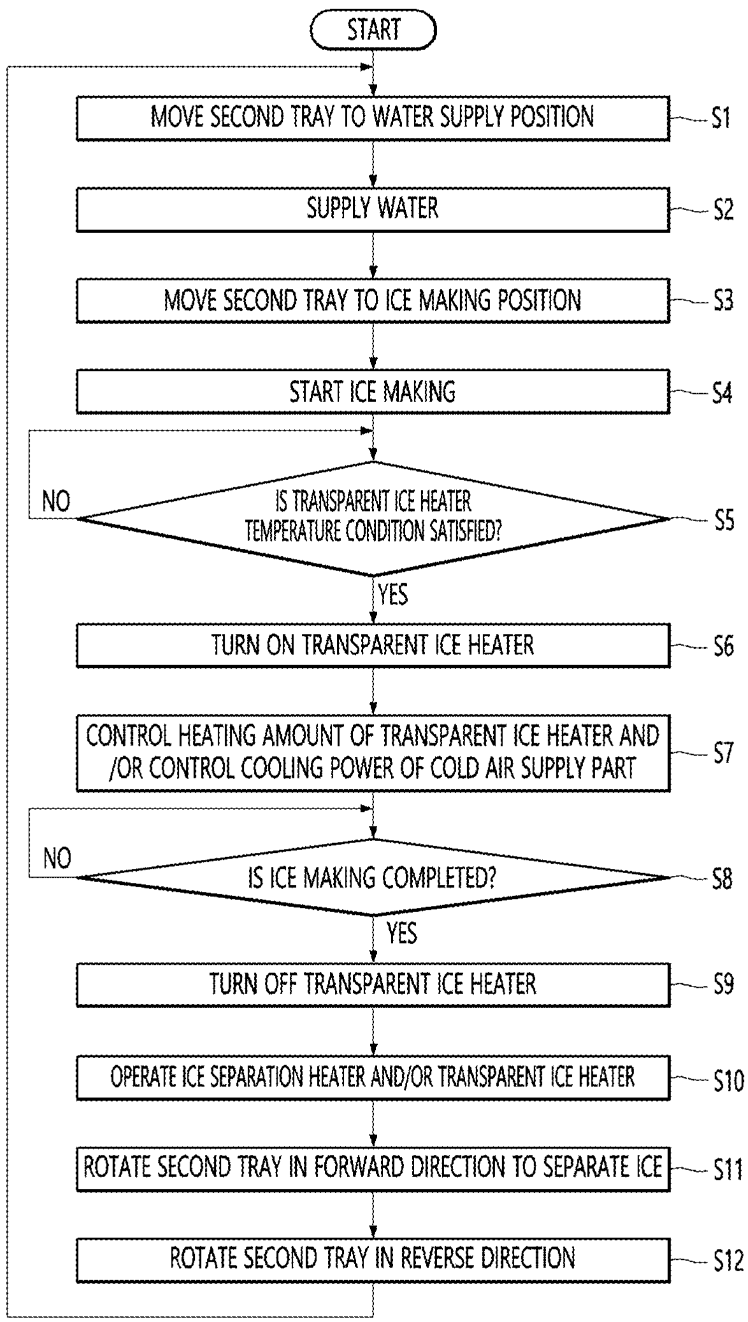

START
MOVE SECOND TRAY TO WATER SUPPLY POSITION
S1
SUPPLY WATER
S2
MOVE SECOND TRAY TO ICE MAKING POSITION
S3
START ICE MAKING
S4
NO
IS TRANSPARENT ICE HEATER
TEMPERATURE CONDITION SATISFIED?
S5
YES
TURN ON TRANSPARENT ICE HEATER
S6
CONTROL HEATING AMOUNT OF TRANSPARENT ICE HEATER AND
/OR CONTROL COOLING POWER OF COLD AIR SUPPLY PART
S7
NO
IS ICE MAKING COMPLETED?
S8
YES
TURN OFF TRANSPARENT ICE HEATER
S9
OPERATE ICE SEPARATION HEATER AND/OR TRANSPARENT ICE HEATER
S10
ROTATE SECOND TRAY IN FORWARD DIRECTION TO SEPARATE ICE
S11
ROTATE SECOND TRAY IN REVERSE DIRECTION
S12

FIG. 52

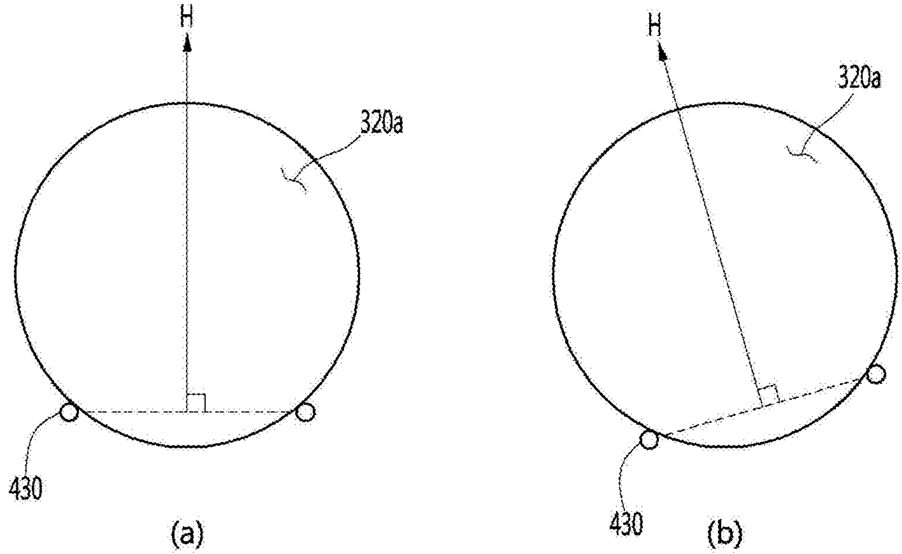

H
320a
430
(a)
H
320a
430
(b)

FIG. 53

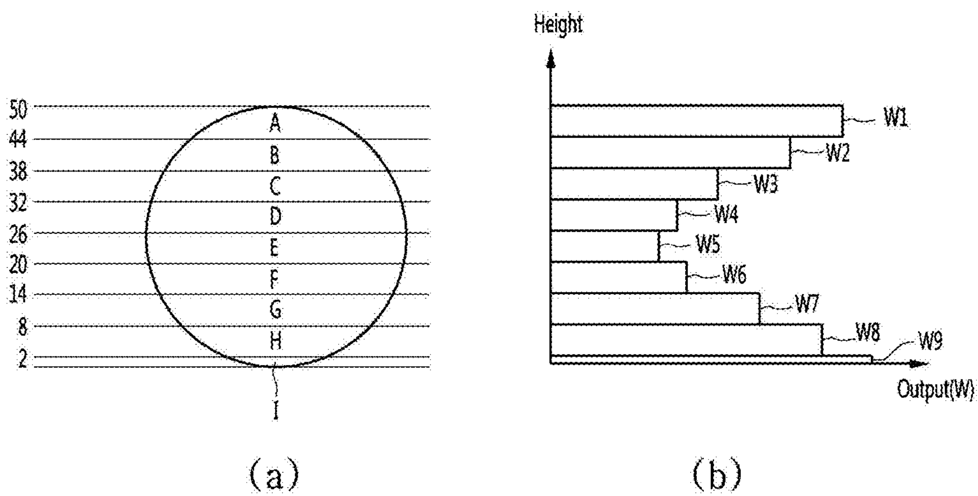

50
44
38
32
26
20
14
8
2
A
B
C
D
E
F
G
H
I
(a)
Height
W1
W2
W3
W4
W5
W6
W7
W8
W9
Output(W)
(b)

FIG. 54

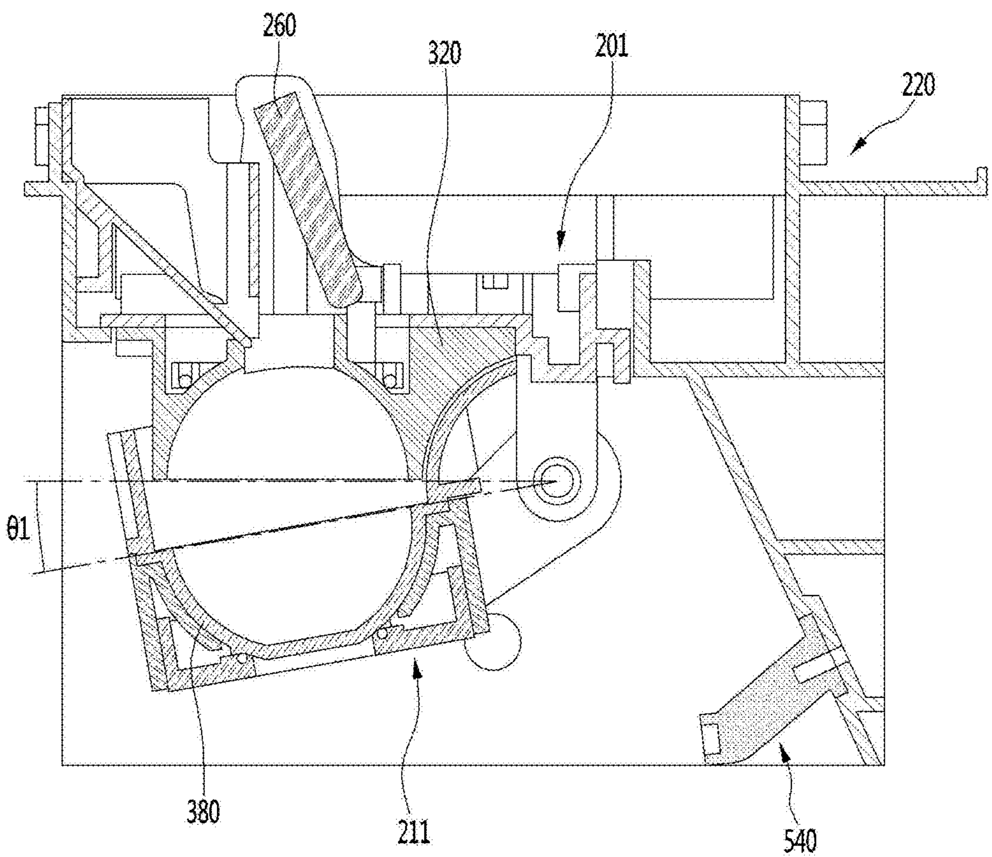
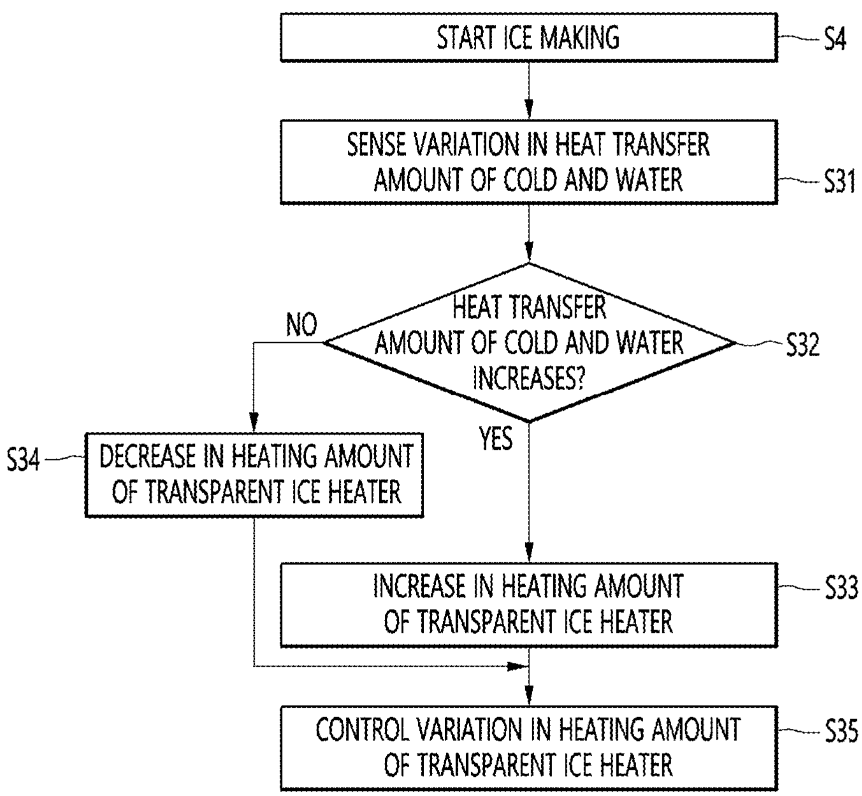


FIG. 55



START ICE MAKING
S4
SENSE VARIATION IN HEAT TRANSFER AMOUNT OF COLD AND WATER
S31
HEAT TRANSFER AMOUNT OF COLD AND WATER INCREASES?
S32
NO
YES
DECREASE IN HEATING AMOUNT OF TRANSPARENT ICE HEATER
S34
INCREASE IN HEATING AMOUNT OF TRANSPARENT ICE HEATER
S33
CONTROL VARIATION IN HEATING AMOUNT OF TRANSPARENT ICE HEATER
S35

FIG. 56

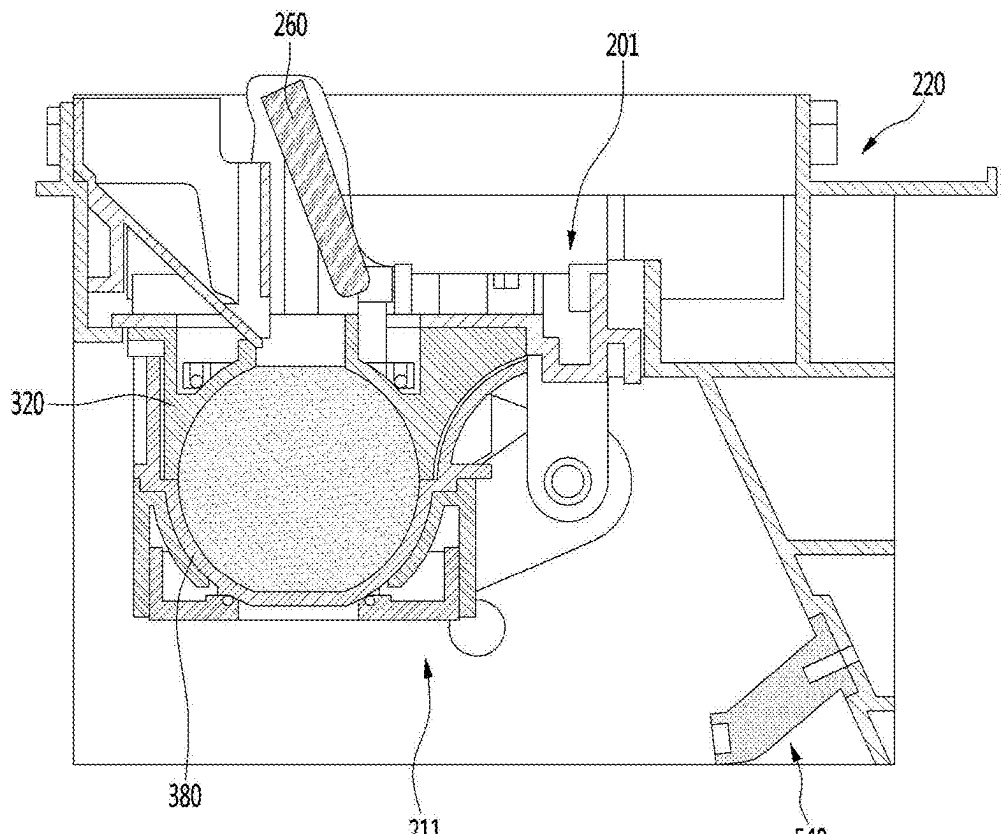


FIG. 57

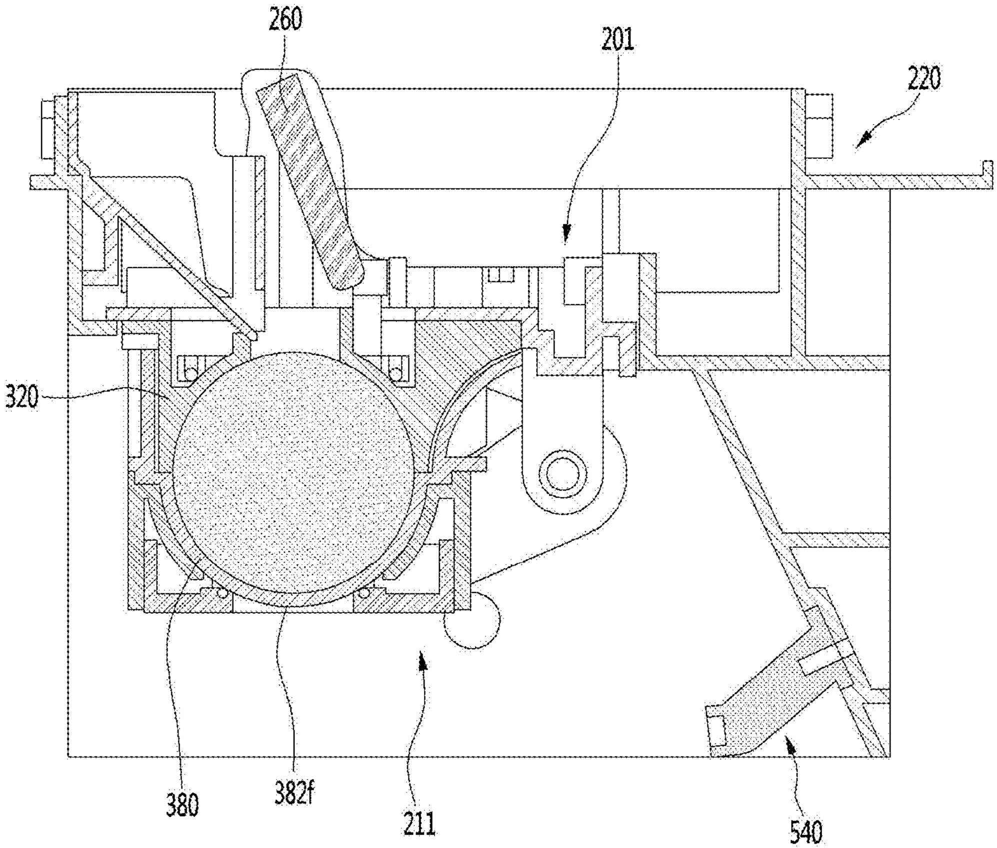


FIG. 58

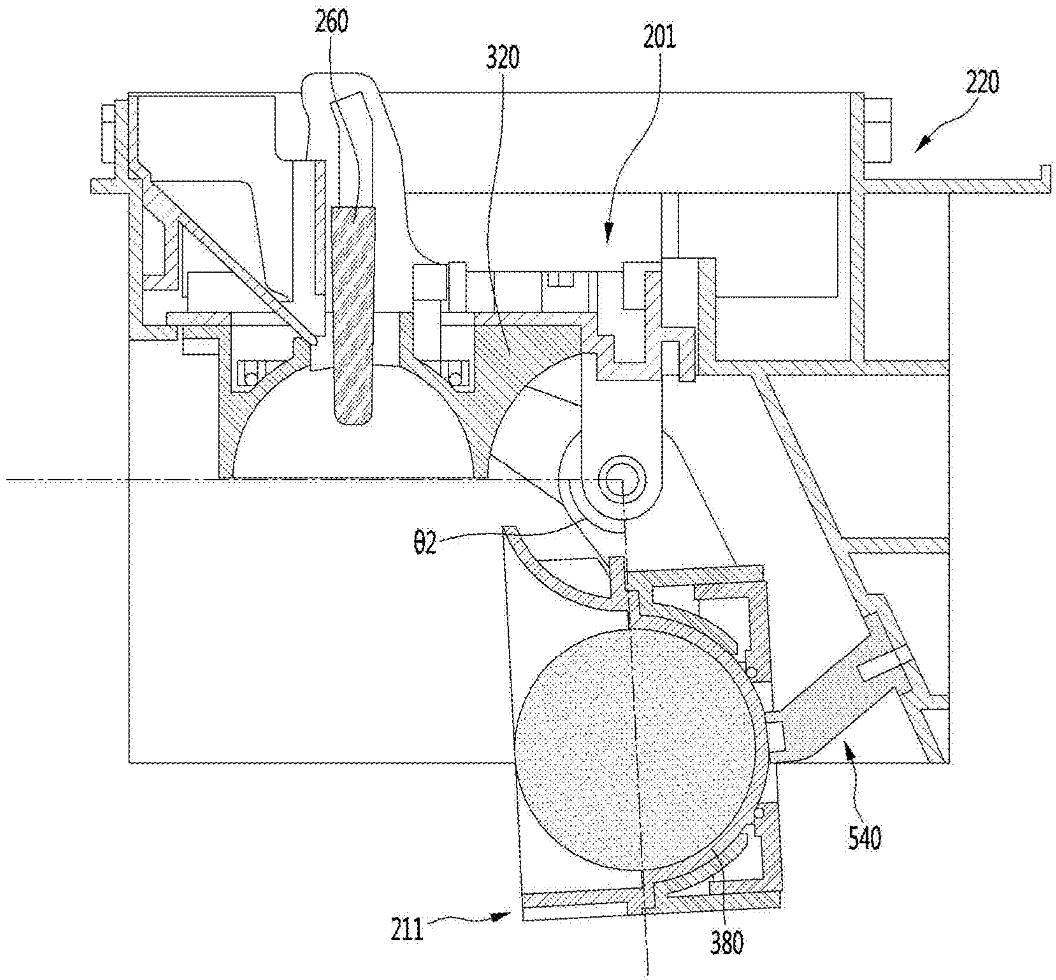


FIG. 59

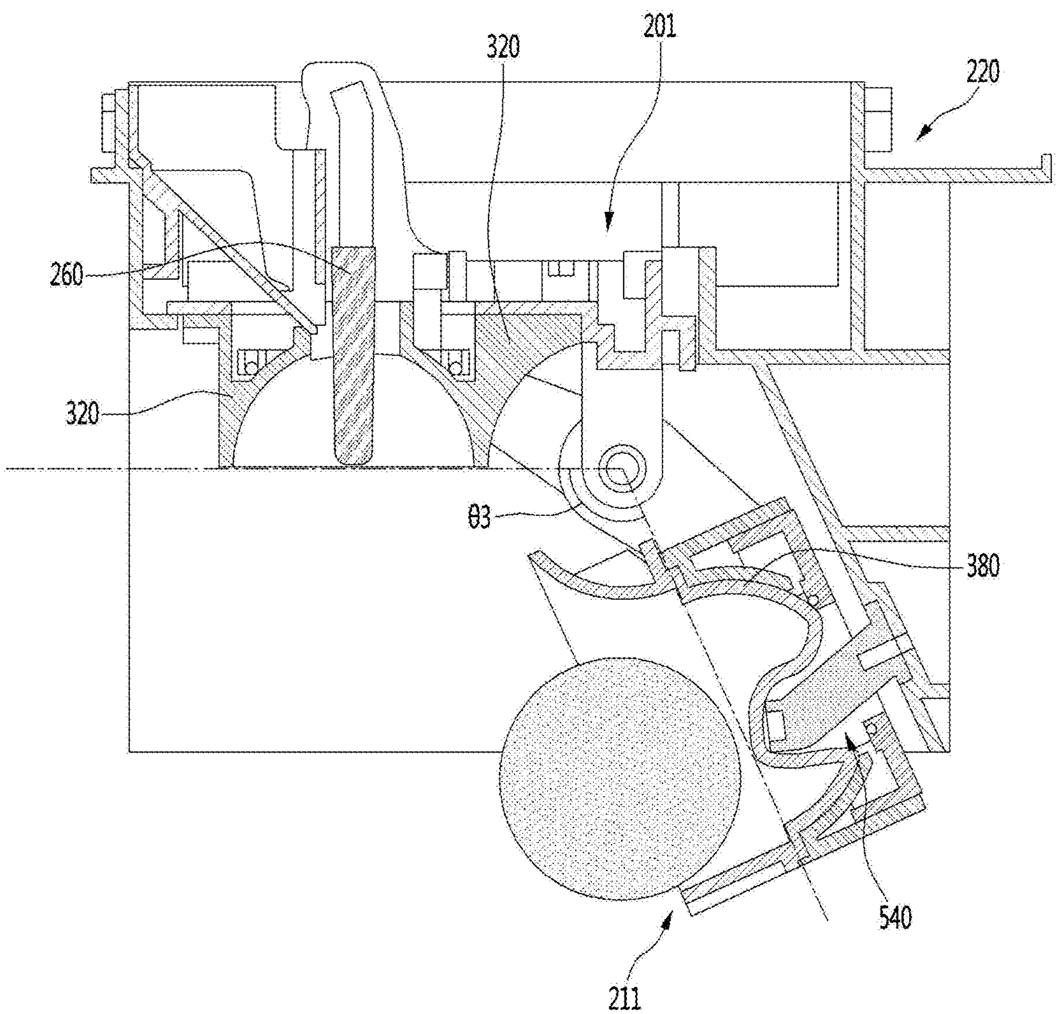


FIG. 60

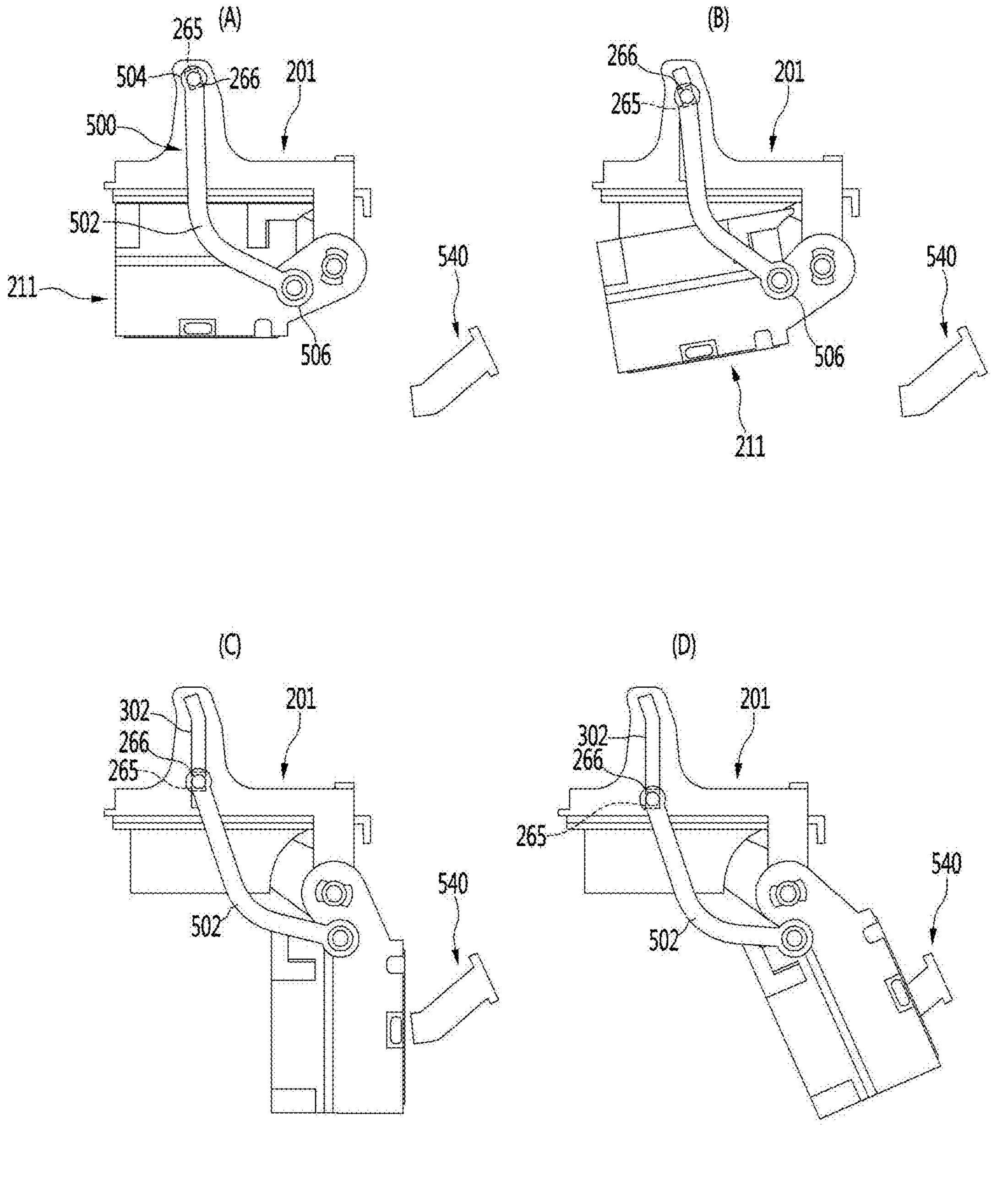

(A)
265
504
266
201
500
502
211
506
540
(B)
266
265
201
506
211
540
(C)
201
302
266
265
540
502
(D)
201
302
266
265
540
502

FIG. 61

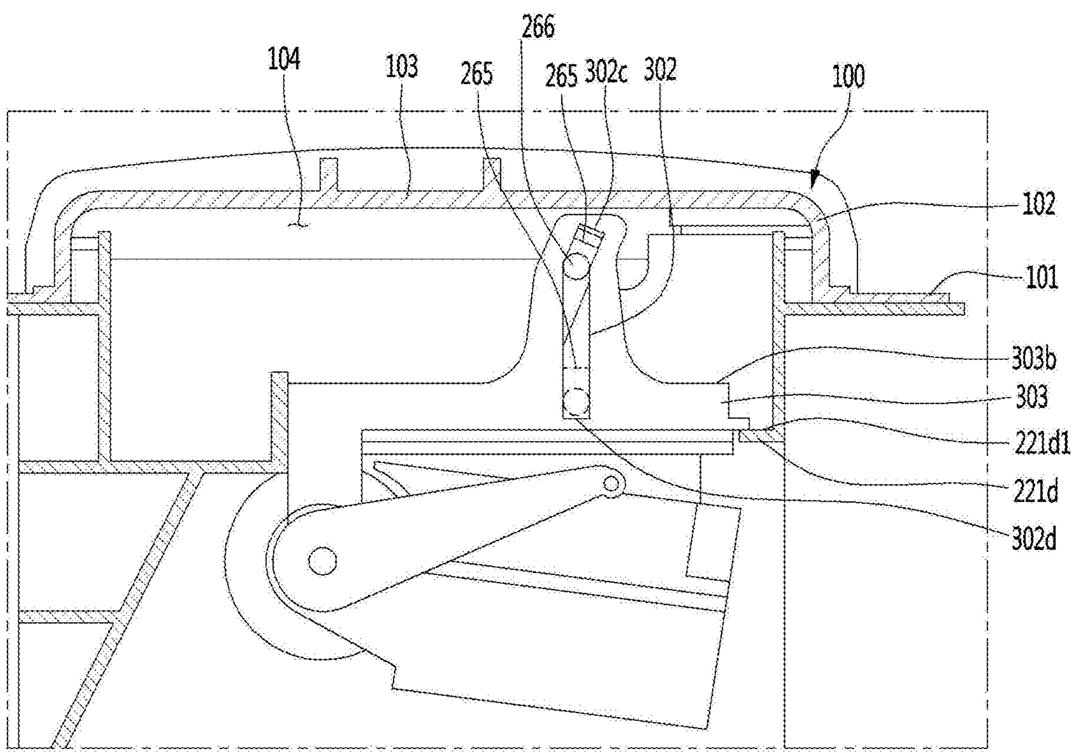

266
104
103
265
265
302c
302
100
102
101
303b
303
221d1
221d
302d

FIG. 62

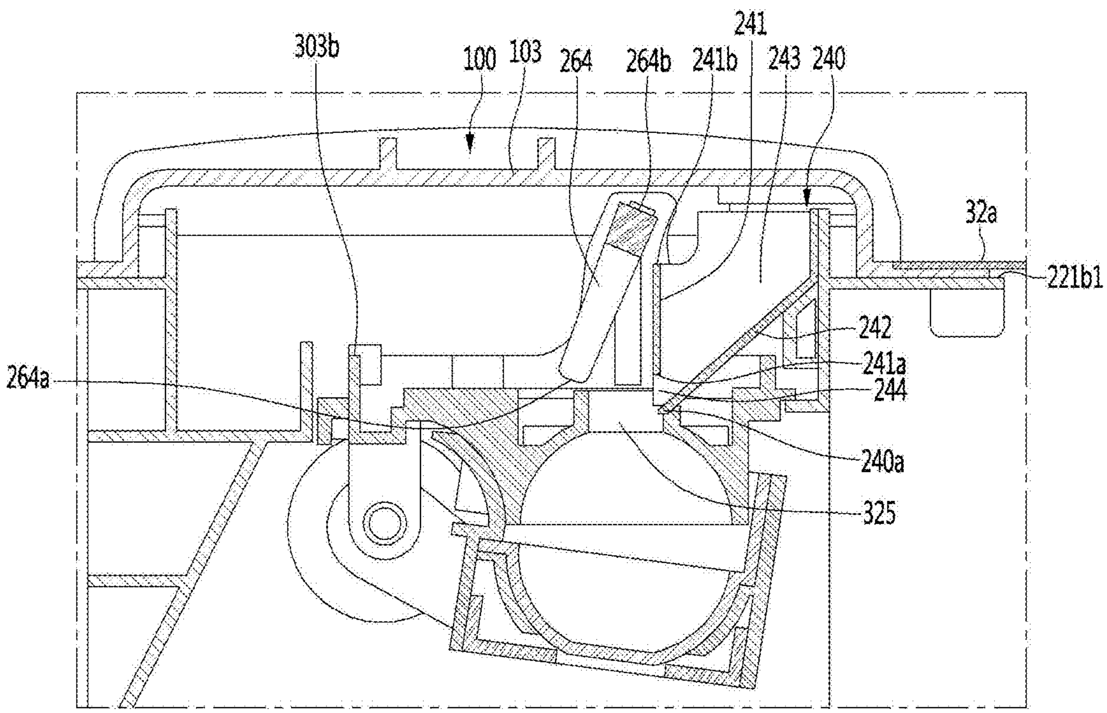

303b
100
103
264
264b
241b
241
243
240
32a
221b1
242
241a
244
264a
240a
325

FIG. 63

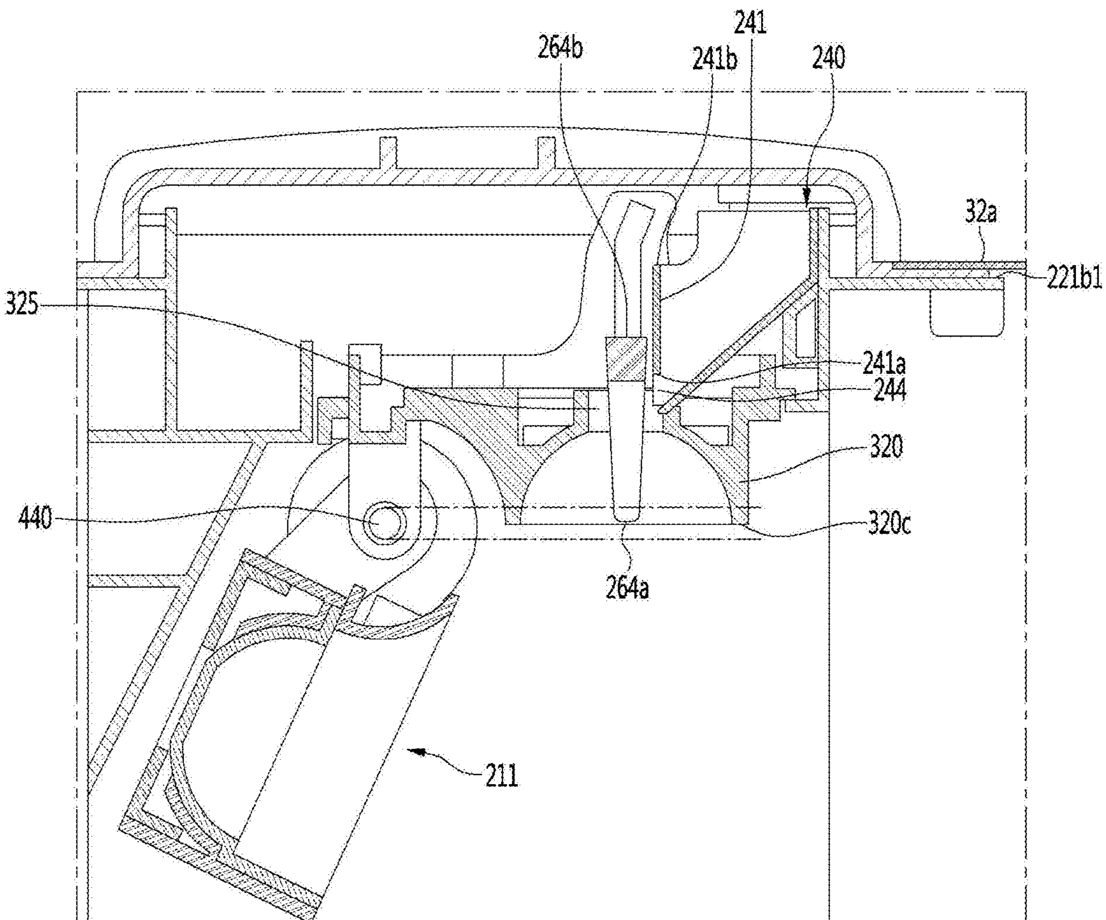


FIG. 64

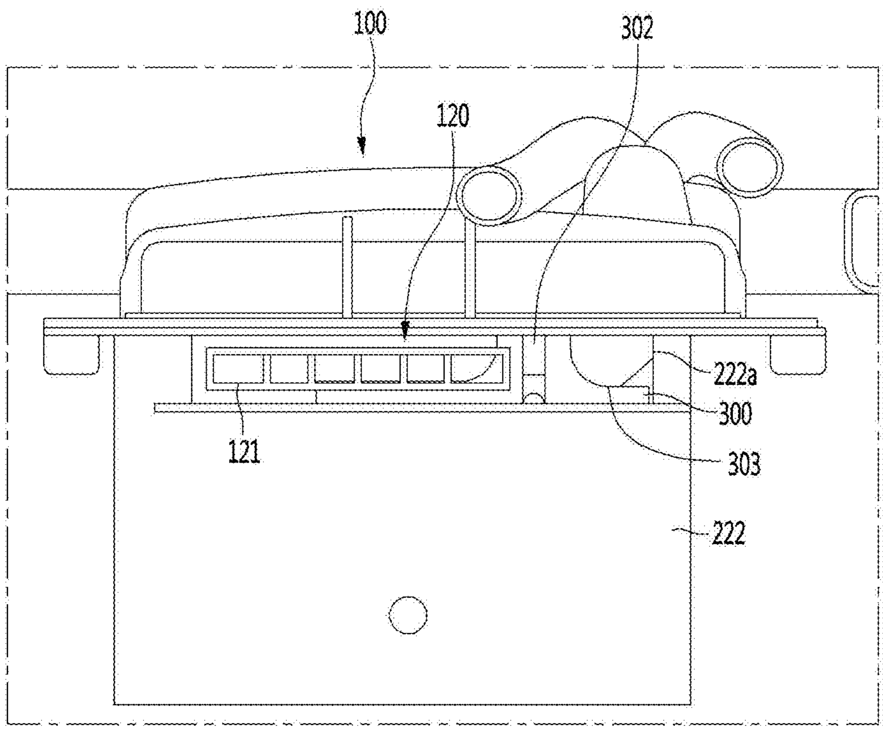
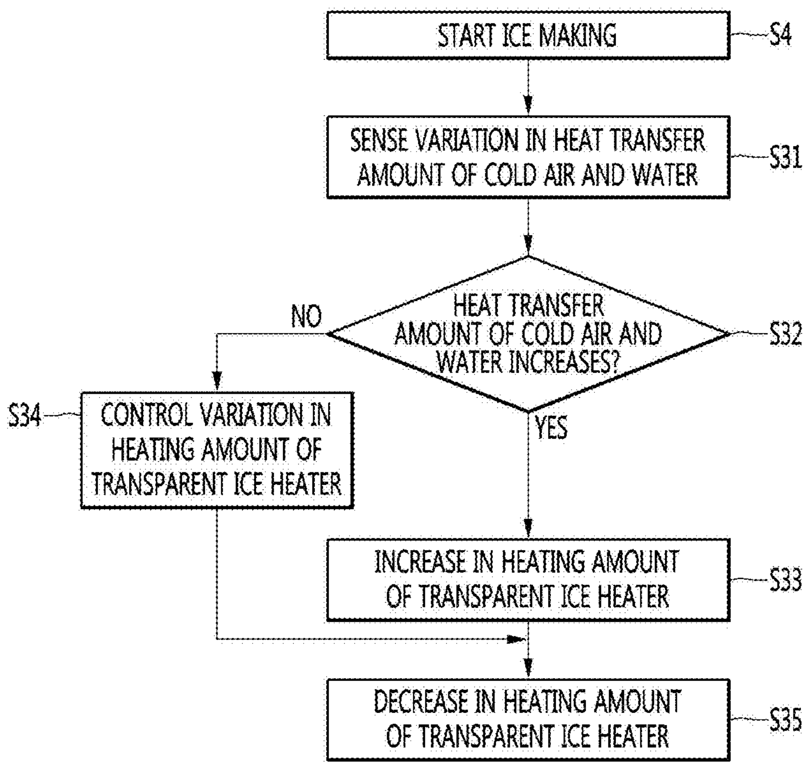


FIG. 65


START ICE MAKING
S4
SENSE VARIATION IN HEAT TRANSFER AMOUNT OF COLD AIR AND WATER
S31
HEAT TRANSFER AMOUNT OF COLD AIR AND WATER INCREASES?
S32
NO
YES
S34
CONTROL VARIATION IN HEATING AMOUNT OF TRANSPARENT ICE HEATER
INCREASE IN HEATING AMOUNT OF TRANSPARENT ICE HEATER
S33
DECREASE IN HEATING AMOUNT OF TRANSPARENT ICE HEATER
S35

REFRIGERATOR

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/386,517, filed Nov. 2, 2023, which is a continuation of U.S. patent application Ser. No. 17/281,950, filed Mar. 31, 2021, now issued as U.S. Pat. No. 11,841,180, which is a U.S. National Stage Application under 35 U.S.C. § 371 of PCT Application No. PCT/KR2019/012977, filed Oct. 2, 2019, which claims priority to Korean Patent Application Nos. 10-2018-0117819, filed Oct. 2, 2018, 10-2018-0117821, filed Oct. 2, 2018, 10-2018-0117822, filed Oct. 2, 2018, 10-2018-0117785, filed Oct. 2, 2018, 10-2018-0142117, filed Nov. 16, 2018, and 10-2019-0081688, filed Jul. 6, 2019, whose entire disclosures are hereby incorporated by reference.

TECHNICAL FIELD

[0002] Embodiments provide a refrigerator.

BACKGROUND ART

[0003] In general, refrigerators are home appliances for storing foods at a low temperature in a storage chamber that is covered by a door. The refrigerator may cool the inside of the storage space by using cold air to store the stored food in a refrigerated or frozen state. Generally, an ice maker for making ice is provided in the refrigerator. The ice maker makes ice by cooling water after accommodating the water supplied from a water supply source or a water tank into a tray. The ice maker may separate the made ice from the ice tray in a heating manner or twisting manner.

[0004] As described above, the ice maker through which water is automatically supplied, and the ice automatically separated may be opened upward so that the mode ice is pumped up.

[0005] As described above, the ice made in the ice maker may have at least one flat surface such as crescent or cubic shape.

[0006] When the ice has a spherical shape, it is more convenient to use the ice, and also, it is possible to provide different feeling of use to a user. Also, even when the made ice is stored, a contact area between the ice cubes may be minimized to minimize a mat of the ice cubes.

[0007] An ice maker is disclosed in Korean Registration No. 10-1850918 (hereinafter, referred to as a "prior art document 1") that is a prior art document.

[0008] The ice maker disclosed in the prior art document 1 includes an upper tray in which a plurality of upper cells, each of which has a hemispherical shape, are arranged, and which includes a pair of link guide parts extending upward from both side ends thereof, a lower tray in which a plurality of upper cells, each of which has a hemispherical shape and which is rotatably connected to the upper tray, a rotation shaft connected to rear ends of the lower tray and the upper tray to allow the lower tray to rotate with respect to the upper tray, a pair of links having one end connected to the lower tray and the other end connected to the link guide part, and an upper ejecting pin assembly connected to each of the pair of links in at state in which both ends thereof are inserted into the link guide part and elevated together with the upper ejecting pin assembly.

[0009] In the prior art document 1, although the spherical ice is made by the hemispherical upper cell and the hemispherical lower cell, since the ice is made at the same time in the upper and lower cells, bubbles containing water are not completely discharged but are dispersed in the water to make opaque ice.

[0010] An ice maker is disclosed in Japanese Patent Laid-Open No. 9-269172 (hereinafter, referred to as a "prior art document 2") that is a prior art document.

[0011] The ice maker disclosed in the prior art document 2 includes an ice making plate and a heater for heating a lower portion of water supplied to the ice making plate.

[0012] In the case of the ice maker disclosed in the prior art document 2, water on one surface and a bottom surface of an ice making block is heated by the heater in an ice making process. Thus, when solidification proceeds on the surface of the water, and also, convection occurs in the water to make transparent ice.

[0013] When growth of the transparent ice proceeds to reduce a volume of the water within the ice making block, the solidification rate is gradually increased, and thus, sufficient convection suitable for the solidification rate may not occur.

[0014] Thus, in the case of the prior art document 2, when about ⅔ of water is solidified, a heating amount of heater increases to suppress an increase in the solidification rate.

[0015] However, the prior art document 2 discloses a feature in which when the volume of water is simply reduced, only the heating amount of heater increases and does not disclose a structure and a heater control logic for making ice having high transparency without reducing the ice making rate.

DISCLOSURE

Technical Problem

[0016] Embodiments provide a refrigerator capable of making ice having uniform transparency by reducing transfer of heat, which is transferred to one tray adjacent to an operating heater, to an ice making cell provided by the other tray in an ice making process.

[0017] Embodiments provide a refrigerator in which transfer of heat of a heater to a portion at which ice is made is reduced to minimize a decrease in ice making rate even while making transparent ice.

[0018] Embodiments provide a refrigerator in which transparency per unit height is uniform even while transparent ice is made.

Technical Solution

[0019] In one embodiment, a refrigerator may include a first tray assembly defining a portion of an ice making cell and a second tray assembly defining another portion of the ice making cell.

[0020] A heater may be disposed on one of the first and second assemblies. The one tray assembly may include a first portion that defines at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion. This configuration may reduce transfer of the heat, which is transferred from the heater to the one tray assembly, to the ice making cell defined by the other tray assembly. A predetermined point of the first portion may be an end of the first part or a point at which the first and second tray assemblies meet each other. The tray assembly may be defined as a tray. The tray assembly may be defined as a tray and a tray case surrounding the tray. The one tray assembly may be closer to the heater than the other tray assembly. The heater may be disposed on the one tray assembly.

[0021] At least a portion of the second portion may extend in a direction away from the ice making cell defined by the other tray assembly. This configuration may reduce transfer of the heat, which is transferred from the heater to the first portion, to the ice making cell defined by the other tray assembly. The direction may be a horizontal direction passing through a center of the ice making cell. The direction may be an upward direction based on a horizontal line passing through the center of the ice making cell. The second portion may include a first part extending in the horizontal direction from the predetermined point and a second part extending in the same direction as the first part. The second part may include a first part extending in a horizontal direction from the predetermined point and a third part extending in a direction different from that of the first part.

[0022] The second portion may include a first part extending in the horizontal direction from the predetermined point and second and third parts branched from the first part. This configuration may induce heat transferred from the heater to the first part so as to be bypassed to the second and third parts, thereby reducing transfer of the heat to the ice making cell defined by the other tray assembly. The first part may further include a portion extending from the predetermined point in a vertical direction. The third part may have a length greater than that of the second part. The third part and the first part may have heights different from each other. The second part may extend in the same direction as the first part. The third part may extend in a direction different from that of the first part. The third part may have a curvature greater than that of the second part. The third part may have a curvature radius less than that of the second part. The third part may have the same curvature radius. At least a portion of the second portion may have the same curvature radius with respect to a rotational center thereof that is connected to the driver to rotate. This configuration may prevent the tray assembly from interfering when the tray assembly rotates by the driver.

[0023] It may be advantageous to design a length of a heat conduction path defined by the second portion as long as possible. This is because the longer the heat conduction path, the greater the heat released to the outside through the heat conduction path, and the heat transmitted to the ice making cell defined by the first tray assembly may be reduced. The second portion may extend up to a point that is equal to or higher than an upper portion of the ice making cell defined by the other tray assembly. The second portion may extend up to a point that is equal to or higher than the uppermost end of the ice making cell defined by the other tray assembly. The refrigerator may further include a rotation shaft around which the second tray assembly rotates, and the second portion may extend up to a point higher than an upper end of the rotation shaft. The heat conduction path defined by the second portion may have a length greater than a distance from the center of the ice making cell to an outer circumferential surface of the ice making cell.

[0024] The tray assembly may include a first portion defining at least a portion of the ice making cell and first and second extension parts of the second portion respectively extending from first and second points of the first portion. The one tray assembly of the first and second tray assemblies may include a first portion defined at least a portion of the ice making cell, a first extension part of a second portion extending from a first point of the first portion, and a second extension part of the second portion extending from a second point of the first portion. This configuration may reduce transfer of heat, which is transferred from the heater to one tray assembly, to the ice making cell defined by the other tray assembly. The first extension part may be disposed at a left side of the ice making cell. The second extension part may be disposed at a right side of the ice making cell. The first and second extension parts may be different in shape or asymmetrical to each other. A length of the second extension part in a horizontal direction passing through a center of the ice making cell may be greater than that of the first extension part in the horizontal direction.

[0025] The refrigerator may further include a bracket defining at least a portion of a space accommodating the first and second tray assemblies. The first extension part may be disposed closer than the second extension part with respect to one of edges of the space defined by the bracket. A length of the second extension part in the horizontal direction may be greater than that of the first extension part in the horizontal direction. This configuration may reduce that the first extension part interferes with the bracket. This is because a heat conduction path defined by the tray assembly is lengthened while minimizing the space in which the tray assembly and the components are installed. The ice making cell may be eccentric with respect to the bracket.

[0026] The refrigerator may further include a rotation shaft connected to the driver so that at least one of the first and second trays is rotatable. The second extension part may be disposed closer to the center of the rotation shaft than the first extension part. A length of the second extension part in the horizontal direction may be greater than that of the first extension part in the horizontal direction. This configuration may increase rotational force of the rotating tray assembly. As described above, it is desirable to increase coupling force of the first and second tray assemblies so as to make ice having a specific shape such as transparent ice or spherical ice. As described above, when ice is made in the state in which the coupling force between the first and second tray assemblies increases, adhesion between the made ice and the tray assembly may also increase. Thus, a component may be needed to allow ice to be more easily separated from the tray assembly during ice separation after ice making is complete. For example, the refrigerator may further include a heater disposed at one side of the tray assembly. The heater may be an ice separation heater. As another example, the refrigerator may further include a pusher capable of pressurizing ice during the ice separation process. When at least one of the pusher or the tray assembly moves, ice may be pressurized in the ice separation process. The movement may be a motion in an axial direction of at least one of the X, Y, or Z axes. The movement may be a motion that rotates about at least one of the X, Y, or Z axes. When the movement is rotational movement, pushing force supplied by the pusher to ice may be greater as a rotation radius is greater with respect to the rotational force that is supplied to at least one of the pusher or the tray assembly by the driver. As the length of the second extension part closer to the rotational center increases, a distance between the rotational centers increases, the pressing force supplied by the pusher to the ice may increase, and the heat conduction path through the second extension part may increase. The second extension part may include a portion having the same curvature with respect to the rotation shaft. As a result, interference during the rotation of the tray assembly may not occur. The first extension part may include a portion extending upward with respect to the horizontal line. The second extension part may extend in a direction away from the ice making cell while extending upward on the horizontal line, whereas the first extension part may extend only in the upward direction with respect to the horizontal line. Due to the shape of the first and second extension parts, the coupling force between the first and second tray assemblies may increase. A rotation angle of the rotating assembly tray assembly may be greater than about 90 degrees and less than about 180 degrees. This may increase the pressing force that is supplied to the ice by the pusher. The rotational center may be eccentric to one side with respect to the bracket.

[0027] The one tray assembly and the other tray assembly may contact each other. The first portion of one tray assembly, which defines the ice making cell, and the third portion of the other tray assembly, which defines the ice making cell, may contact each other. The reason for this is to reduce leakage of water in the ice making cell defied by the first and second tray assemblies. The other tray assembly may include a third portion defining a portion of the ice making cell and a fourth portion extending from a predetermined point of the third portion, and the second portion may be disposed outside the fourth portion. At least a portion of the second portion extending from the predetermined point of the first portion and the fourth portion extending from the predetermined point of the third portion may be spaced apart from each other. This is because transfer of the heat, which is transferred to the second portion, to the fourth portion is capable of being reduced.

[0028] The first tray assembly may include a first tray, and the second tray assembly may include a second tray. One tray of the first and second trays may be disposed to be closer to the heater than the other tray. The one tray may include a first portion that defines at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion. At least a portion of the second portion may extend in a direction away from the ice making cell defined by the other tray. This configuration may reduce transfer of the heat, which is transferred from the heater to the first portion, to the ice making cell defined by the other tray. The second portion may further include a portion extending in the horizontal direction from the predetermined point and a portion extending upward in the horizontal direction passing through the center of the ice making cell (for example, a center of the gravity or a volume center). The second portion may include a first part extending in the horizontal direction from the predetermined point and second and third parts branched from the first part. This configuration may reduce transfer of the heat, which is transferred from the heater to the first part, to the ice making cell defined by the other tray.

[0029] The first portion may include a first region and a second region having a curvature different from that of the first region. The first region may include a shape that is recessed in a direction opposite to a direction in which the ice is expanded. This configuration may induce ice to be made in the direction from the ice making cell defined by the other tray to the ice making cell defined by the one tray assembly after the ice making process starts. The refrigerator may further include a pusher. The first region may include a pressing part that contacts the pusher. The pressing part may be disposed at a portion in which the shape recessed in the opposite to the direction in which the ice is expanded is defined. The pressing part may be a portion or the whole of the portion in which the recessed shape is defined.

[0030] The controller may control the pusher to move from a first point outside the ice making cell so as to contact the pressing part and then to move to a second point inside the ice making cell.

[0031] The refrigerator may further include an outer case and an inner case, and a degree of heat transfer of the first region may be less than that of the outer or inner case.

[0032] The refrigerator may further include an outer case and an inner case, and a degree of restoration of the first region may be greater than that of the case. The tray assembly may further include a tray case. The degree of heat transfer of the first region may also be less than that of the tray case. The degree of restoration of the first region may be greater than that of the tray case. This configuration may allow the first region to be easily deformed and to be easily restored after external force is removed.

[0033] The thermal insulation of the other tray assembly with respect to cold may be less than that of the one tray assembly.

[0034] In another embodiment, a refrigerator includes: a storage chamber configured to store foods; a cooler configured to supply cold into the storage chamber; a first temperature sensor configured to sense a temperature within the storage chamber; a first tray assembly configured to define a portion of an ice making cell that is a space in which water is phase-changed into ice by the cold; a second tray assembly configured to define another portion of the ice making cell, the second tray assembly being connected to a driver to contact the first tray assembly during an ice making process and to be spaced apart from the first tray assembly during an ice separation process; a water supply part configured to supply water into the ice making cell; a second temperature sensor configured to sense a temperature of the water or the ice within the ice making cell; a heater disposed adjacent to at least one of the first tray assembly or the second tray assembly; and a controller configured to control the heater and the driver.

[0035] The controller may control the cooler so that the cold is supplied to the ice making cell after the second tray assembly moves to an ice making position when the water is completely supplied to the ice making cell. The controller may control the second tray assembly so that the second tray assembly moves in a reverse direction after moving to an ice separation position in a forward direction so as to take out the ice in the ice making cell when the ice is completely made in the ice making cell. The controller may control the second tray assembly so that the supply of the water starts after the second tray assembly moves to a water supply position in the reverse direction when the ice is completely separated.

[0036] The controller may control the heater to be turned on in at least partial section while the cooler supplies the cold so that bubbles dissolved in the water within the ice making cell moves from a portion, at which the ice is made, toward the water that is in a liquid state to make transparent ice.

[0037] One of the first tray assembly and the second tray assembly may be disposed closer to the heater than the other tray. The one tray assembly may include a first portion that defines at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion.

[0038] The one tray assembly may include a tray defining the ice making cell. Alternatively, the one tray assembly may include a tray defining the ice making cell and a tray case surrounding the tray.

[0039] A predetermined point of the first portion may be an end of the first part or a point at which the first and second tray assemblies meet each other. At least a portion of the second portion may extend in a direction away from the ice making cell defined by the other tray assembly. The direction away from the ice making cell may be a horizontal direction passing through a center of the ice making cell. The direction away from the ice making cell may be an upward direction with respect to a horizontal line passing through the center of the ice making cell.

[0040] The second portion may include a first part extending in the horizontal direction from the predetermined point and a second part extending in the same direction as the first part.

[0041] The second portion may include a first part extending in the horizontal direction from the predetermined point and a third part extending in a direction different from that of the first part.

[0042] The second portion may include a first part extending in the horizontal direction from the predetermined point and second and third parts branched from the first part.

[0043] The first part may further include a portion extending from the predetermined point in a vertical direction. The third part may have a length greater than that of the second part.

[0044] The second part may extend in the same direction as the first part. The third part may extend in a direction different from that of the first part. The third part may have the same curvature in a longitudinal direction. The third part may have a curvature greater than that of the second part. The third part may have a curvature radius less than that of the second part.

[0045] The second portion may extend up to a point that is equal to or higher than an upper portion of the ice making cell defined by the other tray assembly. The second portion may extend up to a point that is equal to or higher than the uppermost end of the ice making cell defined by the other tray assembly. The second portion may have a length greater than a radius of the ice making cell.

[0046] The refrigerator may further include a rotation shaft connected to the driver so that the second tray assembly rotates. A center of curvature of at least a portion of the second portion may be a center of the rotation shaft. The second portion may extend to a point higher than the uppermost end of the rotation shaft.

[0047] The second part may include a first extension part extending from a first point of the first portion and a second extension part extending from a second point of the first portion.

[0048] The first extension part may be disposed at one side of the vertical central line, and the second extension part may be disposed at the other side of the vertical central line with respect to the vertical central line passing through the center of the ice making cell.

[0049] The first extension part and the second extension part may have different shapes, or the first extension part and the second extension part may be asymmetrical with respect to the vertical central line passing through the center of the ice making cell.

[0050] The second extension part may have a length greater than that of the first extension part with respect to the direction of the horizontal line passing through the center of the ice making cell.

[0051] The refrigerator may further include a bracket defining at least a portion of a space accommodating the first and second tray assemblies. The first extension part may be disposed closer than the second extension part with respect to one of edges of the space defined by the bracket. The horizontal length of the second extension part may be greater than that of the first extension part. The ice making cell may be eccentrically disposed based on a line that bisects the length of the bracket in one direction.

[0052] The second extension part may be disposed closer to the rotational center of the rotation shaft than the first extension part. The second extension part may include a portion having the center of the rotation shaft as the center of curvature.

[0053] The first extension part may include a portion extending upward with respect to the horizontal line passing through the center of the ice making cell. A rotation angle of the second tray assembly may be greater than about 90 degrees and less than about 180 degrees.

[0054] The rotation shaft may be eccentrically disposed based on the line that bisects the length of the bracket in one direction.

[0055] The one tray assembly and the other tray assembly may contact each other at an ice making position.

[0056] The other tray assembly may include a third portion contacting the first portion and defining a portion of the ice making cell. The other tray assembly may further include a fourth portion extending from a predetermined point of the third portion. At least a portion of the second portion may be spaced apart from the fourth portion.

[0057] The heater may be disposed in the one tray assembly. The refrigerator may further include an additional heater disposed in the other tray assembly. A heating amount of the additional heater may be less than that of the heater in at least a section in which the cooler supplies the cold.

[0058] The one tray assembly may be disposed below the other tray assembly.

[0059] The controller may control the heater so that when a heat transfer amount between the cold within the storage chamber and the water of the ice making cell increases, the heating amount of heater increases, and when the heat transfer amount between the cold within the storage chamber and the water of the ice making cell decreases, the heating amount of heater decreases so as to maintain an ice making rate of the water within the ice making cell within a predetermined range that is less than an ice making rate when the ice making is performed in a state in which the heater is turned off.

[0060] The controller may control one or more of cooling power of the cooler and the heating amount of heater to vary according to a mass per unit height of water in the ice making cell.

[0061] The heater may contact the one tray assembly. At least a portion of the portion at which the heater does not contact the one tray assembly may be sealed with a heat insulation material.

[0062] The heat transfer from the heater to a central direction of the ice making cell may be greater than that from the heater to a circumferential direction of the ice making cell.

[0063] The cooler may be disposed so that an amount of cold supplied to the other tray assembly is greater than that of cold supplied to the one tray assembly.

[0064] A distance between the cooler and a portion of the ice making cell, which is defined by the one tray assembly, may be greater than a distance between the cooler and a portion of the ice making cell, which is defined by the other tray assembly.

[0065] The thermal insulation of the other tray assembly with respect to cold may be less than the thermal insulation of the one tray assembly.

[0066] The ice making cell defined by the other tray assembly may include a first region defined close to the heater and a second region defined far from the heater, and the cooler may be disposed so that an amount of cold supplied to the second region is greater than an amount of cold supplied to the first region.

[0067] A distance between the cooler and the first region may be greater than that between the cooler and the second region. The heat insulation degree of the second region with respect to the cold may be less than that of the first region. A through-hole through which the cold is supplied may be defined in the second region.

[0068] The other tray assembly may be provided with an additional heater, and the controller may turn on the additional heater before the second tray assembly moves forward to the ice making position. The heating amount of additional heater may be greater than that of above-described heater.

[0069] A refrigerator according to another aspect may include a first tray assembly and a second tray assembly. The first tray assembly may include a first tray, the second tray assembly may include a second tray, and one of the first tray and the second tray may be disposed closer to the heater than the other tray.

[0070] The one tray may include a first portion that defines at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion. At least a portion of the second portion may extend in a direction away from the ice making cell defined by the other tray.

[0071] The second portion may include a portion extending from the predetermined point in a horizontal direction and a portion extending upward from a horizontal line passing through a center of the ice making cell.

[0072] The second portion may include a first part extending in the horizontal direction from the predetermined point and second and third parts branched from the first part.

[0073] The first portion may include a first region and a second region having a curvature different from that of the first region. The first region may include a shape recessed in a direction opposite to a direction in which ice is expanded in the ice making cell.

[0074] A distance from a center of the ice making cell to a portion in which the recessed shape is disposed in the first region may be less than a distance from the center of the ice making cell to the second region.

[0075] The refrigerator may further include a pusher configured to separate the ice from the ice making cell. The first region may comprise a pressing part that contacts the pusher.

[0076] The controller may control the pusher and a position of the one tray so that the pusher contacts the pressing part at a first point outside the ice making cell to additionally press the pressing part. The pressing unit may be a portion or the whole of the shape recessed in the first region.

[0077] The first portion may include a first region and a second region disposed further away from the heater.

[0078] In a refrigerator according to another aspect, a controller controls a heater so that when a heat transfer amount between the cold cooling an ice making cell and water of the ice making cell increases, a heating amount of heater increases, and when a heat transfer amount between the cold cooling the ice making cell and the water of the ice making cell decreases, wherein one tray of the first tray and the second tray is disposed closer to the heater than the other tray.

[0079] The refrigerator may include a first portion defining at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion so that transfer of heat, which is transferred from the heater to the one tray, to the ice making cell defined by the other tray is reduced.

[0080] At least a portion of the second portion may extend in a direction away from the ice making cell defined by the other tray.

[0081] The second portion may include a portion extending from the predetermined point in a horizontal direction and a portion extending upward from a horizontal line passing through a center of the ice making cell.

[0082] The second portion may include a first part extending in the horizontal direction from the predetermined point and second and third parts branched from the first part.

[0083] A refrigerator according to further another aspect may include a first tray assembly including a first tray, a second tray assembly including a second tray, a heater disposed closer to one tray of the first and second trays than the other tray, and a controller configured to control the heater.

[0084] The controller may control the heater to be turned on in at least partial section while the cooler supplies cold so that bubbles dissolved in the water within the ice making cell moves from a portion, at which the ice is made, toward the water that is in a liquid state to make transparent ice.

[0085] The one tray may include a first portion that defines at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion.

[0086] The first portion may include a first region and a second region having a curvature different from that of the first region so that the ice is made in a direction from the ice making cell defined by the other tray toward the ice making cell defined by the one tray.

[0087] The first region may include a shape recessed in a direction opposite to a direction in which ice is expanded in the ice making cell.

[0088] The refrigerator may further include a pusher configured to separate the ice from the ice making cell The first region may include a pressing part that contacts the pusher.

[0089] The second region may be disposed further away from the heater than the first region.

Advantageous Effects

[0090] According to the embodiments, since the heater is turned on in at least a portion of the sections while the cooler supplies cold, the ice making rate may decrease by the heat of the heater so that the bubbles dissolved in the water inside the ice making cell move toward the liquid water from the portion at which the ice is made, thereby making the transparent ice.

[0091] Also, as the transfer of the heat of the heater to the ice making portion in the ice making cell during the ice making, the transparent ice may be made while minimizing the delay of the ice making rate.

[0092] Also, according to the embodiments, one or more of the cooling power of the cooler and the heating amount of heater may be controlled to vary according to the mass per unit height of water in the ice making cell to make the ice having the uniform transparency as a whole regardless of the shape of the ice making cell.

[0093] Also, the heating amount of transparent ice heater and/or the cooling power of the cooler may vary in response to the change in the heat transfer amount between the water in the ice making cell and the cold air in the storage chamber, thereby making the ice having the uniform transparency as a whole.

BRIEF DESCRIPTION OF DRAWINGS

[0094] FIG. **1** is a front view of a refrigerator according to an embodiment.

[0095] FIG. **2** is a perspective view of an ice maker according to an embodiment.

[0096] FIG. **3** is a front view of the ice maker of FIG. **2**.

[0097] FIG. **4** is a perspective view illustrating a state in which a bracket is removed from the ice maker of FIG. **3**.

[0098] FIG. **5** is an exploded perspective view of the ice maker according to an embodiment.

[0099] FIGS. **6** and **7** are perspective views of the bracket according to an embodiment.

[0100] FIG. **8** is a perspective view of a first tray when viewed from an upper side.

[0101] FIG. **9** is a perspective view of the first tray when viewed from a lower side.

[0102] FIG. **10** is a plan view of the first tray.

[0103] FIG. **11** is a cutaway cross-sectional view taken along line **11-11** of FIG. **8**.

[0104] FIG. **12** is a bottom view of the first tray of FIG. **9**.

[0105] FIG. **13** is a cutaway cross-sectional view taken along line **13-13** of FIG. **11**.

[0106] FIG. **14** is a cutaway cross-sectional view taken along line **14-14** of FIG. **11**.

[0107] FIG. **15** is a cutaway cross-sectional view taken along line **15-15** of FIG. **8**.

[0108] FIG. **16** is a perspective view of the first tray.

[0109] FIG. **17** is a bottom perspective view of a first tray cover.

[0110] FIG. **18** is a plan view of the first tray cover.

[0111] FIG. **19** is a side view of a first tray case.

[0112] FIG. **20** is a perspective view of a first heater case.

[0113] FIG. **21** is a bottom perspective view of the first heater case.

[0114] FIG. **22** is a partial enlarged view of the first heater case.

[0115] FIG. **23** is a cross-sectional view illustrating a coupling relationship between the first heater case and the first tray.

[0116] FIG. **24** is a plan view of a first tray supporter.

[0117] FIG. **25** is a perspective view of a second tray according to an embodiment.

[0118] FIG. **26** is a perspective view of the second tray when viewed from a lower side.

[0119] FIG. **27** is a bottom view of the second tray.

[0120] FIG. **28** is a plan view of the second tray.

[0121] FIG. **29** is a cutaway cross-sectional view taken along line **29-29** of FIG. **25**.

[0122] FIG. **30** is a cutaway cross-sectional view taken along line **30-30** of FIG. **25**.

[0123] FIG. **31** is a cutaway cross-sectional view taken along line **31-31** of FIG. **25**.

[0124] FIG. **32** is a cutaway cross-sectional view taken along line **32-32** of FIG. **28**.

[0125] FIG. **33** is a cutaway cross-sectional view taken along line **33-33** of FIG. **29**.

[0126] FIG. **34** is a perspective view of a second tray cover.

[0127] FIG. **35** is a plan view of the second tray cover.

[0128] FIG. **36** is a top perspective view of a second tray supporter.

[0129] FIG. **37** is a bottom perspective view of the second tray supporter.

[0130] FIG. **38** is a cutaway cross-sectional view taken along line **38-38** of FIG. **36**.

[0131] FIG. **39** is a perspective view of a second heater case.

[0132] FIG. **40** is a view illustrating a state in which a transparent ice heater is coupled to the second heater case.

[0133] FIG. **41** is a cutaway cross-sectional view taken along line **41-41** of FIG. **40**.

[0134] FIG. **42** is a partial enlarged view of the second heater case.

[0135] FIG. **43** is a view of a first pusher according to an embodiment.

[0136] FIG. **44** is a view illustrating a state in which the first pusher is connected to a second tray assembly by a link.

[0137] FIG. **45** is a perspective view of a second pusher according to an embodiment.

[0138] FIGS. **44** to **48** are views illustrating an assembly process of an ice maker according to an embodiment.

[0139] FIG. **49** is a cutaway cross-sectional view taken along line **49-49** of FIG. **2**.

[0140] FIG. **50** is a block diagram illustrating a control of a refrigerator according to an embodiment.

[0141] FIG. **51** is a flowchart for explaining a process of making ice in the ice maker according to an embodiment.

[0142] FIG. **52** is a view for explaining a height reference depending on a relative position of the transparent heater with respect to the ice making cell.

[0143] FIG. **53** is a view for explaining an output of the transparent heater per unit height of water within the ice making cell.

[0144] FIG. **54** is a cross-sectional view illustrating a position relationship between a first tray assembly and a second tray assembly at a water supply position.

[0145] FIG. **55** is a view illustrating a state in which supply of water is complete in FIG. **54**.

[0146] FIG. **56** is a cross-sectional view illustrating a position relationship between a first tray assembly and a second tray assembly at an ice making position.

[0147] FIG. **57** is a view illustrating a state in which a pressing part of the second tray is deformed in a state in which ice making is complete.

[0148] FIG. **58** is a cross-sectional view illustrating a position relationship between a first tray assembly and a second tray assembly in an ice separation process.

[0149] FIG. **59** is a cross-sectional view illustrating the position relationship between the first tray assembly and the second tray assembly at the ice separation position.

[0150] FIG. **60** is a view illustrating an operation of a pusher link when the second tray assembly moves from the ice making position to the ice separation position.

[0151] FIG. **61** is a view illustrating a position of a first pusher at a water supply position at which the ice maker is installed in a refrigerator.

[0152] FIG. **62** is a cross-sectional view illustrating the position of the first pusher at the water supply position at which the ice maker is installed in the refrigerator.

[0153] FIG. **63** is a cross-sectional view illustrating a position of the first pusher at the ice separation position at which the ice maker is installed in the refrigerator.

[0154] FIG. **64** is a view illustrating a position relationship between a through-hole of the bracket and a cold air duct.

[0155] FIG. **65** is a view for explaining a method for controlling a refrigerator when a heat transfer amount between cold air and water vary in an ice making process.

MODE FOR INVENTION

[0156] Hereinafter, some embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. It should be noted that when components in the drawings are designated by reference numerals, the same components have the same reference numerals as far as possible even though the components are illustrated in different drawings. Further, in description of embodiments of the present disclosure, when it is determined that detailed descriptions of well-known configurations or functions disturb understanding of the embodiments of the present disclosure, the detailed descriptions will be omitted.

[0157] Also, in the description of the embodiments of the present disclosure, the terms such as first, second, A, B, (a) and (b) may be used. Each of the terms is merely used to distinguish the corresponding component from other components, and does not delimit an essence, an order or a sequence of the corresponding component. It should be understood that when one component is "connected", "coupled" or "joined" to another component, the former may be directly connected or jointed to the latter or may be "connected", coupled" or "joined" to the latter with a third component interposed therebetween.

[0158] The refrigerator according to an embodiment may include a tray assembly defining a portion of an ice making cell that is a space in which water is phase-changed into ice, a cooler supplying cold air to the ice making cell, a water supply part supplying water to the ice making cell, and a controller. The refrigerator may further include a temperature sensor detecting a temperature of water or ice of the ice making cell. The refrigerator may further include a heater disposed adjacent to the tray assembly. The refrigerator may further include a driver to move the tray assembly. The refrigerator may further include a storage chamber in which food is stored in addition to the ice making cell. The refrigerator may further include a cooler supplying cold to the storage chamber. The refrigerator may further include a temperature sensor sensing a temperature in the storage chamber. The controller may control at least one of the water supply part or the cooler. The controller may control at least one of the heater or the driver.

[0159] The controller may control the cooler so that cold is supplied to the ice making cell after moving the tray assembly to an ice making position. The controller may control the second tray assembly so that the second tray assembly moves to an ice separation position in a forward direction so as to take out the ice in the ice making cell when the ice is completely made in the ice making cell. The controller may control the tray assembly so that the supply of the water supply part after the second tray assembly moves to the water supply position in the reverse direction when the ice is completely separated. The controller may control the tray assembly so as to move to the ice making position after the water supply is completed.

[0160] According to an embodiment, the storage chamber may be defined as a space that is controlled to a predetermined temperature by the cooler. An outer case may be defined as a wall that divides the storage chamber and an external space of the storage chamber (i.e., an external space of the refrigerator). An insulation material may be disposed between the outer case and the storage chamber. An inner case may be disposed between the insulation material and the storage chamber.

[0161] According to an embodiment, the ice making cell may be disposed in the storage chamber and may be defined as a space in which water is phase-changed into ice. A circumference of the ice making cell refers to an outer surface of the ice making cell irrespective of the shape of the ice making cell. In another aspect, an outer circumferential surface of the ice making cell may refer to an inner surface of the wall defining the ice making cell. A center of the ice making cell refers to a center of gravity or volume of the ice making cell. The center may pass through a symmetry line of the ice making cell.

[0162] According to an embodiment, the tray may be defined as a wall partitioning the ice making cell from the inside of the storage chamber. The tray may be defined as a wall defining at least a portion of the ice making cell. The tray may be configured to surround the whole or a portion of the ice making cell. The tray may include a first portion that defines at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion. The tray may be provided in plurality. The plurality of trays may contact each other. For example, the tray disposed at the lower portion may include a plurality of trays. The tray disposed at the upper portion may include a plurality of trays. The refrigerator may include at least one tray disposed under the ice making cell. The refrigerator may further include a tray disposed above the ice making cell. The first portion and the second portion may have a structure inconsideration of a degree of heat transfer of the tray, a degree of cold transfer of the tray, a degree of deformation resistance of the tray, a recovery degree of the tray, a degree of supercooling of the tray, a degree of attachment between the tray and ice solidified in the tray, and coupling force between one tray and the other tray of the plurality of trays.

[0163] According to an embodiment, the tray case may be disposed between the tray and the storage chamber. That is, the tray case may be disposed so that at least a portion thereof surrounds the tray. The tray case may be provided in plurality. The plurality of tray cases may contact each other. The tray case may contact the tray to support at least a portion of the tray. The tray case may be configured to connect components except for the tray (e.g., a heater, a sensor, a power transmission member, etc.). The tray case may be directly coupled to the component or coupled to the component via a medium therebetween. For example, if the wall defining the ice making cell is provided as a thin film, and a structure surrounding the thin film is provided, the thin film may be defined as a tray, and the structure may be defined as a tray case. For another example, if a portion of the wall defining the ice making cell is provided as a thin film, and a structure includes a first portion defining the other portion of the wall defining the ice making cell and a second part surrounding the thin film, the thin film and the first portion of the structure are defined as trays, and the second portion of the structure is defined as a tray case.

[0164] According to an embodiment, the tray assembly may be defined to include at least the tray. According to an embodiment, the tray assembly may further include the tray case.

[0165] According to an embodiment, the refrigerator may include at least one tray assembly connected to the driver to move. The driver is configured to move the tray assembly in at least one axial direction of the X, Y, or Z axis or to rotate about the axis of at least one of the X, Y, or Z axis. The embodiment may include a refrigerator having the remaining configuration except for the driver and the power transmission member connecting the driver to the tray assembly in the contents described in the detailed description. According to an embodiment, the tray assembly may move in a first direction.

[0166] According to an embodiment, the cooler may be defined as a part configured to cool the storage chamber including at least one of an evaporator or a thermoelectric element.

[0167] According to an embodiment, the refrigerator may include at least one tray assembly in which the heater is disposed. The heater may be disposed in the vicinity of the tray assembly to heat the ice making cell defined by the tray assembly in which the heater is disposed. The heater may include a heater to be turned on in at least partial section while the cooler supplies cold so that bubbles dissolved in the water within the ice making cell moves from a portion, at which the ice is made, toward the water that is in a liquid state to make transparent ice. The heater may include a heater (hereinafter referred to as an "ice separation heater") controlled to be turned on in at least a section after the ice making is completed so that ice is easily separated from the tray assembly. The refrigerator may include a plurality of transparent ice heaters. The refrigerator may include a plurality of ice separation heaters. The refrigerator may include a transparent ice heater and an ice separation heater. In this case, the controller may control the ice separation heater so that a heating amount of ice separation heater is greater than that of transparent ice heater.

[0168] According to an embodiment, the tray assembly may include a first region and a second region, which define an outer circumferential surface of the ice making cell. The tray assembly may include a first portion that defines at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion.

[0169] For example, the first region may be defined in the first portion of the tray assembly. The first and second regions may be defined in the first portion of the tray assembly. Each of the first and second regions may be a portion of the one tray assembly. The first and second regions may be disposed to contact each other. The first region may be a lower portion of the ice making cell defined by the tray assembly. The second region may be an upper portion of an ice making cell defined by the tray assembly. The refrigerator may include an additional tray assembly. One of the first and second regions may include a region contacting the additional tray assembly. When the additional tray assembly is disposed in a lower portion of the first region, the additional tray assembly may contact the lower portion of the first region. When the additional tray assembly is disposed in an upper portion of the second region, the additional tray assembly and the upper portion of the second region may contact each other.

[0170] For another example, the tray assembly may be provided in plurality contacting each other. The first region may be disposed in a first tray assembly of the plurality of tray assemblies, and the second region may be disposed in a second tray assembly. The first region may be the first tray assembly. The second region may be the second tray assembly. The first and second regions may be disposed to contact each other. At least a portion of the first tray assembly may be disposed under the ice making cell defined by the first and second tray assemblies. At least a portion of the second tray assembly may be disposed above the ice making cell defined by the first and second tray assemblies.

[0171] The first region may be a region closer to the heater than the second region. The first region may be a region in which the heater is disposed. The second region may be a region closer to a heat absorbing part (i.e., a coolant pipe or a heat absorbing part of a thermoelectric module) of the cooler than the first region. The second region may be a region closer to the through-hole supplying cold to the ice making cell than the first region. To allow the cooler to supply the cold through the through-hole, an additional through-hole may be defined in another component. The second region may be a region closer to the additional through-hole than the first region. The heater may be a transparent ice heater. The heat insulation degree of the second region with respect to the cold may be less than that of the first region.

[0172] The heater may be disposed in one of the first and second tray assemblies of the refrigerator. For example, when the heater is not disposed on the other one, the controller may control the heater to be turned on in at least a sections of the cooler to supply the cold air. For another example, when the additional heater is disposed on the other one, the controller may control the heater so that the heating amount of heater is greater than that of additional heater in at least a section of the cooler to supply the cold air. The heater may be a transparent ice heater.

[0173] The embodiment may include a refrigerator having a configuration excluding the transparent ice heater in the contents described in the detailed description.

[0174] The embodiment may include a pusher including a first edge having a surface pressing the ice or at least one surface of the tray assembly so that the ice is easily separated from the tray assembly. The pusher may include a bar extending from the first edge and a second edge disposed at an end of the bar. The controller may control the pusher so that a position of the pusher is changed by moving at least one of the pusher or the tray assembly. The pusher may be defined as a penetrating type pusher, a non-penetrating type pusher, a movable pusher, or a fixed pusher according to a view point.

[0175] The through-hole through which the pusher moves may be defined in the tray assembly, and the pusher may be configured to directly press the ice in the tray assembly. The pusher may be defined as a penetrating type pusher.

[0176] The tray assembly may be provided with a pressing part to be pressed by the pusher, the pusher may be configured to apply a pressure to one surface of the tray assembly. The pusher may be defined as a non-penetrating type pusher.

[0177] The controller may control the pusher to move so that the first edge of the pusher is disposed between a first point outside the ice making cell and a second point inside the ice making cell.

[0178] The pusher may be defined as a movable pusher. The pusher may be connected to a driver, the rotation shaft of the driver, or the tray assembly that is connected to the driver and is movable.

[0179] The controller may control the pusher to move at least one of the tray assemblies so that the first edge of the pusher is disposed between the first point outside the ice making cell and the second point inside the ice making cell. The controller may control at least one of the tray assemblies to move to the pusher. Alternatively, the controller may control a relative position of the pusher and the tray assembly so that the pusher further presses the pressing part after contacting the pressing part at the first point outside the ice making cell. The pusher may be coupled to a fixed end. The pusher may be defined as a fixed pusher.

[0180] According to an embodiment, the ice making cell may be cooled by the cooler cooling the storage chamber. For example, the storage chamber in which the ice making cell is disposed may be a freezing compartment which is controlled at a temperature lower than 0 degree, and the ice making cell may be cooled by the cooler cooling the freezing compartment.

[0181] The freezing compartment may be divided into a plurality of regions, and the ice making cell may be disposed in one region of the plurality of regions.

[0182] According to an embodiment, the ice making cell may be cooled by a cooler other than the cooler cooling the storage chamber. For example, the storage chamber in which the ice making cell is disposed is a refrigerating compartment which is controlled to a temperature higher than 0 degree, and the ice making cell may be cooled by a cooler other than the cooler cooling the refrigerating compartment. That is, the refrigerator may include a refrigerating compartment and a freezing compartment, the ice making cell may be disposed inside the refrigerating compartment, and the ice maker cell may be cooled by the cooler that cools the freezing compartment.

[0183] The ice making cell may be disposed in a door that opens and closes the storage chamber.

[0184] According to an embodiment, the ice making cell is not disposed inside the storage chamber and may be cooled by the cooler. For example, the entire storage chamber defined inside the outer case may be the ice making cell.

[0185] According to an embodiment, a degree of heat transfer indicates a degree of heat transfer from a high-temperature object to a low-temperature object and is defined as a value determined by a shape including a thickness of the object, a material of the object, and the like. In terms of the material of the object, a high degree of the heat transfer of the object may represent that thermal conductivity of the object is high. The thermal conductivity may be a unique material property of the object. Even when the material of the object is the same, the degree of heat transfer may vary depending on the shape of the object.

[0186] The degree of heat transfer may vary depending on the shape of the object. The degree of heat transfer from a point A to a point B may be influenced by a length of a path through which heat is transferred from the point A to the point B (hereinafter, referred to as a “heat transfer path”). The more the heat transfer path from the point A to the point B increases, the more the degree of heat transfer from the point A to the point B may decrease. The more the heat transfer path from the point A to the point B, the more the degree of heat transfer from the point A to the point B may increase.

[0187] The degree of heat transfer from the point A to the point B may be influenced by a thickness of the path through which heat is transferred from the point A to the point B. The more the thickness in a path direction in which heat is transferred from the point A to the point B decreases, the more the degree of heat transfer from the point A to the point B may decrease. The greater the thickness in the path direction from which the heat from point A to point B is transferred, the more the degree of heat transfer from point A to point B.

[0188] According to an embodiment, a degree of cold transfer indicates a degree of heat transfer from a low-temperature object to a high-temperature object and is defined as a value determined by a shape including a thickness of the object, a material of the object, and the like. The degree of cold transfer is a term defined in consideration of a direction in which cold air flows and may be regarded as the same concept as the degree of heat transfer. The same concept as the degree of heat transfer will be omitted.

[0189] According to an embodiment, a degree of supercooling is a degree of supercooling of a liquid and may be defined as a value determined by a material of the liquid, a material or shape of a container containing the liquid, an external factors applied to the liquid during a solidification process of the liquid, and the like. An increase in frequency at which the liquid is supercooled may be seen as an increase in degree of the supercooling. The lowering of the temperature at which the liquid is maintained in the supercooled state may be seen as an increase in degree of the supercooling. Here, the supercooling refers to a state in which the liquid exists in the liquid phase without solidification even at a temperature below a freezing point of the liquid. The supercooled liquid has a characteristic in which the solidification rapidly occurs from a time point at which the supercooling is terminated. If it is desired to maintain a rate at which the liquid is solidified, it is advantageous to be designed so that the supercooling phenomenon is reduced.

[0190] According to an embodiment, a degree of deformation resistance represents a degree to which an object resists deformation due to external force applied to the object and is a value determined by a shape including a thickness of the object, a material of the object, and the like. For example, the external force may include a pressure applied to the tray assembly in the process of solidifying and expanding water in the ice making cell. In another example, the external force may include a pressure on the ice or a portion of the tray assembly by the pusher for separating the ice from the tray assembly. For another example, when coupled between the tray assemblies, it may include a pressure applied by the coupling.

[0191] In terms of the material of the object, a high degree of the deformation resistance of the object may represent that rigidity of the object is high. The thermal conductivity may be a unique material property of the object. Even when the material of the object is the same, the degree of deformation resistance may vary depending on the shape of the object. The the degree of deformation resistance may be affected by a deformation resistance reinforcement part extending in a direction in which the external force is applied. The more the rigidity of the deformation resistant resistance reinforcement part increases, the more the degree of deformation resistance may increase. The more the height of the extending deformation resistance reinforcement part increase, the more the degree of deformation resistance may increase.

[0192] According to an embodiment, a degree of restoration indicates a degree to which an object deformed by the external force is restored to a shape of the object before the external force is applied after the external force is removed and is defined as a value determined by a shape including a thickness of the object, a material of the object, and the like. For example, the external force may include a pressure applied to the tray assembly in the process of solidifying and expanding water in the ice making cell. In another example, the external force may include a pressure on the ice or a portion of the tray assembly by the pusher for separating the ice from the tray assembly. For another example, when coupled between the tray assemblies, it may include a pressure applied by the coupling force.

[0193] In view of the material of the object, a high degree of the restoration of the object may represent that an elastic modulus of the object is high. The elastic modulus may be a material property unique to the object. Even when the material of the object is the same, the degree of restoration may vary depending on the shape of the object. The degree of restoration may be affected by an elastic resistance reinforcement part extending in a direction in which the external force is applied. The more the elastic modulus of the elastic resistance reinforcement part increases, the more the degree of restoration may increase.

[0194] According to an embodiment, the coupling force represents a degree of coupling between the plurality of tray assemblies and is defined as a value determined by a shape including a thickness of the tray assembly, a material of the tray assembly, magnitude of the force that couples the trays to each other, and the like.

[0195] According to an embodiment, a degree of attachment indicates a degree to which the ice and the container are attached to each other in a process of making ice from water contained in the container and is defined as a value determined by a shape including a thickness of the container, a material of the container, a time elapsed after the ice is made in the container, and the like.

[0196] The refrigerator according to an embodiment includes a first tray assembly defining a portion of an ice making cell that is a space in which water is phase-changed into ice by cold, a second tray assembly defining the other portion of the ice making cell, a cooler supplying cold to the ice making cell, a water supply part supplying water to the ice making cell, and a controller. The refrigerator may further include a storage chamber in addition to the ice making cell. The storage chamber may include a space for storing food. The ice making cell may be disposed in the storage chamber. The refrigerator may further include a first temperature sensor sensing a temperature in the storage chamber. The refrigerator may further include a second temperature sensor sensing a temperature of water or ice of the ice making cell. The second tray assembly may contact the first tray assembly in the ice making process and may be connected to the driver to be spaced apart from the first tray assembly in the ice making process. The refrigerator may further include a heater disposed adjacent to at least one of the first tray assembly or the second tray assembly.

[0197] The controller may control at least one of the heater or the driver. The controller may control the cooler so that the cold is supplied to the ice making cell after the second tray assembly moves to an ice making position when the water is completely supplied to the ice making cell. The controller may control the second tray assembly so that the second tray assembly moves in a reverse direction after moving to an ice separation position in a forward direction so as to take out the ice in the ice making cell when the ice is completely made in the ice making cell. The controller may control the second tray assembly so that the supply of the water supply part after the second tray assembly moves to the water supply position in the reverse direction when the ice is completely separated.

[0198] Transparent ice will be described. Bubbles are dissolved in water, and the ice solidified with the bubbles may have low transparency due to the bubbles. Therefore, in the process of water solidification, when the bubble is guided to move from a freezing portion in the ice making cell to another portion that is not yet frozen, the transparency of the ice may increase.

[0199] A through-hole defined in the tray assembly may affect the making of the transparent ice. The through-hole defined in one side of the tray assembly may affect the making of the transparent ice. In the process of making ice, if the bubbles move to the outside of the ice making cell from the frozen portion of the ice making cell, the transparency of the ice may increase. The through-hole may be defined in one side of the tray assembly to guide the bubbles so as to move out of the ice making cell. Since the bubbles have lower density than the liquid, the through-hole (hereinafter, referred to as an "air exhaust hole") for guiding the bubbles to escape to the outside of the ice making cell may be defined in the upper portion of the tray assembly.

[0200] The position of the cooler and the heater may affect the making of the transparent ice. The position of the cooler and the heater may affect an ice making direction, which is a direction in which ice is made inside the ice making cell.

[0201] In the ice making process, when bubbles move or are collected from a region in which water is first solidified in the ice making cell to another predetermined region in a liquid state, the transparency of the made ice may increase. The direction in which the bubbles move or are collected may be similar to the ice making direction. The predetermined region may be a region in which water is to be solidified lately in the ice making cell.

[0202] The predetermined region may be a region in which the cold supplied by the cooler reaches the ice making cell late. For example, in the ice making process, the through-hole through which the cooler supplies the cold to the ice making cell may be defined closer to the upper portion than the lower part of the ice making cell so as to move or collect the bubbles to the lower portion of the ice making cell. For another example, a heat absorbing part of the cooler (that is, a refrigerant pipe of the evaporator or a heat absorbing part of the thermoelectric element) may be disposed closer to the upper portion than the lower portion of the ice making cell. According to an embodiment, the upper and lower portions of the ice making cell may be defined as an upper region and a lower region based on a height of the ice making cell.

[0203] The predetermined region may be a region in which the heater is disposed. For example, in the ice making process, the heater may be disposed closer to the lower portion than the upper portion of the ice making cell so as to move or collect the bubbles in the water to the lower portion of the ice making cell.

[0204] The predetermined region may be a region closer to an outer circumferential surface of the ice making cell than to a center of the ice making cell. However, the vicinity of the center is not excluded. If the predetermined region is near the center of the ice making cell, an opaque portion due to the bubbles moved or collected near the center may be easily visible to the user, and the opaque portion may remain until most of the ice until the ice is melted. Also, it may be difficult to arrange the heater inside the ice making cell containing water. In contrast, when the predetermined region is defined in or near the outer circumferential surface of the ice making cell, water may be solidified from one side of the outer circumferential surface of the ice making cell toward the other side of the outer circumferential surface of the ice making cell, thereby solving the above limitation. The transparent ice heater may be disposed on or near the outer circumferential surface of the ice making cell. The heater may be disposed at or near the tray assembly.

[0205] The predetermined region may be a position closer to the lower portion of the ice making cell than the upper portion of the ice making cell. However, the upper portion is also not excluded. In the ice making process, since liquid water having greater density than ice drops, it may be advantageous that the predetermined region is defined in the lower portion of the ice making cell.

[0206] At least one of the degree of deformation resistance, the degree of restoration, and the coupling force between the plurality of tray assemblies may affect the making of the transparent ice. At least one of the degree of deformation resistance, the degree of restoration, and the coupling force between the plurality of tray assemblies may affect the ice making direction that is a direction in which ice is made in the ice making cell. As described above, the tray assembly may include a first region and a second region, which define an outer circumferential surface of the ice making cell. For example, each of the first and second regions may be a portion of one tray assembly. For another example, the first region may be a first tray assembly. The second region may be a second tray assembly.

[0207] To make the transparent ice, it may be advantageous for the refrigerator to be configured so that the direction in which ice is made in the ice making cell is constant. This is because the more the ice making direction is constant, the more the bubbles in the water are moved or collected in a predetermined region within the ice making cell. It may be advantageous for the deformation of the portion to be greater than the deformation of the other portion so as to induce the ice to be made in the direction of the other portion in a portion of the tray assembly. The ice tends to be grown as the ice is expanded toward a portion at which the degree of deformation resistance is low. To start the ice making again after removing the made ice, the deformed portion has to be restored again to make ice having the same shape repeatedly. Therefore, it may be advantageous that the portion having the low degree of the deformation resistance has a high degree of the restoration than the portion having a high degree of the deformation resistance.

[0208] The degree of deformation resistance of the tray with respect to the external force may be less than that of the tray case with respect to the external force, or the rigidity of the tray may be less than that of the tray case. The tray assembly allows the tray to be deformed by the external force, while the tray case surrounding the tray is configured to reduce the deformation. For example, the tray assembly may be configured so that at least a portion of the tray is surrounded by the tray case. In this case, when a pressure is applied to the tray assembly while the water inside the ice making cell is solidified and expanded, at least a portion of the tray may be allowed to be deformed, and the other part of the tray may be supported by the tray case to restrict the deformation. In addition, when the external force is removed, the degree of restoration of the tray may be greater than that of the tray case, or the elastic modulus of the tray may be greater than that of the tray case. Such a configuration may be configured so that the deformed tray is easily restored.

[0209] The degree of deformation resistance of the tray with respect to the external force may be greater than that of the gasket of the refrigerator with respect to the external force, or the rigidity of the tray may be greater than that of the gasket. When the degree of deformation resistance of the tray is low, there may be a limitation that the tray is excessively deformed as the water in the ice making cell defined by the tray is solidified and expanded. Such a deformation of the tray may make it difficult to make the desired type of ice. In addition, the degree of restoration of the tray when the external force is removed may be configured to be less than that of the refrigerator gasket with respect to the external force, or the elastic modulus of the tray is less than that of the gasket.

[0210] The deformation resistance of the tray case with respect to the external force may be less than that of the refrigerator case with respect to the external force, or the rigidity of the tray case may be less than that of the refrigerator case. In general, the case of the refrigerator may be made of a metal material including steel. In addition, when the external force is removed, the degree of restoration of the tray case may be greater than that of the refrigerator case with respect to the external force, or the elastic modulus of the tray case is greater than that of the refrigerator case.

[0211] The relationship between the transparent ice and the degree of deformation resistance is as follows.

[0212] The second region may have different degree of deformation resistance in a direction along the outer circumferential surface of the ice making cell. The degree of deformation resistance of one portion of the second region may be greater than that of the other portion of the second region. Such a configuration may be assisted to induce ice to be made in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0213] The first and second regions defined to contact each other may have different degree of deformation resistances in the direction along the outer circumferential surface of the ice making cell. The degree of deformation resistance of one portion of the second region may be greater than that of one portion of the first region. Such a configuration may be assisted to induce ice to be made in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0214] In this case, as the water is solidified, a volume is expanded to apply a pressure to the tray assembly, which induces ice to be made in the other direction of the second region or in one direction of the first region. The degree of deformation resistance may be a degree that resists to deformation due to the external force. The external force may a pressure applied to the tray assembly in the process of solidifying and expanding water in the ice making cell. The external force may be force in a vertical direction (Z-axis direction) of the pressure. The external force may be force acting in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0215] For example, in the thickness of the tray assembly in the direction of the outer circumferential surface of the ice making cell from the center of the ice making cell, one portion of the second region may be thicker than the other of the second region or thicker than one portion of the first region. One portion of the second region may be a portion at which the tray case is not surrounded. The other portion of the second region may be a portion surrounded by the tray case. One portion of the first region may be a portion at which the tray case is not surrounded. One portion of the second region may be a portion defining the uppermost portion of the ice making cell in the second region. The second region may include a tray and a tray case locally surrounding the tray. As described above, when at least a portion of the second region is thicker than the other part, the degree of deformation resistance of the second region may be improved with respect to an external force. A minimum value of the thickness of one portion of the second region may be greater than that of the thickness of the other portion of the second region or greater than that of one portion of the first region. A maximum value of the thickness of one portion of the second region may be greater than that of the thickness of the other portion of the second region or greater than that of one portion of the first region. When the through-hole is defined in the region, the minimum value represents the minimum value in the remaining regions except for the portion in which the through-hole is defined. An average value of the thickness of one portion of the second region may be greater than that of the thickness of the other portion of the second region or greater than that of one portion of the first region. The uniformity of the thickness of one portion of the second region may be less than that of the thickness of the other portion of the second region or less than that of one of the thickness of the first region.

[0216] For another example, one portion of the second region may include a first surface defining a portion of the ice making cell and a deformation resistance reinforcement part extending from the first surface in a vertical direction away from the ice making cell defined by the other of the second region. One portion of the second region may include a first surface defining a portion of the ice making cell and a deformation resistance reinforcement part extending from the first surface in a vertical direction away from the ice making cell defined by the first region. As described above, when at least a portion of the second region includes the deformation resistance reinforcement part, the degree of deformation resistance of the second region may be improved with respect to the external force.

[0217] For another example, one portion of the second region may further include a support surface connected to a fixed end of the refrigerator (e.g., the bracket, the storage chamber wall, etc.) disposed in a direction away from the ice making cell defined by the other of the second region from the first surface. One portion of the second region may further include a support surface connected to a fixed end of the refrigerator (e.g., the bracket, the storage chamber wall, etc.) disposed in a direction away from the ice making cell defined by the first region from the first surface. As described above, when at least a portion of the second region includes a support surface connected to the fixed end, the degree of deformation resistance of the second region may be improved with respect to the external force.

[0218] For another example, the tray assembly may include a first portion defining at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion. At least a portion of the second portion may extend in a direction away from the ice making cell defined by the first region. At least a portion of the second portion may include an additional deformation resistant resistance reinforcement part. At least a portion of the second portion may further include a support surface connected to the fixed end. As described above, when at least a portion of the second region further includes the second portion, it may be advantageous to improve the degree of deformation resistance of the second region with respect to the external force. This is because the additional deformation resistance reinforcement part is disposed at in the second portion, or the second portion is additionally supported by the fixed end.

[0219] For another example, one portion of the second region may include a first through-hole. As described above, when the first through-hole is defined, the ice solidified in the ice making cell of the second region is expanded to the outside of the ice making cell through the first through-hole, and thus, the pressure applied to the second region may be reduced. In particular, when water is excessively supplied to the ice making cell, the first through-hole may be contributed to reduce the deformation of the second region in the process of solidifying the water.

[0220] One portion of the second region may include a second through-hole providing a path through which the bubbles contained in the water in the ice making cell of the second region move or escape. When the second through-hole is defined as described above, the transparency of the solidified ice may be improved.

[0221] In one portion of the second region, a third through-hole may be defined to press the penetrating pusher. This is because it may be difficult for the non-penetrating type pusher to press the surface of the tray assembly so as to remove the ice when the degree of deformation resistance of the second region increases. The first, second, and third through-holes may overlap each other. The first, second, and third through-holes may be defined in one through-hole.

[0222] One portion of the second region may include a mounting part on which the ice separation heater is disposed. The induction of the ice in the ice making cell defined by the second region in the direction of the ice making cell defined by the first region may represent that the ice is first made in the second region. In this case, a time for which the ice is attached to the second region may be long, and the ice separation heater may be required to separate the ice from the second region. The thickness of the tray assembly in the direction of the outer circumferential surface of the ice making cell from the center of the ice making cell may be less than that of the other portion of the second region in which the ice separation heater is mounted. This is because the heat supplied by the ice separation heater increases in amount transferred to the ice making cell. The fixed end may be a portion of the wall defining the storage chamber or a bracket.

[0223] The relation between the coupling force of the transparent ice and the tray assembly is as follows.

[0224] To induce the ice to be made in the ice making cell defined by the second region in the direction of the ice making cell defined by the first region, it may be advantageous to increase in coupling force between the first and second regions arranged to contact each other. In the process of solidifying the water, when the pressure applied to the tray assembly while expanded is greater than the coupling force between the first and second regions, the ice may be made in a direction in which the first and second regions are separated from each other. In the process of solidifying the water, when the pressure applied to the tray assembly while expanded is low, the coupling force between the first and second regions is low, It also has the advantage of inducing the ice to be made so that the ice is made in a direction of the region having the smallest degree of deformation resistance in the first and second regions.

[0225] There may be various examples of a method of increasing the coupling force between the first and second regions. For example, after the water supply is completed, the controller may change a movement position of the driver in the first direction to control one of the first and second regions so as to move in the first direction, and then, the movement position of the driver may be controlled to be additionally changed into the first direction so that the coupling force between the first and second regions increases. For another example, since the coupling force between the first and second regions increase, the degree of deformation resistances or the degree of restorations of the first and second regions may be different from each other with respect to the force applied from the driver so that the driver reduces the change of the shape of the ice making cell by the expanding the ice after the ice making process is started (or after the heater is turned on). For another example, the first region may include a first surface facing the second region. The second region may include a second surface facing the first region. The first and second surfaces may be disposed to contact each other. The first and second surfaces may be disposed to face each other. The first and second surfaces may be disposed to be separated from and coupled to each other. In this case, surface areas of the first surface and the second surface may be different from each other. In this configuration, the coupling force of the first and second regions may increase while reducing breakage of the portion at which the first and second regions contact each other. In addition, there is an advantage of reducing leakage of water supplied between the first and second regions.

[0226] The relationship between transparent ice and the degree of restoration is as follows.

[0227] The tray assembly may include a first portion that defines at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion. The second portion is configured to be deformed by the expansion of the ice made and then restored after the ice is removed. The second portion may include a horizontal extension part provided so that the degree of restoration with respect to the horizontal external force of the expanded ice increases. The second portion may include a vertical extension part provided so that the degree of restoration with respect to the vertical external force of the expanded ice increases. Such a configuration may be assisted to induce ice to be made in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0228] The second region may have different degree of restoration in a direction along the outer circumferential surface of the ice making cell. The first region may have different degree of deformation resistance in a direction along the outer circumferential surface of the ice making cell. The degree of restoration of one portion of the first region may be greater than that of the other portion of the first region. Also, the degree of deformation resistance of one portion may be less than that of the other portion. Such a configuration may be assisted to induce ice to be made in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0229] The first and second regions defined to contact each other may have different degree of restoration in the direction along the outer circumferential surface of the ice making cell. Also, the first and second regions may have different degree of deformation resistances in the direction along the outer circumferential surface of the ice making cell. The degree of restoration of one of the first region may be greater than that of one of the second region. Also, The degree of deformation resistance of one of the first regions may be greater than that of one of the second region. Such a configuration may be assisted to induce ice to be made in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0230] In this case, as the water is solidified, a volume is expanded to apply a pressure to the tray assembly, which induces ice to be made in one direction of the first region in which the degree of deformation resistance decreases, or the degree of restoration increases. Here, the degree of restoration may be a degree of restoration after the external force is removed. The external force may a pressure applied to the tray assembly in the process of solidifying and expanding water in the ice making cell. The external force may be force in a vertical direction (Z-axis direction) of the pressure. The external force may be force acting in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0231] For example, in the thickness of the tray assembly in the direction of the outer circumferential surface of the ice making cell from the center of the ice making cell, one portion of the first region may be thinner than the other of the first region or thinner than one portion of the second region. One portion of the first region may be a portion at which the tray case is not surrounded. The other portion of the first region may be a portion that is surrounded by the tray case. One portion of the second region may be a portion that is surrounded by the tray case. One portion of the first region may be a portion of the first region that defines the lowermost end of the ice making cell. The first region may include a tray and a tray case locally surrounding the tray.

[0232] A minimum value of the thickness of one portion of the first region may be less than that of the thickness of the other portion of the second region or less than that of one of the second region. A maximum value of the thickness of one portion of the first region may be less than that of the thickness of the other portion of the first region or less than that of the thickness of one portion of the second region. When the through-hole is defined in the region, the minimum value represents the minimum value in the remaining regions except for the portion in which the through-hole is defined. An average value of the thickness of one portion of the first region may be less than that of the thickness of the other portion of the first region or may be less than that of one of the thickness of the second region. The uniformity of the thickness of one portion of the first region may be greater than that of the thickness of the other portion of the first region or greater than that of one of the thickness of the second region.

[0233] For another example, a shape of one portion of the first region may be different from that of the other portion of the first region or different from that of one portion of the second region. A curvature of one portion of the first region may be different from that of the other portion of the first region or different from that of one portion of the second region. A curvature of one portion of the first region may be less than that of the other portion of the first region or less than that of one portion of the second region. One portion of the first region may include a flat surface. The other portion of the first region may include a curved surface. One portion of the second region may include a curved surface. One portion of the first region may include a shape that is recessed in a direction opposite to the direction in which the ice is expanded. One portion of the first region may include a shape recessed in a direction opposite to a direction in which the ice is made. In the ice making process, one portion of the first region may be modified in a direction in which the ice is expanded or a direction in which the ice is made. In the ice making process, in an amount of deformation from the center of the ice making cell toward the outer circumferential surface of the ice making cell, one portion of the first region is greater than the other portion of the first region. In the ice making process, in the amount of deformation from the center of the ice making cell toward the outer circumferential surface of the ice making cell, one portion of the first region is greater than one portion of the second region.

[0234] For another example, to induce ice to be made in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region, one portion of the first region may include a first surface defining a portion of the ice making cell and a second surface extending from the first surface and supported by one surface of the other portion of the first region. The first region may be configured not to be directly supported by the other component except for the second surface. The other component may be a fixed end of the refrigerator.

[0235] One portion of the first region may have a pressing surface pressed by the non-penetrating type pusher. This is because when the degree of deformation resistance of the first region is low, or the degree of restoration is high, the difficulty in removing the ice by pressing the surface of the tray assembly may be reduced.

[0236] An ice making rate, at which ice is made inside the ice making cell, may affect the making of the transparent ice. The ice making rate may affect the transparency of the made ice. Factors affecting the ice making rate may be an amount of cold and/or heat, which are/is supplied to the ice making cell. The amount of cold and/or heat may affect the making of the transparent ice. The amount of cold and/or heat may affect the transparency of the ice.

[0237] In the process of making the transparent ice, the transparency of the ice may be lowered as the ice making rate is greater than a rate at which the bubbles in the ice making cell are moved or collected. On the other hand, if the ice making rate is less than the rate at which the bubbles are moved or collected, the transparency of the ice may increase. However, the more the ice making rate decreases, the more a time taken to make the transparent ice may increase. Also, the transparency of the ice may be uniform as the ice making rate is maintained in a uniform range.

[0238] To maintain the ice making rate uniformly within a predetermined range, an amount of cold and heat supplied to the ice making cell may be uniform. However, in actual use conditions of the refrigerator, a case in which the amount of cold is variable may occur, and thus, it is necessary to allow a supply amount of heat to vary. For example, when a temperature of the storage chamber reaches a satisfaction region from a dissatisfaction region, when a defrosting operation is performed with respect to the cooler of the storage chamber, the door of the storage chamber may variously vary in state such as an opened state. Also, if an amount of water per unit height of the ice making cell is different, when the same cold and heat per unit height is supplied, the transparency per unit height may vary.

[0239] To solve this limitation, the controller may control the heater so that when a heat transfer amount between the cold within the storage chamber and the water of the ice making cell increases, the heating amount of transparent ice heater increases, and when the heat transfer amount between the cold within the storage chamber and the water of the ice making cell decreases, the heating amount of transparent ice heater decreases so as to maintain an ice making rate of the water within the ice making cell within a predetermined range that is less than an ice making rate when the ice making is performed in a state in which the heater is turned off.

[0240] The controller may control one or more of a cold supply amount of cooler and a heat supply amount of heater to vary according to a mass per unit height of water in the ice making cell. In this case, the transparent ice may be provided to correspond to a change in shape of the ice making cell.

[0241] The refrigerator may further include a sensor measuring information on the mass of water per unit height of the ice making cell, and the controller may control one of the cold supply amount of cooler and the heat supply amount of heater based on the information inputted from the sensor.

[0242] The refrigerator may include a storage part in which predetermined driving information of the cooler is recorded based on information on mass per unit height of the ice making cell, and the controller may control the cold supply amount of cooler to be changed based on the information.

[0243] The refrigerator may include a storage part in which predetermined driving information of the heater is recorded based on information on mass per unit height of the ice making cell, and the controller may control the heat supply amount of heater to be changed based on the information. For example, the controller may control at least one of the cold supply amount of cooler or the heat supply amount of heater to vary according to a predetermined time based on the information on the mass per unit height of the ice making cell. The time may be a time when the cooler is driven or a time when the heater is driven to make ice. For another example, the controller may control at least one of the cold supply amount of cooler or the heat supply amount of heater to vary according to a predetermined temperature based on the information on the mass per unit height of the ice making cell. The temperature may be a temperature of the ice making cell or a temperature of the tray assembly defining the ice making cell.

[0244] When the sensor measuring the mass of water per unit height of the ice making cell is malfunctioned, or when the water supplied to the ice making cell is insufficient or excessive, the shape of the ice making water is changed, and thus the transparency of the made ice may decrease. To solve this limitation, a water supply method in which an amount of water supplied to the ice making cell is precisely controlled is required. Also, the tray assembly may include a structure in which leakage of the tray assembly is reduced to reduce the leakage of water in the ice making cell at the water supply position or the ice making position. Also, it is necessary to increase the coupling force between the first and second tray assemblies defining the ice making cell so as to reduce the change in shape of the ice making cell due to the expansion force of the ice during the ice making. Also, it is necessary to decrease in leakage in the precision water supply method and the tray assembly and increase in coupling force between the first and second tray assemblies so as to make ice having a shape that is close to the tray shape.

[0245] The degree of supercooling of the water inside the ice making cell may affect the making of the transparent ice. The degree of supercooling of the water may affect the transparency of the made ice.

[0246] To make the transparent ice, it may be desirable to design the degree of supercooling or lower the temperature inside the ice making cell and thereby to maintain a predetermined range. This is because the supercooled liquid has a characteristic in which the solidification rapidly occurs from a time point at which the supercooling is terminated. In this case, the transparency of the ice may decrease.

[0247] In the process of solidifying the liquid, the controller of the refrigerator may control the supercooling release part to operate so as to reduce a degree of supercooling of the liquid if the time required for reaching the specific temperature below the freezing point after the temperature of the liquid reaches the freezing point is less than a reference value. After reaching the freezing point, it is seen that the temperature of the liquid is cooled below the freezing point as the supercooling occurs, and no solidification occurs.

[0248] An example of the supercooling release part may include an electrical spark generating part. When the spark is supplied to the liquid, the degree of supercooling of the liquid may be reduced. Another example of the supercooling release part may include a driver applying external force so that the liquid moves. The driver may allow the container to move in at least one direction among X, Y, or Z axes or to rotate about at least one axis among X, Y, or Z axes. When kinetic energy is supplied to the liquid, the degree of supercooling of the liquid may be reduced. Further another example of the supercooling release part may include a part supplying the liquid to the container. After supplying the liquid having a first volume less than that of the container, when a predetermined time has elapsed or the temperature of the liquid reaches a certain temperature below the freezing point, the controller of the refrigerator may control an amount of liquid to additionally supply the liquid having a second volume greater than the first volume. When the liquid is divided and supplied to the container as described above, the liquid supplied first may be solidified to act as freezing nucleus, and thus, the degree of supercooling of the liquid to be supplied may be further reduced.

[0249] The more the degree of heat transfer of the container containing the liquid increase, the more the degree of supercooling of the liquid may increase. The more the degree of heat transfer of the container containing the liquid decrease, the more the degree of supercooling of the liquid may decrease.

[0250] The structure and method of heating the ice making cell in addition to the heat transfer of the tray assembly may affect the making of the transparent ice. As described above, the tray assembly may include a first region and a second region, which define an outer circumferential surface of the ice making cell. For example, each of the first and second regions may be a portion of one tray assembly. For another example, the first region may be a first tray assembly. The second region may be a second tray assembly.

[0251] The cold supplied to the ice making cell and the heat supplied to the ice making cell have opposite properties. To increase the ice making rate and/or improve the transparency of the ice, the design of the structure and control of the cooler and the heater, the relationship between the cooler and the tray assembly, and the relationship between the heater and the tray assembly may be very important.

[0252] For a constant amount of cold supplied by the cooler and a constant amount of heat supplied by the heater, it may be advantageous for the heater to be arranged to locally heat the ice making cell so as to increase the ice making rate of the refrigerator and/or to increase the transparency of the ice. As the heat transmitted from the heater to the ice making cell is transferred to an area other than the area on which the heater is disposed, the ice making rate may be improved. As the heater heats only a portion of the ice making cell, the heater may move or collect the bubbles to an area adjacent to the heater in the ice making cell, thereby increasing the transparency of the ice.

[0253] When the amount of heat supplied by the heater to the ice making cell is large, the bubbles in the water may be moved or collected in the portion to which the heat is supplied, and thus, the made ice may increase in transparency. However, if the heat is uniformly supplied to the outer circumferential surface of the ice making cell, the ice making rate of the ice may decrease. Therefore, as the heater locally heats a portion of the ice making cell, it is possible to increase the transparency of the made ice and minimize the decrease of the ice making rate.

[0254] The heater may be disposed to contact one side of the tray assembly. The heater may be disposed between the tray and the tray case. The heat transfer through the conduction may be advantageous for locally heating the ice making cell.

[0255] At least a portion of the other side at which the heater does not contact the tray may be sealed with a heat insulation material. Such a configuration may reduce that the heat supplied from the heater is transferred toward the storage chamber.

[0256] The tray assembly may be configured so that the heat transfer from the heater toward the center of the ice making cell is greater than that transfer from the heater in the circumference direction of the ice making cell.

[0257] The heat transfer of the tray toward the center of the ice making cell in the tray may be greater than the that transfer from the tray case to the storage chamber, or the thermal conductivity of the tray may be greater than that of the tray case. Such a configuration may induce the increase in heat transmitted from the heater to the ice making cell via the tray. In addition, it is possible to reduce the heat of the heater is transferred to the storage chamber via the tray case.

[0258] The heat transfer of the tray toward the center of the ice making cell in the tray may be less than that of the refrigerator case toward the storage chamber from the outside of the refrigerator case (for example, an inner case or an outer case), or the thermal conductivity of the tray may be less than that of the refrigerator case. This is because the more the heat or thermal conductivity of the tray increases, the more the supercooling of the water accommodated in the tray may increase. The more the degree of supercooling of the water increase, the more the water may be rapidly solidified at the time point at which the supercooling is released. In this case, a limitation may occur in which the transparency of the ice is not uniform or the transparency decreases. In general, the case of the refrigerator may be made of a metal material including steel.

[0259] The heat transfer of the tray case in the direction from the storage chamber to the tray case may be greater than the that of the heat insulation wall in the direction from the outer space of the refrigerator to the storage chamber, or the thermal conductivity of the tray case may be greater than that of the heat insulation wall (for example, the insulation material disposed between the inner and outer cases of the refrigerator). Here, the heat insulation wall may represent a heat insulation wall that partitions the external space from the storage chamber. If the degree of heat transfer of the tray case is equal to or greater than that of the heat insulation wall, the rate at which the ice making cell is cooled may be excessively reduced.

[0260] The first region may be configured to have a different degree of heat transfer in a direction along the outer circumferential surface. The degree of heat transfer of one portion of the first region may be less than that of the other portion of the first region. Such a configuration may be assisted to reduce the heat transfer transferred through the tray assembly from the first region to the second region in the direction along the outer circumferential surface.

[0261] The first and second regions defined to contact each other may be configured to have a different degree of heat transfer in the direction along the outer circumferential surface. The degree of heat transfer of one portion of the first region may be configured to be less than the degree of heat transfer of one portion of the second region. Such a configuration may be assisted to reduce the heat transfer transferred through the tray assembly from the first region to the second region in the direction along the outer circumferential surface. In another aspect, it may be advantageous to reduce the heat transferred from the heater to one portion of the first region to be transferred to the ice making cell defined by the second region. As the heat transmitted to the second region is reduced, the heater may locally heat one portion of the first region. Thus, it may be possible to reduce the decrease in ice making rate by the heating of the heater. In another aspect, the bubbles may be moved or collected in the region in which the heater is locally heated, thereby improving the transparency of the ice. The heater may be a transparent ice heater.

[0262] For example, a length of the heat transfer path from the first region to the second region may be greater than that of the heat transfer path in the direction from the first region to the outer circumferential surface from the first region. For another example, in a thickness of the tray assembly in the direction of the outer circumferential surface of the ice making cell from the center of the ice making cell, one portion of the first region may be thinner than the other of the first region or thinner than one portion of the second region. One portion of the first region may be a portion at which the tray case is not surrounded. The other portion of the first region may be a portion that is surrounded by the tray case. One portion of the second region may be a portion that is surrounded by the tray case. One portion of the first region may be a portion of the first region that defines the lowest end of the ice making cell. The first region may include a tray and a tray case locally surrounding the tray.

[0263] As described above, when the thickness of the first region is thin, the heat transfer in the direction of the center of the ice making cell may increase while reducing the heat transfer in the direction of the outer circumferential surface of the ice making cell. For this reason, the ice making cell defined by the first region may be locally heated.

[0264] A minimum value of the thickness of one portion of the first region may be less than that of the thickness of the other portion of the second region or less than that of one of the second region. A maximum value of the thickness of one portion of the first region may be less than that of the thickness of the other portion of the first region or less than that of the thickness of one portion of the second region. When the through-hole is defined in the region, the minimum value represents the minimum value in the remaining regions except for the portion in which the through-hole is defined. An average value of the thickness of one portion of the first region may be less than that of the thickness of the other portion of the first region or may be less than that of one of the thickness of the second region. The uniformity of the thickness of one portion of the first region may be greater than that of the thickness of the other portion of the first region or greater than that of one of the thickness of the second region.

[0265] For example, the tray assembly may include a first portion defining at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion. The first region may be defined in the first portion. The second region may be defined in an additional tray assembly that may contact the first portion. At least a portion of the second portion may extend in a direction away from the ice making cell defined by the second region. In this case, the heat transmitted from the heater to the first region may be reduced from being transferred to the second region.

[0266] The structure and method of cooling the ice making cell in addition to the degree of cold transfer of the tray assembly may affect the making of the transparent ice. As described above, the tray assembly may include a first region and a second region, which define an outer circumferential surface of the ice making cell. For example, each of the first and second regions may be a portion of one tray assembly. For another example, the first region may be a first tray assembly. The second region may be a second tray assembly.

[0267] For a constant amount of cold supplied by the cooler and a constant amount of heat supplied by the heater, it may be advantageous to configure the cooler so that a portion of the ice making cell is more intensively cooled to increase the ice making rate of the refrigerator and/or increase the transparency of the ice. The more the cold supplied to the ice making cell by the cooler increases, the more the ice making rate may increase. However, as the cold is uniformly supplied to the outer circumferential surface of the ice making cell, the transparency of the made ice may decrease. Therefore, as the cooler more intensively cools a portion of the ice making cell, the bubbles may be moved or collected to other regions of the ice making cell, thereby increasing the transparency of the made ice and minimizing the decrease in ice making rate.

[0268] The cooler may be configured so that the amount of cold supplied to the second region differs from that of cold supplied to the first region so as to allow the cooler to more intensively cool a portion of the ice making cell. The amount of cold supplied to the second region by the cooler may be greater than that of cold supplied to the first region.

[0269] For example, the second region may be made of a metal material having a high cold transfer rate, and the first region may be made of a material having a cold rate less than that of the metal.

[0270] For another example, to increase the degree of cold transfer transmitted from the storage chamber to the center of the ice making cell through the tray assembly, the second region may vary in degree of cold transfer toward the central direction. The degree of cold transfer of one portion of the second region may be greater than that of the other portion of the second region. A through-hole may be defined in one portion of the second region. At least a portion of the heat absorbing surface of the cooler may be disposed in the through-hole. A passage through which the cold air supplied from the cooler passes may be disposed in the through-hole. The one portion may be a portion that is not surrounded by the tray case. The other portion may be a portion surrounded by the tray case. One portion of the second region may be a portion defining the uppermost portion of the ice making cell in the second region. The second region may include a tray and a tray case locally surrounding the tray. As described above, when a portion of the tray assembly has a high cold transfer rate, the supercooling may occur in the tray assembly having a high cold transfer rate. As described above, designs may be needed to reduce the degree of the supercooling.

[0271] Hereinafter, a specific embodiment of the refrigerator according to an embodiment will be described with reference to the drawings.

[0272] FIG. **1** is a front view of a refrigerator according to an embodiment.

[0273] Referring to FIG. **1**, a refrigerator according to an embodiment may include a cabinet **14** including a storage chamber and a door that opens and closes the storage chamber.

[0274] The storage chamber may include a refrigerating compartment **18** and a freezing compartment **32**. The refrigerating compartment **18** is disposed at an upper side, and the freezing compartment **32** is disposed at a lower side. Each of the storage chamber may be opened and closed individually by each door. For another example, the freezing compartment may be disposed at the upper side and the refrigerating compartment may be disposed at the lower side. Alternatively, the freezing compartment may be disposed at one side of left and right sides, and the refrigerating compartment may be disposed at the other side.

[0275] The freezing compartment **32** may be divided into an upper space and a lower space, and a drawer **40** capable of being withdrawn from and inserted into the lower space may be provided in the lower space.

[0276] The door may include a plurality of doors **10**, **20**, **30** for opening and closing the refrigerating compartment **18** and the freezing compartment **32**. The plurality of doors **10**, **20**, and **30** may include some or all of the doors **10** and **20** for opening and closing the storage chamber in a rotatable manner and the door **30** for opening and closing the storage chamber in a sliding manner.

[0277] The freezing compartment **32** may be provided to be separated into two spaces even though the freezing compartment **32** is opened and closed by one door **30**.

[0278] In this embodiment, the freezing compartment **32** may be referred to as a first storage chamber, and the refrigerating compartment **18** may be referred to as a second storage chamber.

[0279] The freezing compartment **32** may be provided with an ice maker **200** capable of making ice. The ice maker **200** may be disposed, for example, in an upper space of the freezing compartment **32**.

[0280] An ice bin **600** in which the ice made by the ice maker **200** falls to be stored may be disposed below the ice maker **200**. A user may take out the ice bin **600** from the freezing compartment **32** to use the ice stored in the ice bin **600**. The ice bin **600** may be mounted on an upper side of a horizontal wall that partitions an upper space and a lower space of the freezing compartment **32** from each other. Although not shown, the cabinet **14** is provided with a duct supplying cold air to the ice maker **200** (not shown). The duct guides the cold air heat-exchanged with a refrigerant flowing through the evaporator to the ice maker **200**. For example, the duct may be disposed behind the cabinet **14** to discharge the cold air toward a front side of the cabinet **14**. The ice maker **200** may be disposed at a front side of the duct. Although not limited, a discharge hole of the duct may be provided in one or more of a rear wall and an upper wall of the freezing compartment **32**.

[0281] Although the above-described ice maker **200** is provided in the freezing compartment **32**, a space in which the ice maker **200** is disposed is not limited to the freezing compartment **32**. For example, the ice maker **200** may be disposed in various spaces as long as the ice maker **200** receives the cold air.

[0282] Therefore, hereinafter, the ice maker **200** will be described as being disposed in a storage chamber.

[0283] FIG. **2** is a perspective view of the ice maker according to an embodiment, and FIG. **3** is a front view of the ice maker of FIG. **2**. FIG. **4** is a perspective view illustrating a state in which a bracket is removed from the ice maker of FIG. **3**, and FIG. **5** is an exploded perspective view of the ice maker according to an embodiment.

[0284] Referring to FIGS. **2** to **5**, each component of the ice maker **200** may be provided inside or outside the bracket **220**, and thus, the ice maker **200** may constitute one assembly.

[0285] The ice maker **200** may include a first tray assembly and a second tray assembly.

[0286] The first tray assembly may include a first tray **320**, a first tray case, or all of the first tray **320** and a second tray case. The second tray assembly may include a second tray **380**, a second tray case, or all of the second tray **380** and a second tray case.

[0287] The bracket **220** may define at least a portion of a space that accommodates the first tray assembly and the second tray assembly.

[0288] The bracket **220** may be installed at, for example, the upper wall of the freezing compartment **32**. The bracket **220** may be provided with a water supply part **240**. The water supply part **240** may guide water supplied from the upper side to the lower side of the water supply part **240**. A water supply pipe (not shown) to which water is supplied may be installed above the water supply part **240**. The water supplied to the water supply part **240** may move downward. The water supply part **240** may prevent the water discharged from the water supply pipe from dropping from a high position, thereby preventing the water from splashing. Since the water supply part **240** is disposed below the water supply pipe, the water may be guided downward without splashing up to the water supply part **240**, and an amount of splashing water may be reduced even if the water moves downward due to the lowered height.

[0289] The ice maker **200** may include an ice making cell (see **320***a* in FIG. **49**) in which water is phase-changed into ice by the cold air.

[0290] The first tray **320** may constitute at least a portion of the ice making cell **320***a*. The second tray **380** may include a second tray **380** defining the other portion of the ice making cell **320***a*.

[0291] The second tray **380** may be disposed to be relatively movable with respect to the first tray **320**. The second tray **380** may linearly move or rotate. Hereinafter, the rotation of the second tray **380** will be described as an example.

[0292] For example, in an ice making process, the second tray **380** may move with respect to the first tray **320** so that the first tray **320** and the second tray **380** contact each other. When the first tray **320** and the second tray **380** contact each other, the complete ice making cell **320***a* may be defined.

[0293] On the other hand, the second tray **380** may move with respect to the first tray **320** during the ice making process after the ice making is completed, and the second tray **380** may be spaced apart from the first tray **320**.

[0294] In this embodiment, the first tray **320** and the second tray **380** may be arranged in a vertical direction in a state in which the ice making cell **320***a* is formed. Accordingly, the first tray **320** may be referred to as an upper tray, and the second tray **380** may be referred to as a lower tray.

[0295] A plurality of ice making cells **320***a* may be defined by the first tray **320** and the second tray **380**. Hereinafter, in the drawing, three ice making cells **320***a* are provided as an example.

[0296] When water is cooled by cold air while water is supplied to the ice making cell **320***a*, ice having the same or similar shape as that of the ice making cell **320***a* may be made.

[0297] In this embodiment, for example, the ice making cell **320***a* may be provided in a spherical shape or a shape similar to a spherical shape. The ice making cell **320***a* may have a rectangular parallelepiped shape or a polygonal shape.

[0298] For example, the first tray case may include the first tray supporter **340** and the first tray cover **320**. The first tray supporter **340** and the first tray cover **320** may be integrally provided or coupled to each other with each other after being manufactured in separate configurations. For example, at least a portion of the first tray cover **300** may be disposed above the first tray **320**. At least a portion of the first tray supporter **340** may be disposed under the first tray **320**. The first tray cover **300** may be manufactured as a separate part from the bracket **220** and then may be coupled to the bracket **220** or integrally formed with the bracket **220**. That is, the first tray case may include the bracket **220**.

[0299] The ice maker **200** may further include a first heater case **280**. An ice separation heater (see **290** of FIG. **21**) may be installed in the first heater case **280**. The heater case **280** may be integrally formed with the first tray cover **300** or may be separately formed. The ice separation heater **290** may be disposed at a position adjacent to the first tray **320**. The ice separation heater **290** may be, for example, a wire type heater. For example, the ice separation heater **290** may be installed to contact the first tray **320** or may be disposed at a position spaced a predetermined distance from the first tray **320**.

[0300] In some case, the ice separation heater **290** may supply heat to the first tray **320**, and the heat supplied to the first tray **320** may be transferred to the ice making cell **320***a*.

[0301] The first tray cover **300** may be provided to correspond to a shape of the ice making cell **320***a* of the first tray **320** and may contact a lower portion of the first tray **320**.

[0302] The ice maker **200** may include a first pusher **260** separating the ice during an ice separation process. The first pusher **260** may receive power of the driver **480** to be described later.

[0303] The first tray cover **300** may be provided with a guide slot **302** guiding movement of the first pusher **260**. The guide slot **302** may be provided in a portion extending upward from the first tray cover **300**. A guide connection part of the first pusher **260** to be described later may be inserted into the guide slot **302**. Thus, the guide connection part may be guided along the guide slot **302**.

[0304] The first pusher **260** may include at least one pushing bar **264**. For example, the first pusher **260** may include a pushing bar **264** provided with the same number as the number of ice making cells **320***a*, but is not limited thereto. The pushing bar **264** may push out the ice disposed in the ice making cell **320***a* during the ice separation process. For example, the pushing bar **264** may be inserted into the ice making cell **320***a* through the first tray cover **300**. Therefore, the first tray cover **300** may be provided with an opening **304** (or through-hole) through which a portion of the first pusher **260** passes.

[0305] The first pusher **260** may be coupled to a pusher link **500**. In this case, the first pusher **260** may be coupled to the pusher link **500** so as to be rotatable. Therefore, when the pusher link **500** moves, the first pusher **260** may also move along the guide slot **302**.

[0306] The second tray case may include, for example, a second tray cover **360** and a second tray supporter **400**.

[0307] The second tray cover **360** and the second tray supporter **400** may be integrally formed or coupled to each other with each other after being manufactured in separate configurations. For example, at least a portion of the second tray cover **360** may be disposed above the second tray **380**. At least a portion of the second tray supporter **400** may be disposed below the second tray **380**. The second tray supporter **400** may be disposed at a lower side of the second tray to support the second tray **380**. For example, at least a portion of the wall defining a second cell **320***c* of the second tray **380** may be supported by the second tray supporter **400**.

[0308] A spring **402** may be connected to one side of the second tray supporter **400**. The spring **402** may provide elastic force to the second tray supporter **400** to maintain a state in which the second tray **380** contacts the first tray **320**.

[0309] The second tray **380** may include a circumferential wall **387** surrounding a portion of the first tray **320** in a state of contacting the first tray **320**. The second tray cover **360** may cover at least a portion of the circumferential wall **387**.

[0310] The ice maker **200** may further include a second heater case **420**. A transparent ice heater **430** to be described later may be installed in the second heater case **420**. The second heater case **420** may be integrally formed with the second tray supporter **400** or may be separately provided to be coupled to the second tray supporter **400**.

[0311] The ice maker **200** may further include a driver **480** that provides driving force. The second tray **380** may relatively move with respect to the first tray **320** by receiving the driving force of the driver **480**. The first pusher **260** may move by receiving the driving force of the driving force **480**.

[0312] A through-hole **282** may be defined in an extension part **281** extending downward in one side of the first tray cover **300**. A through-hole **404** may be defined in the extension part **403** extending in one side of the second tray supporter **400**. At least a portion of the through-hole **404** may be disposed at a position higher than a horizontal line passing through a center of the ice making cell **320***a*. The ice maker **200** may further include a shaft **440** (or a rotation shaft) that passes through the through-holes **282** and **404** together.

[0313] A rotation arm **460** may be provided at each of both ends of the shaft **440**. The shaft **440** may rotate by receiving rotational force from the driver **480**. One end of the rotation arm **460** may be connected to one end of the spring **402**, and thus, a position of the rotation arm **460** may move to an initial value by restoring force when the spring **402** is tensioned.

[0314] The driver **480** may include a motor and a plurality of gears. A full ice detection lever **520** may be connected to the driver **480**. The full ice detection lever **520** may also rotate by the rotational force provided by the driver **480**. The full ice detection lever **520** may have a 'L' shape as a whole. For example, the full ice detection lever **520** may include a first lever **521** and a pair of second levers **522** extending in a direction crossing the first lever **521** at both ends of the first lever **521**. One of the pair of second levers **522** may be coupled to the driver **480**, and the other may be coupled to the bracket **220** or the first tray cover **300**. The full ice detection lever **520** may rotate to detect ice stored in the ice bin **600**.

[0315] The driver **480** may further include a cam that rotates by the rotational power of the motor. The ice maker **200** may further include a sensor that senses the rotation of the cam.

[0316] For example, the cam is provided with a magnet, and the sensor may be a hall sensor detecting magnetism of the magnet during the rotation of the cam. The sensor may output first and second signals that are different outputs according to whether the sensor senses a magnet. One of the first signal and the second signal may be a high signal, and the other may be a low signal.

[0317] The controller **800** to be described later may determine a position of the second tray **380** (or the second tray assembly) based on the type and pattern of the signal outputted from the sensor. That is, since the second tray **380** and the cam rotate by the motor, the position of the second tray **380** may be indirectly determined based on a detection signal of the magnet provided in the cam. For example, a water supply position, an ice making position, and an ice separation position, which will be described later, may be distinguished and determined based on the signals outputted from the sensor.

[0318] The ice maker **200** may further include a second pusher **540**. The second pusher **540** may be installed, for example, on the bracket **220**. The second pusher **540** may include at least one pushing bar **544**. For example, the second pusher **540** may include a pushing bar **544** provided with the same number as the number of ice making cells **320***a*, but is not limited thereto. The pushing bar **544** may push out the ice disposed in the ice making cell **320***a*. For example, the pushing bar **544** may pass through the second tray supporter **400** to contact the second tray **380** defining the ice making cell **320***a* and then press the contacting second tray **380**.

[0319] The first tray cover **300** may be rotatably coupled to the second tray supporter **400** with respect to the second tray supporter **400** and then be disposed to change in angle about the shaft **440**.

[0320] In this embodiment, the second tray **380** may be made of a non-metal material. For example, when the second tray **380** is pressed by the second pusher **540**, the second tray **380** may be made of a flexible or soft material which is deformable. Although not limited, the second tray **380** may be made of, for example, a silicon material.

[0321] Therefore, while the second tray **380** is deformed while the second tray **380** is pressed by the second pusher **540**, pressing force of the second pusher **540** may be transmitted to ice. The ice and the second tray **380** may be separated from each other by the pressing force of the second pusher **540**.

[0322] When the second tray **380** is made of the non-metal material and the flexible or soft material, the coupling force or attaching force between the ice and the second tray **380** may be reduced, and thus, the ice may be easily separated from the second tray **380**.

[0323] Also, if the second tray **380** is made of the non-metallic material and the flexible or soft material, after the shape of the second tray **380** is deformed by the second pusher **540**, when the pressing force of the second pusher **540** is removed, the second tray **380** may be easily restored to its original shape.

[0324] For another example, the first tray **320** may be made of a metal material. In this case, since the coupling force or the attaching force between the first tray **320** and the ice is strong, the ice maker **200** according to this embodiment may include at least one of the ice separation heater (see **290** in FIG. **21**) or the first pusher **260**.

[0325] For another example, the first tray **320** may be made of a non-metallic material. When the first tray **320** is made of the non-metallic material, the ice maker **200** may include only one of the ice separation heater **290** and the first pusher **260**. Alternatively, the ice maker **200** may not include the ice separation heater **290** and the first pusher **260**.

[0326] Although not limited, the first tray **320** may be made of, for example, a silicon material. That is, the first tray **320** and the second tray **380** may be made of the same material. When the first tray **320** and the second tray **380** are made of the same material, the first tray **320** and the second tray **380** may have different hardness to maintain sealing performance at the contact portion between the first tray **320** and the second tray **380**. In this embodiment, since the second tray **380** is pressed by the second pusher **540** to be deformed, the second tray **380** may have hardness less than that of the first tray **320** to facilitate the deformation of the second tray **380**.

[0327] FIGS. **6** and **7** are perspective views of the bracket according to an embodiment.

[0328] Referring to FIGS. **6** and **7**, the bracket **220** may be fixed to at least one surface of the storage chamber or to a cover member (to be described later) fixed to the storage chamber.

[0329] The bracket **220** may include a first wall **221** having a through-hole **221**_a_ defined therein. At least a portion of the first wall **221** may extend in a horizontal direction. The first wall **221** may include a first fixing wall **221**_b_ to be fixed to one surface of the storage chamber or the cover member. At least a portion of the first fixing wall **221**_b_ may extend in the horizontal direction. The first fixing wall **221**_b_ may also be referred to as a horizontal fixing wall. One or more fixing protrusions **221**_c_ may be provided on the first fixing wall **221**_b_. A plurality of fixing protrusions **221**_c_ may be provided on the first fixing wall **221**_b_ to firmly fix the bracket **220**.

[0330] The first wall **221** may further include a second fixing wall **221**_e_ to be fixed to one surface of the storage chamber or the cover member. At least a portion of the second fixing wall **221**_e_ may extend in a vertical direction. The second fixing wall **221**_e_ may also be referred to as a vertical fixing wall. The second fixing wall **221**_e_ may extend upward from the first fixing wall **221**_b_. The second fixing wall **221**_e_ may include a fixing rib **221**_e_**1** and/or a hook **221**_e_**2**.

[0331] In this embodiment, the first wall **221** may include at least one of the first fixing wall **221**_b_ or the second fixing wall **221**_e_ to fix the bracket **220**.

[0332] The first wall **221** may be provided in a shape in which a plurality of walls are stepped in the vertical direction. In one example, a plurality of walls may be arranged with a height difference in the horizontal direction, and the plurality of walls may be connected by a vertical connection wall.

[0333] The first wall **221** may further include a support wall **221**_d_ supporting the first tray assembly. At least a portion of the support wall **221**_d_ may extend in the horizontal direction. The support wall **221**_d_ may be disposed at the same height as the first fixing wall **221**_b_ or disposed at a different height. In FIG. **6**, for example, the support wall **221**_d_ is disposed at a position lower than that of the first fixing wall **221**_b_.

[0334] The bracket **220** may further include a second wall **222** having a through-hole **222**_a_ through which cold air generated by a cooling part passes. The second wall **222** may extend from the first wall **221**. At least a portion of the second wall **222** may extend in the vertical direction. At least a portion of the through-hole **222**_a_ may be disposed at a position higher than that of the support wall **221**_d_. In FIG. **6**, for example, the lowermost end of the through-hole **222**_a_ is disposed at a position higher than that of the support wall **221**_d_.

[0335] The bracket **220** may further include a third wall **223** on which the driver **480** is installed. The third wall **223** may extend from the first wall **221**. At least a portion of the third wall **223** may extend in the vertical direction. At least a portion of the third wall **223** may be disposed to face the second wall **222** while being spaced apart from the second wall **222**. At least a portion of the ice making cell (see **320**_a_ in FIG. **49**) may be disposed between the second wall **222** and the second wall **223**.

[0336] The driver **480** may be installed on the third wall **223** between the second wall **222** and the third wall **223**. Alternatively, the driver **480** may be installed on the third wall **223** so that the third wall **223** is disposed between the second wall **222** and the driver **480**. In this case, a shaft hole **223**_a_ through which a shaft of the motor constituting the driver **480** passes may be defined in the third wall **223**. FIG. **7** illustrates that the shaft hole **223**_a_ is defined in the third wall **223**.

[0337] The bracket **220** may further include a fourth wall **224** to which the second pusher **540** is fixed. The fourth wall **224** may extend from the first wall **221**. The fourth wall **224** may connect the second wall **222** to the third wall **223**. The fourth wall **224** may be inclined at an angle with respect to the horizontal line and the vertical line. For example, the fourth wall **224** may be inclined in a direction away from the shaft hole **223**_a_ from the upper side to the lower side. The fourth wall **224** may be provided with a mounting groove **224**_a_ in which the second pusher **540** is mounted. The mounting groove **224**_a_ may be provided with a coupling hole **224**_b_ through which a coupling part coupled to the second pusher **540** passes.

[0338] The second tray **380** and the second pusher **540** may contact each other while the second tray assembly rotates while the second pusher **540** is fixed to the fourth wall **224**. Ice may be separated from the second tray **380** while the second pusher **540** presses the second tray **380**.

[0339] When the second pusher **540** presses the second tray **380**, the ice also presses the second pusher **540** before the ice is separated from the second tray **380**. Force for pressing the second pusher **540** may be transmitted to the fourth wall **224**. Since the fourth wall **224** is provided in a thin plate shape, a strength reinforcement member **224**_c_ may be provided on the fourth wall **224** to prevent the fourth wall **224** from being deformed or broken.

[0340] For example, the strength reinforcement member **224**_c_ may include ribs disposed in a lattice form. That is, the strength reinforcement member **224**_c_ may include a first rib extending in the first direction and a second rib extending in a second direction crossing the first direction.

[0341] In this embodiment, two or more of the first to fourth walls **221** to **224** may define a space in which the first and second tray assemblies are disposed.

[0342] FIG. **8** is a perspective view of the first tray when viewed from an upper side, and FIG. **9** is a perspective view of the first tray when viewed from a lower side. FIG. **10** is a plan view of the first tray. FIG. **11** is a cutaway cross-sectional view taken along line **11-11** of FIG. **8**.

[0343] Referring to FIGS. **8** to **10**, the first tray **320** may define a first cell **321***a* that is a portion of the ice making cell **320***a*.

[0344] The first tray **320** may include a first tray wall **321** defining a portion of the ice making cell **320***a*. For example, the first tray **320** may define a plurality of first cells **321***a*. For example, the plurality of first cells **321***a* may be arranged in a line. The plurality of first cells **321***a* may be arranged in an X-axis direction in FIG. **9**. For example, the first tray wall **321** may define the plurality of first cells **321***a*.

[0345] The first tray wall **321** may include a plurality of first cell walls **3211** that respectively define the plurality of first cells **321***a*, and a connection wall **3212** connecting the plurality of first cell walls **3211** to each other. The first tray wall **321** may be a wall extending in the vertical direction.

[0346] The first tray **320** may include an opening **324**. The opening **324** may communicate with the first cell **321***a*. The opening **324** may allow the cold air to be supplied to the first cell **321***a*. The opening **324** may allow water for making ice to be supplied to the first cell **321***a*. The opening **234** may provide a passage through which a portion of the first pusher **260** passes. For example, in the ice separation process, a portion of the first pusher **260** may be inserted into the ice making cell **320***a* through the opening **234**.

[0347] The first tray **320** may include a plurality of openings **324** corresponding to the plurality of first cells **321***a*. One of the plurality of openings **324 324***a* may provide a passage of the cold air, a passage of the water, and a passage of the first pusher **260**. In the ice making process, the bubbles may escape through the opening **324**.

[0348] The first tray **320** may include a case accommodation part **321***b*. For example, a portion of the first tray wall **321** may be recessed downward to provide the case accommodation part **321***b*. At least a portion of the case accommodation part **321***b* may be disposed to surround the opening **324**. A bottom surface of the case accommodation part **321***b* may be disposed at a position lower than that of the opening **324**. Therefore, the ice separation heater **290** and the second temperature sensor **700** may be disposed at positions lower than that of a support surface on which the first tray **320** supports the first tray cover **300**.

[0349] The first tray **320** may further include an auxiliary storage chamber **325** communicating with the ice making cell **320***a*. For example, the auxiliary storage chamber **325** may store water overflowed from the ice making cell **320***a*. The ice expanded in a process of phase-changing the supplied water may be disposed in the auxiliary storage chamber **325**. That is, the expanded ice may pass through the opening **304** and be disposed in the auxiliary storage chamber **325**. The auxiliary storage chamber **325** may be defined by a storage chamber wall **325***a*. The storage chamber wall **325***a* may extend upwardly around the opening **324**. The storage chamber wall **325***a* may have a cylindrical shape or a polygonal shape. Substantially, the first pusher **260** may pass through the opening **324** after passing through the storage chamber wall **325***a*.

[0350] The storage chamber wall **325***a* may define the auxiliary storage chamber **325** and also reduce deformation of the periphery of the opening **324** in the process in which the first pusher **260** passes through the opening **324** during the ice separation process.

[0351] When the first tray **320** defines a plurality of first cells **321***a*, at least one **325***b* of the plurality of storage chamber walls **325***a* may support the water supply part **240**.

[0352] The storage chamber wall **325***b* supporting the water supply part **240** may have a polygonal shape. For example, the storage chamber wall **325***b* may include a round part rounded in a horizontal direction and a plurality of straight portions.

[0353] For example, the storage chamber wall **325***b* may include a round wall **325***b***1**, a pair of straight walls **325***b***2** and **325***b***3** extending side by side from both ends of the round wall **325***b*, and a connection wall **325***b***4** connecting the pair of straight walls **325***b***2** to each other. The connection wall **325***b***4** may be a rounded wall or a straight wall.

[0354] An upper end of the connection wall **325***b***4** may be disposed at a position lower than that of an upper end of the remaining walls **325***b***1**, **325***b***2**, and **325***b***3**. The connection wall **325***b***4** may support the water supply part **240**. An opening **324***a* corresponding to the storage chamber wall **325***b* supporting the water supply part **240** may also be defined in the same shape as the storage chamber wall **325***b*.

[0355] The first tray **320** may further include a heater accommodation part **321***c*.

[0356] The ice separation heater **290** may be accommodated in the heater accommodation part **321***c*. The ice separation heater **290** may contact a bottom surface of the heater accommodation part **321***c*. The heater accommodation part **321***c* may be provided on the first tray wall **321** as an example. The heater accommodation part **321***c* may be recessed downward from the case accommodation part **321***b*. The heater accommodation part **321***c* may be disposed to surround the periphery of the first cell **321***a*. For example, at least a portion of the heater accommodation part **321***c* may be rounded in the horizontal direction. The bottom surface of the heater accommodating portion **321***c* may be disposed at a position lower than that of the opening **324**.

[0357] The first tray **320** may include a first contact surface **322***c* contacting the second tray **380**. The bottom surface of the heater accommodating portion **321***c* may be disposed between the opening **324** and the first contact surface **322***c*. At least a portion of the heater accommodation part **321***c* may be disposed to overlap the ice making cell **320***a* (or the first cell **321***a* in the vertical direction).

[0358] The first tray **320** may further include a first extension wall **327** extending in the horizontal direction from the first tray wall **321**. For example, the first extension wall **327** may extend in the horizontal direction around an upper end of the first extension wall **327**. One or more first coupling holes **327***a* may be provided in the first extension wall **327**. Although not limited, the plurality of first coupling holes **327***a* may be arranged in one or more axes of the X axis and the Y axis. An upper end of the storage chamber wall **325***b* may be disposed at the same height or higher than an top surface of the first extension wall **327**.

[0359] Referring to FIG. **10**, the first extension wall **327** may include a first edge line **327***b* and a second edge line **327***c*, which are spaced apart from each other in a Y direction with respect to a central line **C1** (or the vertical central line) in the Z axis direction in the ice making cell

320*a*. In this specification, the "central line" is a line passing through a volume center of the ice making cell **320***a* or a center of gravity of water or ice in the ice making cell **320***a* regardless of the axial direction. The first edge line **327***b* and the second edge line **327***c* may be parallel to each other.

[0360] A distance L**1** from the central line C**1** to the first edge line **327***b* is longer than a distance L**2** from the central line C**1** to the first edge line **327***b*.

[0361] The first extension wall **327** may include a third edge line **327***d* and a fourth edge line **327***e*, which are spaced apart from each other in the X direction in the ice making cell **320***a*. The third edge line **327***d* and the fourth edge line **327***e* may be parallel to each other. A length of each of the third edge line **327***d* and the fourth edge line **327***e* may be shorter than a length of each of the first edge line **327***b* and the second edge line **327***c*.

[0362] The length of the first tray **320** in the X-axis direction may be referred to as a length of the first tray, the length of the first tray **320** in the Y-axis direction may be referred to as a width of the first tray, and the length of the first tray **320** in the Z-axis direction may be referred to as a height of the first tray **320**.

[0363] In this embodiment, an X-Y-axis cutting surface may be a horizontal plane.

[0364] When the first tray **320** includes the plurality of first cells **321***a*, the length of the first tray **320** may be longer, but the width of the first tray **320** may be shorter than the length of the first tray **320** to prevent the volume of the first tray **320** from increasing.

[0365] FIG. **12** is a bottom view of the first tray of FIG. **9**, FIG. **13** is a cutaway cross-sectional view taken along line **13-13** of FIG. **11**, and FIG. **14** is a cutaway cross-sectional view taken along line **14-14** of FIG. **11**.

[0366] Referring to FIGS. **11** to **14**, the first tray **320** may include a first portion **322** that defines a portion of the ice making cell **320***a*. For example, the first portion **322** may be a portion of the first tray wall **321**.

[0367] The first portion **322** may include a first cell surface **322***b* (or an outer circumferential surface) defining the first cell **321***a*. The first cell **321** may be divided into a first region defined close to the transparent ice heater **430** and a second region defined far from the transparent ice heater **430** in the Z axis direction. The first region may include the first contact surface **322***c*, and the second region may include the opening **324**. The first portion **322** may be defined as an area between two dotted lines in FIG. **11**.

[0368] The first portion **322** may include the opening **324**. Also, the first portion **322** may include the heater accommodation part **321***c*.

[0369] In a degree of deformation resistance from the center of the ice making cell **320***a* in the circumferential direction, at least a portion of the upper portion of the first portion **322** is greater than at least a portion of the lower portion. The degree of deformation resistance of at least a portion of the upper portion of the first portion **322** is greater than that of the lowermost end of the first portion **322**. The upper and lower portions of the first portion **322** may be divided based on the extension direction of the central line C**1**. The lowermost end of the first portion **322** is the first contact surface **322***c* contacting the second tray **380**.

[0370] The first tray **320** may further include a second portion **323** extending from a predetermined point of the first portion **322**. The predetermined point of the first portion **322** may be one end of the first portion **322**. Alternatively, the predetermined point of the first portion **322** may be one point of the first contact surface **322***c*.

[0371] A portion of the second portion **323** may be defined by the first tray wall **321**, and the other portion of the second portion **323** may be defined by the first extension wall **327**.

[0372] At least a portion of the second portion **323** may extend in a direction away from the transparent ice heater **430**. At least a portion of the second portion **323** may extend upward from the first contact surface **322***c*. At least a portion of the second portion **323** may extend in a direction away from the central line C**1**. For example, the second portion **323** may extend in both directions along the Y axis from the central line C**1**.

[0373] The second portion **323** may be disposed at a position higher than or equal to the uppermost end of the ice making cell **320***a*. The uppermost end of the ice making cell **320***a* is a portion at which the opening **324** is defined.

[0374] The second portion **323** may include a first extension part **323***a* and a second extension part **323***b*, which extend in different directions with respect to the central line C**1**. The first tray wall **321** may include one portion of the second extension part **323***b* of each of the first portion **322** and the second portion **323**. The first extension wall **327** may include the other portion of each of the first extension part **323***a* and the second extension part **323***b*.

[0375] Referring to FIG. **11**, the first extension part **323***a* may be disposed at the left side with respect to the central line C**1**, and the second extension part **323***b* may be disposed at the right side with respect to the central line C**1**.

[0376] The first extension part **323***a* and the second extension part **323***b* may have different shapes based on the central line C**1**. The first extension part **323***a* and the second extension part **323***b* may be provided in an asymmetrical shape with respect to the central line C**1**. A length of the second extension part **323***b* in the Y-axis direction may be greater than that of the first extension part **323***a*. Therefore, while the ice is made and grown from the upper side in the ice making process, the degree of deformation resistance of the second extension part **323***b* may increase.

[0377] The first extension part **323***a* may be disposed closer to an edge part that is disposed at a side opposite to the portion of the second wall **222** or the third wall **223** of the bracket **220**, which is connected to the fourth wall **224**, than the second extension part **323***a*. The second extension part **323***b* may be disposed closer to the shaft **440** that provides a center of rotation of the second tray assembly than the first extension part **323***a*.

[0378] In this embodiment, since the length of the second extension part **323***b* in the Y-axis direction is greater than that of the first extension part **323***a*, the second tray assembly including the second tray **380** contacting the first tray **320** may increase in radius of rotation. When the rotation radius of the second tray assembly increases, centrifugal force of the second tray assembly may increase. Thus, in the ice separation process, separating force for separating the ice from the second tray assembly may increase to improve ice separation performance.

[0379] Referring to FIGS. **11** to **14**, the thickness of the first tray wall **321** is minimized at a side of the first contact surface **322***c*. At least a portion of the first tray wall **321** may increase in thickness from the first contact surface **322***c* toward the upper side.

[0380] FIG. **13** illustrates a thickness of the first tray wall **321** at a first height H**1** from the first contact surface **322***c*, and FIG. **14** illustrates a thickness of the first tray wall **321** at a second height H2 from the first contact surface **322***c*.

[0381] Each of the thicknesses t2 and t3 of the first tray wall **321** at the first height H1 from the first contact surface **322***c* may be greater than the thickness t1 at the first contact surface **322***c* of the first tray wall **321**. The thicknesses t2 and t3 of the first tray wall **321** at the first height H1 from the first contact surface **322***c* may not be constant in the circumferential direction.

[0382] At the first height H1 from the first contact surface **322***c*, the first tray wall **321** further includes a portion of the second portion **323**. Thus, the thickness t3 of the portion at which the second extension part **323***b* is disposed may be greater than the thickness t2 on the opposite side of the second extension part **323***b* with respect to the central line C1.

[0383] The thicknesses t4 and t5 of the first tray wall **321** at the second height H2 from the first contact surface **322***c* may be greater than the thicknesses t2 and t3 of the first tray **321** at the first height H1 of the first tray wall **321**. The thicknesses t4 and t5 of the first tray wall **321** at the second height H2 from the first contact surface **322***c* may not be constant in the circumferential direction.

[0384] At the second height H2 from the first contact surface **322***c*, the first tray wall **321** further includes a portion of the second portion **323**. Thus, the thickness t5 of the portion at which the second extension part **323***b* is disposed may be greater than the thickness t4 on the opposite side of the second extension part **323***b* with respect to the central line C1.

[0385] At least a portion of the outer line of the first tray wall **321** may have a non-zero curvature with respect to the X-Y axis cutting surface of the first tray wall **321**, and thus, the curvature may vary. In this embodiment, the line represents a straight line having zero curvature. A curvature greater than zero represents a curve.

[0386] Referring to FIG. **12**, a circumference of an outer line at the first contact surface **322***c* of the first tray wall **321** may have a constant curvature. That is, an amount of change in curvature around the outer line of the first tray wall **321** on the first contact surface **322***c* may be zero.

[0387] Referring to FIG. **13**, at the first height H1 from the first contact surface **322***c*, an amount of change in curvature of at least a portion of the outer line of the first tray wall **321** may be greater than zero. That is, at the first height H1 from the first contact surface **322***c*, a curvature of at least a portion of the outer line of the first tray wall **321** may vary in the circumferential direction.

[0388] For example, at the first height H1 from the first contact surface **322***c*, the curvature of the outer line **323***b***1** of the second portion **323** may be greater than that of the outer line of the first portion **322**.

[0389] Referring to FIG. **14**, at the second height H2 from the first contact surface **322***c*, an amount of change in curvature of the outer line of the first tray wall **321** may be greater than zero. That is, at the second height H2 from the first contact surface **322***c*, the curvature of the outer line of the first tray wall **321** may vary in the circumferential direction.

[0390] For example, at the second height H2 from the first contact surface **322***c*, the curvature of the outer line **323***b***2** of the second portion **323** may be greater than the curvature of the outer line of the first portion **322**. A curvature of at least a portion of the outer line **323***b***2** of the second portion **323** at the second height H2 from the first contact surface **322***c* is greater than that of at least a portion of the outer line **323***b***1** of the second portion **323** at the first height H1 from the first contact surface **322***c*.

[0391] Referring to FIG. **11**, the curvature of the outer line **322***e* of the first extension part **323***a* in the first portion **322** may be zero in the Y-Z axis cutting surface with respect to the central line C1. In the Y-Z axis cutting surface with respect to the central line C1, the curvature of the outer line **323***d* of the second extension part **323***b* of the second part **323** may be greater than zero. For example, the outer line **323***d* of the second extension part **323***b* uses the shaft **440** as a center of curvature.

[0392] FIG. **15** is a cutaway cross-sectional view taken along line **15-15** of FIG. **8**.

[0393] Referring to FIGS. **8**, **10**, and **15**, the first tray **320** may further include a sensor accommodation part **321***e* in which the second temperature sensor **700** (or the tray temperature sensor) is accommodated. The second temperature sensor **700** may sense a temperature of water or ice of the ice making cell **320***a*.

[0394] The second temperature sensor **700** may be disposed adjacent to the first tray **320** to sense the temperature of the first tray **320**, thereby indirectly determining the water temperature or the ice temperature of the ice making cell **320***a*. In this embodiment, the water temperature or the ice temperature of the ice making cell **320***a* may be referred to as an internal temperature of the ice making cell **320***a*. The sensor accommodation part **321***e* may be recessed downward from the case accommodation part **321***b*. Here, a bottom surface of the sensor accommodation part **321***e* may be disposed at a position lower than that of the bottom surface of the heater accommodation part **321***c* to prevent the second temperature sensor **700** from interfering with the ice separation heater **290** in a state in which the second temperature sensor **700** is accommodated in the sensor accommodation part **321***e*.

[0395] The bottom surface of the sensor accommodating portion **321***e* may be disposed closer to the first contact surface **322***c* of the first tray **320** than the bottom surface of the heater accommodating portion **321***c*.

[0396] The sensor accommodation part **321***e* may be disposed between two adjacent ice making cells **320***a*. For example, the sensor accommodation part **321***e* may be disposed between two adjacent first cells **321***a*. When the sensor accommodation part **321***e* is disposed between the two ice making cells **320***a*, the second temperature sensor **700** may be easily installed without increasing the volume of the second tray **250**. Also, when the sensor accommodation part **321***e* is disposed between the two ice making cells **320***a*, the temperatures of at least two ice making cells **320***a* may be affected. Thus, the temperature sensor may be disposed so that the temperature sensed by the second temperature sensor maximally approaches an actual temperature inside the cell **320***a*.

[0397] Referring to FIG. **10**, the sensor accommodation part **321***e* may be disposed between the two adjacent first cells **321***a* among the three first cells **321***a* arranged in the X-axis direction. The sensor accommodation part **321***e* may be disposed between the right first cell and the central first cell of both the left and right sides among the three first cells **321***a*. Here, a distance D2 between the right first cell and the central first cell on the first contact surface **322***c* may be greater than that D1 between the central first cell and the left first cell so that a space in which the sensor accommodation part **321***e* is disposed may be secured between the right first cell and the central first cell.

[0398] The connection wall **3212** may be provided in plurality to improve the uniformity of the ice making direction between the plurality of ice making cells **320***a*.

[0399] For example, the connection wall **3212** may include a first connection wall **3212***a* and a second connection wall **3212***b*. The second connection wall **3212***b* may be disposed far from the through-hole **222***a* of the bracket **220** than the first connection wall **3212***a*. The first connection wall **3212***a* may include a first region and a second region having a thicker cross-section than the first region. The ice may be made in the direction from the ice making cell **320***a* defined by the first region to the ice making cell **320***a* defined by the second region. The second connection wall **3212***b* may include a first region and a second region including a sensor accommodation part **321***e* in which the second temperature sensor **700** is disposed.

[0400] FIG. **16** is a perspective view of the first tray, FIG. **17** is a bottom perspective view of the first tray cover, FIG. **18** is a plan view of the first tray cover, and FIG. **19** is a side view of the first tray case.

[0401] Referring to FIGS. **16** to **19**, the first tray cover **300** may include an upper plate **301** contacting the first tray **320**.

[0402] A bottom surface of the upper plate **301** may be coupled to contact an upper side of the first tray **320**. For example, the upper plate **301** may contact at least one of a top surface of the first portion **322** and a top surface of the second portion **323** of the first tray **320**. A plate opening **304** (or through-hole) may be defined in the upper plate **301**. The plate opening **304** may include a straight portion and a curved portion. Water may be supplied from the water supply part **240** to the first tray **320** through the plate opening **304**. Also, the extension part **264** of the first pusher **260** may pass through the plate opening **304** to separate ice from the first tray **320**. Also, cold air may pass through the plate opening **304** to contact the first tray **320**.

[0403] A first case coupling part **301***b* extending upward may be disposed at a side of the straight portion of the plate opening **304** in the upper plate **301**. The first case coupling part **301***b* may be coupled to the first coupling parts **285** and **286** (see FIG. **20**) of the first heater case **280** that will be described later.

[0404] The first tray cover **300** may further include a circumferential wall **303** extending upward from an edge of the upper plate **301**. The circumferential wall **303** may include two pairs of walls facing each other. For example, the pair of walls may be spaced apart from each other in the X-axis direction, and another pair of walls may be spaced apart from each other in the Y-axis direction.

[0405] The circumferential walls **303** spaced apart from each other in the Y-axis direction of FIG. **16** may include an extension wall **302***e* extending upward. The extension wall **302***e* may extend upward from a top surface of the circumferential wall **303**.

[0406] The first tray cover **300** may include a pair of guide slots **302** guiding the movement of the first pusher **260**. A portion of the guide slot **302** may be defined in the extension wall **302***e*, and the other portion may be defined in the circumferential wall **303** disposed below the extension wall **302***e*. A lower portion of the guide slot **302** may be defined in the circumferential wall **303**. The guide slot **302** may extend in the Z-axis direction of FIG. **16**.

[0407] The first pusher **260** may be inserted into the guide slot **302** to move. Also, the first pusher **260** may move up and down along the guide slot **302**.

[0408] The guide slot **302** may include a first slot **302***a* extending perpendicular to the upper plate **301** and a second slot **302***b* that is bent at an angle from an upper end of the first slot **302***a*. Alternatively, the guide slot **302** may include only the first slot **302***a* extending in the vertical direction.

[0409] The lower end **302***d* of the first slot **302***a* may be disposed lower than the upper end of the circumferential wall **303**. Also, the upper end **302***c* of the first slot **302***a* may be disposed higher than the upper end of the circumferential wall **303**. The portion bent from the first slot **302***a* to the second slot **302***b* may be disposed at a position higher than the circumferential wall **303**. A length of the first slot **302***a* may be greater than that of the second slot **302***b*. The second slot **302***b* may be bent toward the horizontal extension part **305**.

[0410] When the first pusher **260** moves upward along the guide slot **302**, the first pusher **260** rotates or is tilted at a predetermined angle in the portion moving along the second slot **302***b*.

[0411] When the first pusher **260** rotates, the pushing bar **264** of the first pusher **260** may rotate so that the pushing bar **264** is spaced apart vertically above the opening **324** of the first tray **320**.

[0412] When the first pusher **260** moves along the second slot **302***b* that is bent and extended, the end of the pushing bar **264** may be spaced apart so as not to contact with water supplied when water is supplied to the pushing bar. Thus, the water may be cooled at the end of **264** to prevent the pushing bar **264** from being inserted into the opening **324** of the first tray **320**.

[0413] The first tray cover **300** may include a plurality of coupling parts **301***a* coupling the first tray **320** to the first tray supporter **340** (see FIG. **24**) to be described later. The plurality of coupling parts **301***a* may be disposed on the upper plate **301**. The plurality of coupling parts **301***a* may be spaced apart from each other in the X-axis and/or Y-axis directions. The coupling part **301***a* may protrude upward from the top surface of the upper plate **301**. For example, a portion of the plurality of coupling parts **301***a* may be connected to the circumferential wall **303**.

[0414] The coupling part **301***a* may be coupled to a coupling member to fix the first tray **320**. The coupling member coupled to the coupling part **301***a* may be, for example, a bolt. The coupling member may pass through the coupling hole **341***a* of the first tray supporter **340** and the first coupling hole **327***a* of the first tray **320** at the bottom surface of the first tray supporter **340** and then be coupled to the coupling part **301***a*.

[0415] A horizontal extension part **305** extending horizontally form the circumferential wall **303** may be disposed on one circumferential wall **3030** of the circumferential walls **303** spaced apart from and facing each other in the Y-axis direction of FIG. **16**. The horizontal extension part **305** may extend from the circumferential wall **303** in a direction away from the plate opening **304** so as to be supported by the support wall **221***d* of the bracket **220**.

[0416] A plurality of vertical coupling parts **303***a* may be provided on the other one of the circumferential walls **303** spaced apart from and facing each other in the Y-axis direction.

[0417] The vertical coupling part **303***a* may be coupled to the first wall **221** of the bracket **220**. The vertical coupling parts **303***a* may be arranged to be spaced apart from each other in the X-axis direction.

[0418] The upper plate **301** may be provided with a lower protrusion **306** protruding downward. The lower protrusion **306** may extend along the length of the upper plate **301** and may be disposed around the circumferential wall **303** of the other of the circumferential walls **303** spaced apart from each other in the Y-axis direction. A step portion **306***a* may be disposed on the lower protrusion **306**. The step portion **306***a* may be disposed between a pair of extension parts **281** described later. Thus, when the second tray **380** rotates, the second tray **380** and the first tray cover **300** may not interfere with each other.

[0419] The first tray cover **300** may further include a plurality of hooks **307** coupled to the first wall **221** of the bracket **220**. For example, the hooks **307** may be provided on the horizontal protrusion **306**. The plurality of hooks **307** may be spaced apart from each other in the X-axis direction. The plurality of hooks **307** may be disposed between the pair of extension parts **281**.

[0420] Each of the hooks **307** may include a first portion **307***a* horizontally extending from the circumferential wall **303** in the opposite direction to the upper plate **301** and a second portion **307***b* bent from an end of the first portion **307***a* to extend vertically downward.

[0421] The first tray cover **300** may further include a pair of extension parts **281** to which the shaft **440** is coupled. For example, the pair of extension parts **281** may extend downward from the lower protrusion **306**. The pair of extension parts **281** may be spaced apart from each other in the X-axis direction. Each of the extension parts **281** may include a through-hole **282** through which the shaft **440** passes.

[0422] The first tray cover **300** may further include an upper wire guide part **310** guiding a wire connected to the ice separation heater **290**, which will be described later. The upper wire guide part **310** may, for example, extend upward from the upper plate **301**. The upper wire guide part **310** may include a first guide **312** and a second guide **314**, which are spaced apart from each other. For example, the first guide **312** and the second guide **314** may extend vertically upward from the upper plate **310**.

[0423] The first guide **312** may include a first portion **312***a* extending from one side of the plate opening **304** in the Y-axis direction, a second portion **312***b* bent and extending from the first portion **312***a*, and a third portion **312***c* bent from the second portion **312***b* to extend in the X-axis direction. The third portion **312***c* may be connected to one circumferential wall **303**.

[0424] A first protrusion **313** may be disposed on an upper end of the second portion **312***b* to prevent the wire from being separated.

[0425] The second guide **314** may include a first extension part **314***a* disposed to face the second portion **312***b* of the first guide **312** and a second extension part **314***b* bent to extend from the first extension part **314***a* and disposed to face the third portion **312***c*. The second portion **312***b* of the first guide **312** and the first extension part **314***a* of the second guide **314** and also the third portion **312***c* of the first guide **312** and the second extension part **314***b* of the second guide **314** may be parallel to each other. A second protrusion **315** may be disposed on an upper end of the first extension part **314***a* to prevent the wire from being separated.

[0426] The wire guide slots **313***a* and **315***a* may be defined in the upper plate **310** to correspond to the first and second protrusions **313** and **315**, and a portion of the wire may be the wire guide slots **313***a* and **315***a* to prevent the wire from being separated.

[0427] FIG. **20** is a perspective view of a first heater case, FIG. **21** is a bottom perspective view of the first heater case, FIG. **22** is a partial enlarged view of the first heater case, and FIG. **23** is a cross-sectional view illustrating a coupling relationship between the first heater case and the first tray.

[0428] Referring to FIGS. **20** to **23**, an ice separation heater **290** providing heat to the ice making cell **320***a* may be fixed to the first heater case **280**. The ice separation heater **290** may be, for example, a wire type heater. Thus, the ice separation heater **290** may be bent.

[0429] The ice separation heater **290** may be disposed to surround the plurality of ice making cells **320***a* so that the heat of the ice making heater **290** is evenly transferred to each of the plurality of ice making cells **320***a*. For example, the ice separation heater **290** may be disposed to surround the first cell **321***a*.

[0430] The first heater case **280** may be coupled to the first tray cover **300** at an upper end of the first tray **320**. Alternatively, the first heater case **280** may be integrally formed with the first tray cover **300**.

[0431] The first heater case **280** may include a first heater accommodation part **283** accommodating the ice separation heater **290**. The first heater accommodation part **283** may include a curved portion **283***a* and a straight portion **283***b*. The ice separation heater **290** may be accommodated in the first heater accommodation part **283** from a lower side of the first heater accommodation part **283**.

[0432] The first heater accommodation part **283** may include a plurality of curved portions **283***a* corresponding to the upper portion of the first tray **320** defining the plurality of ice making cells **320***a* and a straight portion **283***b* between the plurality of curved portions **283***a*. A plurality of separation prevention protrusions **283***c* preventing the ice separation heater **290** from being separated when the ice separation heater **290** is accommodated may be disposed on the curved portion **283***a* and the straight portion **283***b*.

[0433] The first heater accommodation part **283** may be provided with a plurality of separation prevention grooves **283***d* defined to correspond to the separation prevention protrusions **283***c*. The separation prevention protrusion **238***c* may extend to the lower portion of the first heater case **280**, and the end thereof may be bent in the horizontal direction.

[0434] For example, the separation prevention protrusion **283***c* may include a first protrusion portion extending vertically from the lower portion of the first heater accommodation part **283** and a second protrusion portion extending in the horizontal direction from the first protrusion. The second protrusion portion of the separation prevention protrusion **283***c* may extend toward the central portion of the first heater case **280**. For example, the separation prevention groove **283***d* may be defined in the curved portion **283***a* and/or the straight portion **283***b* corresponding to the size of one of the separation prevention protrusions **283***c*. For another example, the separation prevention groove **283***d* may have a size corresponding to the plurality of separation prevention protrusions **283***c*.

[0435] Since a portion of the ice separation heater **290** is inserted into the separation prevention groove **283***d*, the ice separation heater **290** may be prevented from being separated or being disconnected by the separation prevention protrusion **283***c* that prevent the ice separation heater **290** from being separated.

[0436] A plurality of first coupling parts **285** and **286** coupled to the first tray cover **300** may be provided on the straight portion **283***b*. The first coupling parts **285** and **286** may extend vertically upward from the straight portion **283***b*. The plurality of first coupling parts **285** and **286** may be spaced apart from each other in the Y-axis direction and face each other.

[0437] The first coupling parts **285** and **286** may include first protrusions **285***a* and **286***a* that protrude outwardly from the first heater case **280** and second protrusions **285***a* and **286***a* protruding with a radius that is less than that of the first protrusions **285***a* and **286***a*. For example, the second protrusions **285***b* and **286***b* may be coupled to the first case coupling part **301***b* of the first tray cover **300**.

[0438] A portion of the first heater case **280** may be inserted into the plate opening **304** of the first tray cover **300**, and a portion of the first coupling parts **285** and **586** may protrude upward from the plate opening **304** and be coupled to the first case coupling part **301***b* of the first tray cover **300**.

[0439] A temperature sensor accommodation part **284** accommodating the second temperature sensor **700** may be provided between the pair of first heater coupling parts **285** among the first coupling parts **285** and **286**. The temperature sensor accommodation part **284** may be accommodated from the lower side of the heater case **280** to contact the first tray **320** without contacting the moving ice heater **290**. The temperature sensor accommodation part **284** may include a temperature sensor accommodation space in a lower side thereof to prevent the ice separation heater **290** from contacting and to secure a space in which the second temperature sensor **700** is accommodated. An interference prevention part **284***a* may be disposed on the temperature sensor accommodation part **284** to prevent an interference with the first tray **320**.

[0440] The first case coupling part **285** connected to the temperature sensor accommodation part **284** may have a relatively short length that extends upward when compared to the other first coupling part **286** because the temperature sensor accommodation part **284** further protrudes upward.

[0441] Referring to FIG. **23**, a portion of the first tray **320** may be inserted into a through-opening **288** of the first heater case **280**, and the ice separation heater **290** accommodated in the first heater case **280** may contact the first tray **320** to apply heat to the first tray **320**.

[0442] The second temperature sensor **700** may be accommodated in the sensor accommodation part **321***e* of the first tray **320**. In the state in which the second temperature sensor **700** is accommodated in the sensor accommodation part **321***e*, the second temperature sensor **700** is spaced apart from the ice separation heater **290**. For example, since the ice separation heater **290** contacts the top surface of the second protrusion portion on the separation prevention protrusion **283***a*, the ice separation heater **290** may be spaced apart from the second temperature sensor **700** by the second protrusion portion.

[0443] An upper end of at least one of the ice separation heater **290** and the second temperature sensor **700** may be disposed at a position lower than that of the support wall **221***d* of the bracket **220**. An upper end of at least one of the ice separation heater **290** and the second temperature sensor **700** may be disposed below an upper end of the auxiliary storage chamber **325**.

[0444] FIG. **24** is a plan view of a first tray supporter.

[0445] Referring to FIG. **24**, the first tray supporter **340** may be coupled to the first tray cover **300** to support the first tray **320**.

[0446] The first tray supporter **340** includes a horizontal portion **341** contacting a bottom surface of the upper end of the first tray **320** and an insertion opening **342** through which a lower portion of the first tray **320** is inserted into a center of the horizontal portion **341**.

[0447] The horizontal portion **341** may have a size corresponding to the upper plate **301** of the first tray cover **300**. The horizontal portion **341** may include a plurality of coupling holes **341***a* engaged with the coupling parts **301***a* of the first tray cover **300**. The plurality of coupling holes **341***a* may be spaced apart from each other in the X-axis and/or Y-axis direction of FIG. **24** to correspond to the coupling part **301***a* of the first tray cover **300**.

[0448] When the first tray cover **300**, the first tray **320**, and the first tray supporter **340** are coupled to each other, the upper plate **301** of the first tray cover **300**, the first extension wall **327** of the first tray **320**, and the horizontal portion **341** of the first tray supporter **340** may sequentially contact each other.

[0449] The bottom surface of the upper plate **301** of the first tray cover **300** and the top surface of the first extension wall **327** of the first tray **320** may contact each other, and the bottom surface of the first extension wall **327** of the first tray **320** and the top surface of the horizontal part **341** of the first tray supporter **340** may contact each other.

[0450] FIG. **25** is a perspective view of a second tray according to an embodiment, and FIG. **26** is a perspective view of the second tray when viewed from a lower side. FIG. **27** is a bottom view of the second tray, and FIG. **28** is a plan view of the second tray.

[0451] Referring to FIGS. **25** to **28**, the second tray **380** may define a second cell **381***a* which is another portion of the ice making cell **320***a*. The second tray **380** may include a second tray wall **381** defining a portion of the ice making cell **320***a*.

[0452] For example, the second tray **380** may define a plurality of second cells **381***a*. For example, the plurality of second cells **381***a* may be arranged in a line. Referring to FIG. **28**, the plurality of second cells **381***a* may be arranged in the X-axis direction. For example, the second tray wall **381** may define the plurality of second cells **381***a*.

[0453] The second tray wall **381** may include a plurality of second cell walls **3811** which respectively define the plurality of second cells **381***a*. The two adjacent second cell walls **3811** may be connected to each other.

[0454] The second tray **380** may include a circumferential wall **387** extending along a circumference of an upper end of the second tray wall **381**. The circumferential wall **387** may be formed integrally with the second tray wall **381** and may extend from an upper end of the second tray wall **381**. For another example, the circumferential wall **387** may be provided separately from the second tray wall **381** and disposed around the upper end of the second tray wall **381**. In this case, the circumferential wall **387** may contact the second tray wall **381** or be spaced apart from the third tray wall **381**. In any case, the circumferential wall **387** may surround at least a portion of the first tray **320**.

[0455] If the second tray **380** includes the circumferential wall **387**, the second tray **380** may surround the first tray **320**. When the second tray **380** and the circumferential wall **387** are provided separately from each other, the circumferential wall **387** may be integrally formed with the second tray case or may be coupled to the second tray case.

[0456] For example, one second tray wall may define a plurality of second cells **381***a*, and one continuous circumferential wall **387** may surround the first tray **250**. The circumferential wall **387** may include a first extension wall **387***b* extending in the horizontal direction and a second extension wall **387***c* extending in the vertical direction. The first extension wall **387***b* may be provided with one or more second coupling holes **387***a* to be coupled to the second tray case. The plurality of second coupling holes **387***a* may be arranged in at least one axis of the X axis or the Y axis.

[0457] The second tray **380** may include a second contact surface **382***c* contacting the first contact surface **322***c* of the first tray **320**.

[0458] The first contact surface **322***c* and the second contact surface **382***c* may be horizontal planes. Each of the first contact surface **322***c* and the second contact surface **382***c* may be provided in a ring shape. When the ice making cell **320***a* has a spherical shape, each of the first contact surface **322***c* and the second contact surface **382***c* may have a circular ring shape.

[0459] FIG. **29** is a cutaway cross-sectional view taken along line **29-29** of FIG. **25**, FIG. **30** is a cutaway cross-sectional view taken along line **30-30** of FIG. **25**, FIG. **31** is a cutaway cross-sectional view taken along line **31-31** of FIG. **25**, FIG. **32** is a cutaway cross-sectional view taken along line **32-32** of FIG. **28**, and FIG. **33** is a cutaway cross-sectional view taken along line **33-33** of FIG. **29**.

[0460] FIG. **29** illustrates a Y-Z cutting surface passing through the central line C**1**.

[0461] Referring to FIGS. **29** to **33**, the second tray **380** may include a first portion or circumferential wall **382** that defines at least a portion of the ice making cell **320***a*. For example, the first portion **382** may be a portion or the whole of the second tray wall **381**.

[0462] In this specification, the first portion **322** of the first tray **320** may be referred to as a third portion so as to be distinguished from the first portion **382** of the second tray **380**. Also, the second portion **323** of the first tray **320** may be referred to as a fourth portion so as to be distinguished from the second portion **383** of the second tray **380**.

[0463] The first portion **382** may include a second cell surface **382***b* (or an outer circumferential surface) defining the second cell **381***a* of the ice making cell **320***a*. The first portion **382** may be defined as an area between two dotted lines in FIG. **29**. The uppermost end of the first portion **382** is the second contact surface **382***c* contacting the first tray **320**.

[0464] The second tray **380** may further include a second portion **383**. The second portion **383** may reduce transfer of heat, which is transferred from the transparent ice heater **430** to the second tray **380**, to the ice making cell **320***a* defined by the first tray **320**. That is, the second portion **383** serves to allow the heat conduction path to move in a direction away from the first cell **321***a*. The second portion **383** may be a portion or the whole of the circumferential wall **387**.

[0465] The second portion **383** may extend from a predetermined point of the first portion **382**. In the following description, for example, the second portion **383** is connected to the first portion **382**. The predetermined point of the first portion **382** may be one end of the first portion **382**. Alternatively, the predetermined point of the first portion **382** may be one point of the second contact surface **382***c*.

[0466] The second portion **383** may include the other end that does not contact one end contacting the predetermined point of the first portion **382**. The other end of the second portion **383** may be disposed farther from the first cell **321***a* than one end of the second portion **383**.

[0467] At least a portion of the second portion **383** may extend in a direction away from the first cell **321***a*. At least a portion of the second portion **383** may extend in a direction away from the second cell **381***a*.

[0468] At least a portion of the second portion **383** may extend upward from the second contact surface **382***c*. At least a portion of the second portion **383** may extend horizontally in a direction away from the central line C**1**.

[0469] A center of curvature of at least a portion of the second portion **383** may coincide with a center of rotation of the shaft **440** which is connected to the driver **480** to rotate.

[0470] The second part **383** may include a first part **384***a* extending from one point of the first portion **382**.

[0471] The second portion **383** may further include a second part **384***b* extending in the same direction as the extending direction with the first part **384***a*. Alternatively, the second portion **383** may further include a third part **384***b* extending in a direction different from the extending direction of the first part **384***a*. Alternatively, the second portion **383** may further include a second part **384***b* and a third part **384***c* branched from the first part **384***a*.

[0472] For example, the first part **384***a* may extend in the horizontal direction from the first part **382**. A portion of the first part **384***a* may be disposed at a position higher than that of the second contact surface **382***c*. That is, the first part **384***a* may include a horizontally extension part and a vertically extension part. The first part **384***a* may further include a portion extending in the vertical direction from the predetermined point. For example, a length of the third part **384***c* may be greater than that of the second part **384***b*.

[0473] The extension direction of at least a portion of the first part **384***a* may be the same as that of the second part **384***b*. The extension directions of the second part **384***b* and the third part **384***c* may be different from each other. The extension direction of the third part **384***c* may be different from that of the first part **384***a*.

[0474] The third part **384***a* may have a constant curvature based on the Y-Z cutting surface. That is, the same curvature radius of the third part **384***a* may be constant in the longitudinal direction. The curvature of the second part **384***b* may be zero. When the second part **384***b* is not a straight line, the curvature of the second part **384***b* may be less than that of the third part **384***a*. The curvature radius of the second part **384***b* may be greater than that of the third part **384***a*.

[0475] At least a portion of the second portion **383** may be disposed at a position higher than or equal to that of the uppermost end of the ice making cell **320***a*. In this case, since the heat conduction path defined by the second portion **383** is long, the heat transfer to the ice making cell **320***a* may be reduced.

[0476] A length of the second portion **383** may be greater than the radius of the ice making cell **320***a*. The second portion **383** may extend up to a point higher than the center of rotation C**4** of the shaft **440**. For example, the second portion **383** may extend up to a point higher than the uppermost end of the shaft **440**.

[0477] The second portion **383** may include a first extension part **383***a* extending from a first point of the first portion **382** and a second extension part **383***b* extending from a second point of the first portion **382** so that transfer of the heat of the transparent ice heater **430** to the ice making cell **320***a* defined by the first tray **320** is reduced. For example, the first extension part **383***a* and the second extension part **383***b* may extend in different directions with respect to the central line C**1**.

[0478] Referring to FIG. **29**, the first extension part **383***a* may be disposed at the left side with respect to the central line C**1**, and the second extension part **383***b* may be disposed at the right side with respect to the central line C**1**. The first extension part **383***a* and the second extension part **383***b* may have different shapes based on the central line C**1**. The first extension part **383***a* and the second extension part **383***b* may be provided in an asymmetrical shape with respect to the central line C**1**. A length (horizontal length) of the second extension part **383***b* in the Y-axis direction may be longer than the length (horizontal length) of the first extension part **383***a*.

[0479] The first extension part **383***a* may be disposed closer to an edge part that is disposed at a side opposite to the portion of the second wall **222** or the third wall **223** of the bracket **220**, which is connected to the fourth wall **224**, than the second extension part **383***a*.

[0480] The second extension part **383***b* may be disposed closer to the shaft **440** that provides a center of rotation of the second tray assembly than the first extension part **383***a*. In the this embodiment, a length of the second extension part **383***b* in the Y-axis direction may be greater than that of the first extension part **383***a*. In this case, the heat conduction path may increase while reducing the width of the bracket **220** relative to the space in which the ice maker **200** is installed. Since the length of the second extension part **383***b* in the Y-axis direction is greater than that of the first extension part **383***a*, the second tray assembly including the second tray **380** contacting the first tray **320** may increase in radius of rotation. When the rotation radius of the second tray assembly increases centrifugal force of the second tray assembly may increase. Thus, in the ice separation process, separating force for separating the ice from the second tray assembly may increase to improve ice separation performance.

[0481] The center of curvature of at least a portion of the second extension part **383***b* may be a center of curvature of the shaft **440** which is connected to the driver **480** to rotate.

[0482] A distance between an upper portion of the first extension part **383***a* and an upper portion of the second extension part **383***b* may be greater than that between a lower portion of the first extension part **383***a* and a lower portion of the second extension part **383***b* with respect to the Y-Z cutting surface passing through the central line C**1**. For example, a distance between the first extension part **383***a* and the second extension part **383***b* may increase upward.

[0483] Each of the first extension part **383***a* and the third extension part **383***b* may include first to third parts **384***a*, **384***b*, and **384***c*.

[0484] In another aspect, the third part **384***c* may also be described as including the first extension part **383***a* and the second extension part **383***b* extending in different directions with respect to the central line C**1**.

[0485] At least a portion of the X-Y cutting surface of the second extension part **383***b* has a curvature greater than zero, and also, the curvature may vary.

[0486] A first horizontal area **386***a* including a point at which a first extension part C**2** passing through the central line C**1** in the Y-axis direction and the second extension part **383***b* meet each other may have a curvature different from that of a second horizontal area **386***b* of the third part **383***b*, which is spaced apart from the first horizontal area **386***a*. For example, the curvature of the first horizontal area **386***a* may be greater than that of the second horizontal area **386***b*. In the third part **383***b*, the curvature of the first horizontal area **386***a* may be maximized

[0487] A third horizontal area **386***c* including a point at which a second extension part C**3** passing through the central line C**1** in the X-axis direction and the third part **384***c* meet each other may have a curvature different from that of the second horizontal area **386***b* of the third part **383***b*, which is spaced apart from the second horizontal area **386***b*. The curvature of the second horizontal area **386***b* may be greater than that of the third horizontal area **386***c*. In the third part **383***b*, the curvature of the third horizontal area **386***c* may be minimized.

[0488] The second extension part **383***b* may include an inner line **383***b***1** and an outer line **383***b***2**. A curvature of the inner line **383***b***1** may be greater than zero with respect to the X-Y cutting surface. A curvature of the outer line **383***b***2** may be equal to or greater than zero.

[0489] The second extension part **383***b* may be divided into an upper portion and a lower portion in a height direction. An amount of change in curvature of the inner line **383***b***1** of the upper portion of the second extension part **383***b* may be greater than zero with respect to the X-Y cutting surface. An amount of change in curvature of the inner line **383***b***1** of the lower portion of the second extension part **383***b* may be greater than zero. The maximum curvature change amount of the inner line **383***b***1** of the upper portion of the second extension part **383***b* may be greater than that of the inner line **383***b***1** of the lower portion of the second extension part **383***b*. An amount of change in curvature of the outer line **383***b***2** of the upper portion of the second extension part **383***b* may be greater than zero with respect to the X-Y cutting surface. An amount of change in curvature of the outer line **383***b***2** of the lower portion of the second extension part **383***b* may be greater than zero.

[0490] The minimum curvature change amount of the outer line **383***b***2** of the upper portion of the second extension part **383***b* may be greater than that of the outer line **383***b***2** of the lower portion of the second extension part **383***b*. The outer line of the lower portion of the second extension part **383***b* may include a straight portion **383***b***3**. The third part **384***c* may include a plurality of first extension parts **383***a* and a plurality of second extension parts **383***b*, which correspond to the plurality of ice making cells **320***a*.

[0491] The third part **384***c* may include a first connection part **385***a* connecting two adjacent first extension parts **383***a* to each other. The third part **384***c* may include a second connection part **385***b* connecting two adjacent second extension parts **383***b* to each other. In this embodiment, when the ice maker includes three ice making cells **320***a*, the third part **384***c* may include two first connection parts **385***a*.

[0492] As described above, widths (which are lengths in the X-axis direction) W**1** of the two first connection parts **385***a* may be different from each other according to the formation of the sensor accommodation part **321***e*. For example, the second connection part **385***b* may include an inner line **385***b***1** and an outer line **385***b***2**. In this embodiment, when the ice maker includes three ice making cells **320***a*, the third part **384***c* may include two second connection parts **385***b*.

[0493] As described above, widths (which are lengths in the X-axis direction) W**2** of the two second connection parts **385***b* may be different from each other according to the formation of the sensor accommodation part **321***e*.

[0494] Here, the width of the second connection part **385***b* disposed close to the second temperature sensor **700** among the two second connection parts **385***b* may be larger than that of the remaining second connection part **385***b*.

[0495] The width W**1** of the first connection part **385***a* may be larger than the width W**3** of the connection part of two adjacent ice making cells **320***a*. The width W**2** of the second connection part **385***b* may be larger than the width W**3** of the connection part of two adjacent ice making cells **320***a*. The first portion **382** may have a variable radius in the Y-axis direction.

[0496] The first portion **382** may include a first region **382***d* (see region A in FIG. **29**) and a second region **382***e*. The curvature of at least a portion of the first region **382***d* may be different from that of at least a portion of the second region **382***e*. The first region **382***d* may include the lowermost end of the ice making cell **320***a*. The second region **382***e* may have a diameter greater than that of the first region **382***d*. The first region **382***d* and the second region **382***e* may be divided vertically.

[0497] The transparent ice heater **430** may contact the first region **382***d*. The first region **382***d* may include a heater contact surface **382***g* contacting the transparent ice heater **430**. The heater contact surface **382***g* may be, for example, a horizontal plane. The heater contact surface **382***g* may be disposed at a position higher than that of the lowermost end of the first portion **382**. The second region **382***e* may include the second contact surface **382***c*.

[0498] The first region **382***d* may have a shape recessed in a direction opposite to a direction in which ice is expanded in the ice making cell **320***a*. A distance from the center of the ice making cell **320***a* to the second region **382***e* may be less than that from the center of the ice making cell **320***a* to the portion at which the shape recessed in the first area **382***d* is disposed.

[0499] For example, the first region **382***d* may include a pressing part **382***f* that is pressed by the second pusher **540** during the ice separation process. When pressing force of the second pusher **540** is applied to the pressing part **382***f*, the pressing part **382***f* is deformed, and thus, ice is separated from the first portion **382**. When the pressing force applied to the pressing part **382***f* is removed, the pressing part **382***f* may return to its original shape. The central line C**1** may pass through the first region **382***d*. For example, the central line C**1** may pass through the pressing part **382***f*.

[0500] The heater contact surface **382***g* may be disposed to surround the pressing unit **382***f*. The heater contact surface **382***g* may be disposed at a position higher than that of the lowermost end of the pressing part **382***f*. At least a portion of the heater contact surface **382***g* may be disposed to surround the central line C**1**. Accordingly, at least a portion of the transparent ice heater **430** contacting the heater contact surface **382***g* may be disposed to surround the central line C**1**.

[0501] Therefore, the transparent ice heater **430** may be prevented from interfering with the second pusher **540** while the second pusher **540** presses the pressing unit **382***f*.

[0502] A distance from the center of the ice making cell **320***a* to the pressing part **382***f* may be different from that from the center of the ice making cell **320***a* to the second region **382***e*.

[0503] FIG. **34** is a perspective view of the second tray cover, and FIG. **35** is a plan view of the second tray cover.

[0504] Referring to FIGS. **34** and **35**, the second tray cover **360** includes an opening **362** (or through-hole) into which a portion of the second tray **380** is inserted. For example, when the second tray **380** is inserted below the second tray cover **360**, a portion of the second tray **380** may protrude upward from the second tray cover **360** through the opening **362**.

[0505] The second tray cover **360** may include a vertical wall **361** and a curved wall **363** surrounding the opening **362**. The vertical wall **361** may define three surfaces of the second tray cover **360**, and the curved wall **363** may define the other surface of the second tray cover **360**. The vertical wall **361** may be a wall extending vertically upward, and the curved wall **363** may be a wall rounded away from the opening **362** upward.

[0506] The vertical walls **361** and the curved walls **363** may be provided with a plurality of coupling parts **361***a*, **361***c*, and **363***a* to be coupled to the second tray **380** and the second tray case **400**. The vertical wall **361** and the curved wall **363** may further include a plurality of coupling grooves **361***b*, **361***d*, and **363***b* corresponding to the plurality of coupling parts **361***a*, **361***c*, and **363***a*. A coupling member may be inserted into the plurality of coupling parts **361***a*, **361***c*, and **363***a* to pass through the second tray **380** and then be coupled to the coupling parts **401***a*, **401***b*, and **401***c* of the second tray supporter **400**. Here, the coupling part may protrude upward from the vertical wall **361** and the curved wall **363** through the plurality of coupling grooves **361***b*, **361***d*, and **363***b* to prevent an interference with other components.

[0507] A plurality of first coupling parts **361***a* may be provided on the wall facing the curved wall **363** of the vertical wall **361**. The plurality of first coupling parts **361***a* may be spaced apart from each other in the X-axis direction of FIG. **34**. A first coupling groove **361***b* corresponding to each of the first coupling parts **361***a* may be provided. For example, the first coupling groove **361***b* may be defined by recessing the vertical wall **361**, and the first coupling part **361***a* may be provided in the recessed portion of the first coupling groove **361***b*.

[0508] The vertical wall **361** may further include a plurality of second coupling parts **361***c*. The plurality of second coupling parts **361***c* may be provided on the vertical walls **361** that are spaced apart from each other in the X-axis direction. The plurality of second coupling parts **361***c* may be disposed closer to the first coupling parts **361***a* than the third coupling parts **363***a*, which will be described later. This is done for preventing the interference with the extension **403** of the second tray supporter **400** when being coupled to a second tray supporter **400** that will be described later. For example, the vertical wall **361** in which the plurality of second coupling parts **361***c* are disposed may further include a second coupling groove **361***d* defined by spacing portions except for the second coupling parts **361***c* apart from each other.

[0509] The curved wall **363** may be provided with a plurality of third coupling parts **363***a* to be coupled to the second tray **380** and the second tray supporter **400**. For example, the plurality of third coupling parts **363***a* may be spaced apart from each other in the X-axis direction of FIG. **34**. The curved wall **363** may be provided with a third coupling groove **363***b* corresponding to each of the third coupling parts **363***a*. For example, the third coupling groove **363***b* may be defined by vertically recessing the curved wall **363**, and the third coupling part **363***a* may be provided in the recessed portion of the third coupling groove **363***b*.

[0510] The second tray cover **360** may support at least a portion of the second portion **383** of the second tray **380**. For example, the second tray cover **360** may support the first extension part **383***a* and the second extension part **383***b* of the second part **383**.

[0511] FIG. **36** is a top perspective view of a second tray supporter, and FIG. **37** is a bottom perspective view of the second tray supporter. FIG. **38** is a cutaway cross-sectional view taken along line **38-38** of FIG. **36**.

[0512] Referring to FIGS. **36** to **38**, the second tray supporter **400** may include a support body **407** on which a lower portion of the second tray **380** is seated.

[0513] The support body **407** may include an accommodation space **406***a* in which a portion of the second tray **380** is accommodated. The accommodation space **406***a* may be defined corresponding to the first portion **382** of the second tray **380**, and a plurality of accommodation spaces **406***a* may be provided.

[0514] The support body **407** may include a lower opening **406***b* (or a through-hole) through which a portion of the second pusher **540** passes. For example, three lower openings **406***b* may be provided in the support body **407** to correspond to the three accommodation spaces **406***a*.

[0515] A portion of the lower portion of the second tray **380** may be exposed by the lower opening **406***b*. At least a portion of the second tray **380** may be disposed in the lower opening **406***b*. A portion of the second tray **380** may contact the support body **404** by the lower opening **406***b*.

[0516] In the first portion **382** of the second tray **380** defining the ice making cell, a surface area of the area contacting the support body **407** may be greater than that of the non-contact area.

[0517] A top surface **407***a* of the support body **407** may extend in the horizontal direction. The second tray supporter **400** may include a top surface **407***a* of the support body **407** and a stepped lower plate **401**. The lower plate **401** may be disposed at a position higher than that of the top surface **407***a* of the support body **407**.

[0518] The lower plate **401** may include a plurality of coupling parts **401***a*, **401***b*, and **401***c* to be coupled to the second tray cover **360**. The second tray **380** may be inserted and coupled between the second tray cover **360** and the second tray supporter **400**. For example, the second tray **380** may be disposed below the second tray cover **360**, and the second tray **380** may be accommodated above the second tray supporter **400**. The first extension wall **387***b* of the second tray **380** may be coupled to the coupling parts **361***a*, **361***b*, and **361***c* of the second tray cover **360** and the coupling parts **400***a*, **401***b*, and **401***c* of the second tray supporter **400**. The plurality of first coupling parts **401***a* may be spaced apart from each other in the X-axis direction of FIG. **36**. Also, the first coupling part **401***a* and the second and third coupling parts **401***b* and **401***c* may be spaced apart from each other in the Y-axis direction. The third coupling part **401***c* may be disposed farther from the first coupling part **401***a* than the second coupling part **401***b*.

[0519] The second tray supporter **400** may further include a vertical extension wall **405** extending vertically downward from an edge of the lower plate **401**. One surface of the vertical extension wall **405** may be provided with a pair of extension parts **403** coupled to the shaft **440** to allow the second tray **380** to rotate. The pair of extension parts **403** may be spaced apart from each other in the X-axis direction of FIG. **36**. Also, each of the extension parts **403** may further include a through-hole **404**. The shaft **440** may pass through the through-hole **404**, and the extension part **281** of the first tray cover **300** may be disposed inside the pair of extension parts **403**. The through-hole **404** may further include a central portion **404***a* and an extension hole **404***b* extending symmetrically to the central portion **404***a*.

[0520] The second tray supporter **400** may further include a spring coupling part **402***a* to which a spring **402** is coupled. The spring coupling part **402***a* may provide a ring to be hooked with a lower end of the spring **402**. One of the walls spaced apart from and facing each other in the X-axis direction of the vertical extension wall **405** is provided with a guide hole **408** guiding the transparent ice heater **430** to be described later or the wire connected to the transparent ice heater **430**.

[0521] The second tray supporter **400** may further include a link connection part **405***a* to which the pusher link **500** is coupled. For example, the link connection part **405***a* may protrude from the vertical extension wall **405** in the X-axis direction. The link connection part **405***a* may be disposed on an area between the center line CL**1** and the through-hole **404** with respect to FIG. **38**. The lower plate **401** may further include a plurality of second heater coupling parts **409** coupled to the second heater case **420** (see FIG. **39**) that will be described later. The plurality of second heater coupling parts **409** may be arranged to be spaced apart from each other in the X-axis direction and/or the Y-axis direction.

[0522] Referring to FIG. **38**, the second tray supporter **400** may include a first portion **411** supporting the second tray **380** defining at least a portion of the ice making cell **320***a*. In FIG. **38**, the first portion **411** may be an area between two dotted lines. For example, the support body **407** may define the first portion **411**. The second tray supporter **400** may further include a second portion **413** extending from a predetermined point of the first portion **411**.

[0523] The second portion **413** may reduce transfer of heat, which is transfer from the transparent ice heater **430** to the second tray supporter **400**, to the ice making cell **320***a* defined by the first tray **320**. At least a portion of the second portion **413** may extend in a direction away from the first cell **321***a* defined by the first tray **320**. The direction away from the first cell **321** may be a horizontal direction passing through the center of the ice making cell **320***a*. The direction away from the first cell **321** may be a downward direction with respect to a horizontal line passing through the center of the ice making cell **320***a*.

[0524] The second part **413** may include a first part **414***a* extending in the horizontal direction from the predetermined point and a second part **414***b* extending in the same direction as the first part **414***a*.

[0525] The second part **413** may include a first part **414***a* extending in the horizontal direction from the predetermined point, and a third part **414**c extending in a direction different from that of the first part **414**a.

[0526] The second part **413** may include a first part **414**a extending in the horizontal direction from the predetermined point, and a second part **414**b and a third part **414**c, which are branched from the first part **414**a. A top surface **407**a of the support body **407** may provide, for example, the first part **414**a.

[0527] The first part **414**a may further include a fourth part **414**d extending in the vertical line direction. The lower plate **401** may provide, for example, the fourth part **414**d. The vertical extension wall **405** may provide, for example, the third part **414**c. A length of the third part **414**c may be greater than that of the second part **414**b.

[0528] The second part **414**b may extend in the same direction as the first part **414**a. The third part **414**c may extend in a direction different from that of the first part **414**a.

[0529] The second portion **413** may be disposed at the same height as the lowermost end of the first cell **321**a or extend up to a lower point.

[0530] The second portion **413** may include a first extension part **413**a and a second extension part **413**b which are disposed opposite to each other with respect to the center line CL**1** corresponding to the center line C**1** of the ice making cell **320**a. Referring to FIG. **38**, the first extension part **413**a may be disposed at a left side with respect to the center line CL**1**, and the second extension part **413**b may be disposed at a right side with respect to the center line CL**1**.

[0531] The first extension part **413**a and the second extension part **413**b may have different shapes with respect to the center line CL**1**. The first extension part **413**a and the second extension part **413**b may have shapes that are asymmetrical to each other with respect to the center line CL**1**.

[0532] A length of the second extension part **413**b may be greater than that of the first extension part **413**a in the horizontal direction. That is, a length of the thermal conductivity of the second extension **413**b is greater than that of the first extension part **413**a.

[0533] The first extension part **413**a may be disposed closer to an edge part that is disposed at a side opposite to the portion of the second wall **222** or the third wall **223** of the bracket **220**, which is connected to the fourth wall **224**, than the second extension part **413**b.

[0534] The second extension part **413**b may be disposed closer to the shaft **440** that provides a center of rotation of the second tray assembly than the first extension part **413**a.

[0535] In this embodiment, since the length of the second extension part **413**b in the Y-axis direction is greater than that of the first extension part **413**a, the second tray assembly including the second tray **380** contacting the first tray **320** may increase in radius of rotation.

[0536] A center of curvature of at least a portion of the second extension part **413**a may coincide with a center of rotation of the shaft **440** which is connected to the driver **480** to rotate.

[0537] The first extension part **413**a may include a portion **414**e extending upwardly with respect to the horizontal line. The portion **414**e may surround, for example, a portion of the second tray **380**.

[0538] In another aspect, the second tray supporter **400** may include a first region **415**a including the lower opening **406**b and a second region **415**b having a shape corresponding to the ice making cell **320**a to support the second tray **380**. For example, the first region **415**a and the second region **415**b may be divided vertically. In FIG. **38**, for example, the first region **415**a and the second region **415**b are divided by a dashed-dotted line. The first region **415**a may support the second tray **380**.

[0539] The controller controls the ice maker to allow the second pusher **540** to move from a first point outside the ice making cell **320**a to a second point inside the second tray supporter **400** via the lower opening **406**b.

[0540] A degree of deformation resistance of the second tray supporter **400** may be greater than that of the second tray **380**. A degree of restoration of the second tray supporter **400** may be less than that of the second tray **380**.

[0541] In another aspect, the second tray supporter **400** includes a first region **415**a including a lower opening **406**b and a second region **415**b disposed farther from the transparent ice heater **430** than the first region **415**a.

[0542] FIG. **39** is a perspective view of the second heater case, FIG. **40** is a view illustrating a state in which the transparent ice heater is coupled to the second heater case, FIG. **41** is a cutaway cross-sectional view taken along line **41-41** of FIG. **40**, and FIG. **42** is a partial enlarged view of the second heater case.

[0543] The ice maker according to this embodiment may further include a transparent ice heater **430** applying heat to the second tray **380** during the ice making process.

[0544] The second heater case **420** may include a second heater accommodation part **425** accommodating the transparent ice heater **430** transferring heat from the lower side of the second tray **380**. The second heater case **420** may be integrally formed with the second tray supporter **400** or separately provided and then be coupled to the second tray supporter **400**.

[0545] The second heater case **420** may include a second heater plate **421** in which the second heater accommodation part **425** is provided and a second heater vertical wall **424** extending vertically upward from an edge of the second heater plate **421**. The second heater accommodation part **425** includes a plurality of curved portions **425**a contacting a lower end of the second tray **380** and a plurality of straight portions **425**b connecting the plurality of curved portions **425**a to each other.

[0546] An opening **422** into which a portion of the lower end of the second tray **380** is inserted may be provided in a center of the curved portions **425**a. The inside of the plurality of curved portions **425**a surrounding the opening **422** may correspond to a shape of the lower end of the second tray **380**.

[0547] For example, the inside of the curved portion **425**a may be lower than the outside of the curved portion **425**a, and the inside of the curved portion **425**a may be inclined.

[0548] A heater support wall **425**c may be disposed along a circumference of the second heater accommodation part **425** on a bottom surface of the heater plate **421**. The heater support wall **425**c may prevent the transparent ice heater **430** accommodated in the second heater accommodation part **425** from being separated from the second heater accommodation part **425**. A separation prevention protrusion **425**cl may be provided at each of both ends of the plurality of curved portions **425**a in the X-axis direction of FIG. **38** to prevent the transparent ice heater **430** from being separated.

[0549] One of both the ends of the curved portion **425**a may be provided with a guide groove **425**d guiding the transparent ice heater **430** to the outside. A guide wall **423** extending vertically may be disposed on the second heater plate **421** in the direction in which the guide groove **425***d* is defined. The guide wall **423** may guide the wire connected to the transparent ice heater **430** or the transparent ice heater **430** guided along the guide groove **425***d* to the outside of the second heater case **420**. The guide wall **423** may correspond to a shape of the adjacent curved portion **425***a*. The guide wall **423** may be provided in plurality that are spaced apart from each other with respect to the guide groove **425***d*. A plurality of separation prevention protrusions **426***c* may be disposed on the plurality of straight portions **425***b*.

[0550] The second heater plate **421** may be provided with a plurality of second heater coupling parts **421***a* to be coupled to the second heater coupling part **409** of the second tray supporter **400**.

[0551] The plurality of second heater coupling parts **421***a* may be spaced apart from each other in a direction of an arrow A and/or in a direction of an arrow B of FIG. **38**.

[0552] A coupling member may be coupled to the lower side of the second heater case **420** through the second heater coupling part **421***a* and the second heater coupling part **409**. A cutoff part **424***a* may be provided on a portion of one surface of four surfaces of the second heater vertical wall **424**. For example, the transparent ice heater **430** guided through the guide wall **423** or the wire connected to the transparent ice heater **430** may be connected to the outside through the cutoff part **424***a*. The cutout **424***a* may be disposed to correspond to the guide hole **408** of the second tray supporter **400**.

[0553] The transparent ice heater **430** will be described in detail.

[0554] The controller **800** according to this embodiment may control the transparent ice heater **430** so that heat is supplied to the ice making cell **320***a* in at least partial section while cold air is supplied to the ice making cell **320***a* to make the transparent ice.

[0555] An ice making rate may be delayed so that bubbles dissolved in water within the ice making cell **320***a* may move from a portion at which ice is made toward liquid water by the heat of the transparent ice heater **430**, thereby making transparent ice in the ice maker **200**. That is, the bubbles dissolved in water may be induced to escape to the outside of the ice making cell **320***a* or to be collected into a predetermined position in the ice making cell **320***a*.

[0556] When a cold air supply part **900** to be described later supplies cold air to the ice making cell **320***a*, if the ice making rate is high, the bubbles dissolved in the water inside the ice making cell **320***a* may be frozen without moving from the portion at which the ice is made to the liquid water, and thus, transparency of the ice may be reduced.

[0557] On the contrary, when the cold air supply part **900** supplies the cold air to the ice making cell **320***a*, if the ice making rate is low, the above limitation may be solved to increase in transparency of the ice. However, there is a limitation in which an making time increases.

[0558] Accordingly, the transparent ice heater **430** may be disposed at one side of the ice making cell **320***a* so that the heater locally supplies heat to the ice making cell **320***a*, thereby increasing in transparency of the made ice while reducing the ice making time.

[0559] When the transparent ice heater **430** is disposed on one side of the ice making cell **320***a*, the transparent ice heater **430** may be made of a material having thermal conductivity less than that of the metal to prevent heat of the transparent ice heater **430** from being easily transferred to the other side of the ice making cell **320***a*.

[0560] Alternatively, at least one of the first tray **320** and the second tray **380** may be made of a resin including plastic so that the ice attached to the trays **320** and **380** is separated in the ice making process.

[0561] At least one of the first tray **320** or the second tray **380** may be made of a flexible or soft material so that the tray deformed by the pushers **260** and **540** is easily restored to its original shape in the ice separation process.

[0562] The transparent ice heater **430** may be disposed at a position adjacent to the second tray **380**. The transparent ice heater **430** may be, for example, a wire type heater.

[0563] For example, the transparent ice heater **430** may be installed to contact the second tray **380** or may be disposed at a position spaced a predetermined distance from the second tray **380**.

[0564] For another example, the second heater case **420** may not be separately provided, but the transparent heater **430** may be installed on the second tray supporter **400**.

[0565] In some cases, the transparent ice heater **430** may supply heat to the second tray **380**, and the heat supplied to the second tray **380** may be transferred to the ice making cell **320***a*.

[0566] FIG. **43** is a view of the first pusher according to an embodiment, wherein FIG. **43**(*a*) is a perspective view of the first pusher, and FIG. **43**(*b*) is a side view of the first pusher.

[0567] Referring to FIG. **43**, the first pusher **260** may include a pushing bar **264**. The pushing bar **264** may include a first edge **264***a* on which a pressing surface pressing ice or a tray in the ice separation process is disposed and a second edge **264***b* disposed at a side opposite to the first edge **264***a*. For example, the pressing surface may be flat or curved surface.

[0568] The pushing bar **264** may extend in the vertical direction and may be provided in a straight line shape or a curved shape in which at least a portion of the pushing bar **264** is rounded. A diameter of the pushing bar **264** is less than that of the opening **324** of the first tray **320**. Accordingly, the pushing bar **264** may be inserted into the ice making cell **320***a* through the opening **324**. Thus, the first pusher **260** may be referred to as a penetrating type passing through the ice making cell **320***a*.

[0569] When the ice maker includes a plurality of ice making cells **320***a*, the first pusher **260** may include a plurality of pushing bars **264**. Two adjacent pushing bars **264** may be connected to each other by the connection part **263**. The connection part **263** may connect upper ends of the pushing bars **264** to each other. Thus, the second edge **264***a* and the connection part **263** may be prevented from interfering with the first tray **320** while the pushing bar **264** is inserted into the ice making cell **320***a*.

[0570] The first pusher **260** may include a guide connection part **265** passing through the guide slot **302**. For example, the guide connection part **265** may be provided at each of both sides of the first pusher **260**. A vertical cross-section of the guide connection part **265** may have a circular, oval, or polygonal shape. The guide connection part **265** may be disposed in the guide slot **302**. The guide connection part **265** may move in a longitudinal direction along the guide slot **302** in a state of being disposed in the guide slot **302**. For example, the guide connection part **265** may move in the vertical direction.

[0571] Although the guide slot **302** has been described as being provided in the first tray cover **300**, it may be alternatively provided in the wall defining the bracket **220** or the storage chamber.

[0572] The guide connection part **265** may further include a link connection part **266** to be coupled to the pusher link **500**. The link connection part **266** may be disposed at a position lower than that of the second edge **264***b*. The link connection part **266** may be provided in a cylindrical shape so that the link connection part **266** rotates in the state in which the link connection part **266** is coupled to the pusher link **500**.

[0573] FIG. **44** is a view illustrating a state in which the first pusher is connected to the second tray assembly by the link.

[0574] Referring to FIG. **44**, the pusher link **500** may connect the first pusher **500** to the second tray assembly. For example, the pusher link **500** may be connected to the first pusher **260** and the second tray case.

[0575] The pusher link **500** may include a link body **502**. The link body **502** may have a rounded shape. As the link body **502** is provided in a round shape, the pusher link **500** may allow the first pusher **260** to rotate and also to vertically move while the second tray assembly rotates.

[0576] The pusher link **500** may include a first connection part **504** provided at one end of the link body **502** and a second connection part **506** provided at the other end of the link body **502**. The first connection part **504** may include a first coupling hole **504***a* to which the link connection part **266** is coupled. The link connection part **266** may be connected to the first connection part **504** after passing through the guide slot **302**.

[0577] The second connection part **506** may be coupled to the second tray supporter **400**. The second connection part **506** may include a second coupling hole **506***a* to which the link connection part **405***a* provided on the second tray supporter **400** is coupled.

[0578] The second connection part **504** may be connected to the second tray supporter **400** at a position spaced apart from the rotation center C4 of the shaft **440** or the rotation center C4 of the second tray assembly.

[0579] Therefore, according to this embodiment, the pusher link **500** connected to the second tray assembly rotates together by the rotation of the second tray assembly. While the pusher link **500** rotates, the first pusher **260** connected to the pusher link **500** moves vertically along the guide slot **302**. The pusher link **502** may serve to convert rotational force of the second tray assembly into vertical movement force of the first pusher **260**. Accordingly, the first pusher **260** may also be referred to as a movable pusher.

[0580] FIG. **45** is a perspective view of the second pusher according to an embodiment.

[0581] Referring to FIG. **45**, the second pusher **540** according to this embodiment may include a pushing bar **544**. The pushing bar **544** may include a first edge **544***a* on which a pressing surface pressing the second tray **380** is disposed and a second edge **544***b* disposed at a side opposite to the first edge **544***a*.

[0582] The pushing bar **544** may have a curved shape to increase in time taken to press the second tray **380** without interfering with the second tray **380** that rotates in the ice separation process. The first edge **544***a* may be a plane and include a vertical surface or an inclined surface.

[0583] The second edge **544***b* may be coupled to the fourth wall **224** of the bracket **220**, or the second edge **544***b* may be coupled to the fourth wall **224** of the bracket **220** by the coupling plate **542**. The coupling plate **542** may be seated in the mounting groove **224***a* defined in the fourth wall **224** of the bracket **220**.

[0584] When the ice maker **200** includes the plurality of ice making cells **320***a*, the second pusher **540** may include a plurality of pushing bars **544**. The plurality of pushing bars **544** may be connected to the coupling plate **542** while being spaced apart from each other in the horizontal direction. The plurality of pushing bars **544** may be integrally formed with the coupling plate **542** or coupled to the coupling plate **542**.

[0585] The first edge **544***a* may be disposed to be inclined with respect to the center line C**1** of the ice making cell **320***a*. The first edge **544***a* may be inclined in a direction away from the center line C**1** of the ice making cell **320***a* from an upper end toward a lower end. An angle of the inclined surface defined by the first edge **544***a* with respect to the vertical line may be less than that of the inclined surface defined by the second edge **544***b*.

[0586] The direction in which the pushing bar **544** extends from the center of the first edge **544***a* toward the center of the second edge **544***a* may include at least two directions. For example, the pushing bar **544** may include a first portion extending in a first direction and a second portion extending in a direction different from the second portion. At least a portion of the line connecting the center of the second edge **544***a* to the center of the first edge **544***a* along the pushing bar **544** may be curved.

[0587] The first edge **544***a* and the second edge **544***b* may have different heights. The first edge **544***a* may be disposed to be inclined with respect to the second edge **544***b*.

[0588] FIGS. **44** to **48** are views illustrating an assembly process of the ice maker according to an embodiment.

[0589] FIGS. **46** to **48** are views sequentially illustrating an assembling process, i.e., illustrating a process of coupling components to each other.

[0590] First, the first tray assembly and the second tray assembly may be assembled.

[0591] To assemble the first tray assembly, the ice separation heater **290** may be coupled to the first heater case **280**, and the first heater case **280** may be assembled to the first tray case. For example, the first heater case may be assembled to the first tray cover **300**.

[0592] Alternatively, when the first heater case **280** is integrally formed with the first tray cover **300**, the ice separation heater **290** may be coupled to the first tray cover **300**.

[0593] The first tray **320** and the first tray case may be coupled to each other. For example, the first tray cover **300** is disposed above the first tray **320**, the first tray supporter **340** may be disposed below the first tray **320**, and then the coupling member is used to couple the first tray cover **300**, the first tray **320**, and the first tray supporter **340** to each other.

[0594] To assemble the second tray assembly, the transparent ice heater **430** and the second heater case **420** may be coupled to each other.

[0595] The second heater case **420** may be coupled to the second tray case. For example, the second heater case **420** may be coupled to the second tray supporter **400**.

[0596] Alternatively, when the second heater case **420** is integrally formed with the second tray supporter **400**, the transparent ice heater **430** may be coupled to the second tray supporter **400**.

[0597] The second tray **380** and the second tray case may be coupled to each other. For example, the second tray cover **360** is disposed above the second tray **380**, the second tray supporter **400** may be disposed below the second tray **380**, and then the coupling member is used to couple the second tray cover **360**, the second tray **380**, and the second tray supporter **400** to each other.

[0598] The assembled first tray assembly and the second tray assembly may be aligned in a state of contacting each other.

[0599] The power transmission part connected to the driver **480** may be coupled to the second tray assembly. For example, the shaft **440** may pass through the pair of extension parts **403** of the second tray assembly.

[0600] The shaft **440** may also pass through the extension part **281** of the first tray assembly. That is, the shaft **440** may simultaneously pass through the extension part **281** of the first tray assembly and the extension part **403** of the second tray assembly.

[0601] In this case, a pair of extension parts **281** of the first tray assembly may be disposed between the pair of extension parts **403** of the second tray assembly.

[0602] The rotation arm **460** may be connected to the shaft **440**. The spring may be connected to the rotation arm **460** and the second tray assembly.

[0603] The first pusher **260** may be connected to the second tray assembly by the pusher link **500**. The first pusher **260** may be connected to the pusher link **500** in a state in which the first pusher **260** is disposed to be movable in the first tray assembly.

[0604] One end of the pusher link **500** may be connected to the first pusher **260**, and the other end may be connected to the second tray assembly. The first pusher **260** may be disposed to contact the first tray case.

[0605] The assembled first tray assembly may be installed on the bracket **220**. For example, the first tray assembly may be coupled to the bracket **220** in a state in which the first tray assembly is disposed in the through-hole **221***a* of the first wall **221**. For another example, the bracket **220** and the first tray cover may be integrally formed.

[0606] Then, the first tray assembly may be assembled by coupling the bracket **220** to which the first tray cover is integrated, the first tray **320**, and the first tray supporter to each other.

[0607] A water supply part **240** may be coupled to the bracket **220**. For example, the water supply part **240** may be coupled to the first wall **221**.

[0608] The driver **480** may be mounted on the bracket **220**. For example, the driver **480** may be mounted to the third wall **223**.

[0609] FIG. **49** is a cutaway cross-sectional view taken along line **49**-**49** of FIG. **2**.

[0610] Referring to FIG. **49**, the ice maker **200** may include a first tray assembly **201** and a second tray assembly **211**, which are connected to each other.

[0611] The second tray assembly **211** may include a first portion **212** defining at least a portion of the ice making cell **320***a* and a second portion **213** extending from a predetermined point of the first portion **212**. The second portion **213** may reduce transfer of heat from the transparent ice heater **430** to the ice making cell **320***a* defined by the first tray assembly **201**. The first portion **212** may be an area disposed between two vertical dotted lines in FIG. **49**.

[0612] The predetermined point of the first portion **212** may be an end of the first portion **212** or a point at which the first tray assembly **201** and the second tray assembly **211** meet each other. At least a portion of the first portion **212** may extend in a direction away from the ice making cell **320***a* defined by the first tray assembly **201**.

[0613] At least two portions of the second portion **213** may be branched to reduce heat transfer in the direction extending to the second portion **213**.

[0614] A portion of the second portion **213** may extend in the horizontal direction passing through the center of the ice making cell **320***a*. A portion of the second portion **213** may extend in an upward direction with respect to a horizontal line passing through the center of the ice making chamber **320***a*.

[0615] The second portion **213** includes a first part **213***c* extending in the horizontal direction passing through the center of the ice making cell **320***a*, a second part **213***d* extending upward with respect to the horizontal line passing through the center of the ice making cell **320***a*, a third part **213***e* extending downward.

[0616] The first portion **212** may have different degree of heat transfer in a direction along the outer circumferential surface of the ice making cell **320***a* to reduce transfer of heat, which is transferred from the transparent ice heater **430** to the second tray assembly **211**, to the ice making cell **320***a* defined by the first tray assembly **201**. The transparent ice heater **430** may be disposed to heat both sides with respect to the lowermost end of the first portion **212**.

[0617] The first portion **212** may include a first region **214***a* and a second region **214***b*. In FIG. **49**, the first region **214***a* and the second region **214***b* are divided by a horizontal dashed-dotted line. The second region **214***b* may be a region defined above the first region **214***a*. The degree of heat transfer of the second region **214***b* may be greater than that of the first region **214***a*.

[0618] The first region **214***a* may include a portion at which the transparent ice heater **430** is disposed. That is, the transparent ice heater **430** may be disposed in the first region **214***a*. The lowermost end **214***al* of the ice making cell **320***a* in the first region **214***a* may have a heat transfer rate less than that of the other portion of the first region **214***a*.

[0619] The second region **214***b* may include a portion in which the first tray assembly **201** and the second tray assembly **211** contact each other. The first region **214***a* may provide a portion of the ice making cell **320***a*. The second region **214***b* may provide the other portion of the ice making cell **320***a*. The second region **214***b* may be disposed farther from the transparent ice heater **430** than the first region **214***a*.

[0620] Part of the first region **214***a* may have the degree of heat transfer less than that of the other part of the first region **214***a* to reduce transfer of heat, which is transferred from the transparent ice heater **430** to the first region **314***a*, to the ice making cell **320***a* defined by the second region **214***b*.

[0621] To make ice in the direction from the ice making cell **320***a* defined by the first region **214***a* to the ice making cell **320***a* defined by the second region **214***b*, a portion of the first region **214***a* may have a degree of deformation resistance less than that of the other portion of the first region

214*a* and a degree of restoration greater than that of the other portion of the first region **214***a*.

[0622] A portion of the first region **214***a* may be thinner than the other portion of the first region **214***a* in the thickness direction from the center of the ice making cell **320***a* to the outer circumferential surface direction of the ice making cell **320***a*.

[0623] For example, the first region **214***a* may include a second tray case surrounding at least a portion of the second tray **380** and at least a portion of the second tray **380**.

[0624] An average cross-sectional area or average thickness of the first tray assembly **201** may be greater than that of the second tray assembly **211** with respect to the Y-Z cutting surface. A maximum cross-sectional area or maximum thickness of the first tray assembly **201** may be greater than that of the second tray assembly **211** with respect to the Y-Z cutting surface. A minimum cross-sectional area or minimum thickness of the first tray assembly **201** may be greater than that of the second tray assembly **211** with respect to the Y-Z cutting surface. Uniformity of a minimum cross-sectional area or minimum thickness of the first tray assembly **201** may be greater than that of the second tray assembly **211**.

[0625] The rotation center C4 may be eccentric with respect to a line bisecting the length in the Y-axis direction of the bracket **220**. The ice making cell **320***a* may be eccentric with respect to a line bisecting a length in the Y-axis direction of the bracket **200**. The rotation center C4 may be disposed closer to the second pusher **540** than to the ice making cell **320***a*.

[0626] The second portion **213** may include a first extension part **213***a* and a second extension part **323***b*, which are disposed at sides opposite to each other with respect to the central line C**1**. The first extension part **213***a* may be disposed at a left side of the center line C**1** in FIG. **49**, and the second extension part **213***b* may be disposed at a right side of the center line C**1** in FIG. **49**. The water supply part **240** may be disposed close to the first extension part **213***a*. The first tray assembly **301** may include a pair of guide slots **302**, and the water supply part **240** may be disposed in a region between the pair of guide slots **302**. A length of the guide slot **320** may be greater than the sum of a radius of the ice making cell **320***a* and a height of the auxiliary storage chamber **325**.

[0627] FIG. **50** is a block diagram illustrating a control of a refrigerator according to an embodiment.

[0628] Referring to FIG. **50**, the refrigerator according to this embodiment may include a cooler supplying a cold to the freezing compartment **32** (or the ice making cell). In FIG. **50**, for example, the cooler includes a cold air supply part **900**.

[0629] The cold air supply part **900** may supply cold air to the freezing compartment **32** using a refrigerant cycle.

[0630] For example, the cold air supply part **900** may include a compressor compressing the refrigerant. A temperature of the cold air supplied to the freezing compartment **32** may vary according to the output (or frequency) of the compressor.

[0631] Alternatively, the cold air supply part **900** may include a fan blowing air to an evaporator. An amount of cold air supplied to the freezing compartment **32** may vary according to the output (or rotation rate) of the fan. Alternatively, the cold air supply part **900** may include a refrigerant valve controlling an amount of refrigerant flowing through the refrigerant cycle. An amount of refrigerant flowing through the refrigerant cycle may vary by adjusting an opening degree by the refrigerant valve, and thus, the temperature of the cold air supplied to the freezing compartment **32** may vary.

[0632] Therefore, in this embodiment, the cold air supply part **900** may include one or more of the compressor, the fan, and the refrigerant valve.

[0633] The cold air supply part **900** may further include the evaporator exchanging heat between the refrigerant and the air. The cold air heat-exchanged with the evaporator may be supplied to the ice maker **200**.

[0634] The refrigerator according to this embodiment may further include a controller **800** that controls the cold air supply part **900**. The refrigerator may further include a water supply valve **242** controlling an amount of water supplied through the water supply part **240**.

[0635] The controller **800** may control a portion or all of the ice separation heater **290**, the transparent ice heater **430**, the driver **480**, the cold air supply part **900**, and the water supply valve **242**.

[0636] In this embodiment, when the ice maker **200** includes both the ice separation heater **290** and the transparent ice heater **430**, an output of the ice separation heater **290** and an output of the transparent ice heater **430** may be different from each other. When the outputs of the ice separation heater **290** and the transparent ice heater **430** are different from each other, an output terminal of the ice separation heater **290** and an output terminal of the transparent ice heater **430** may be provided in different shapes, incorrect connection of the two output terminals may be prevented. Although not limited, the output of the ice separation heater **290** may be set larger than that of the transparent ice heater **430**. Accordingly, ice may be quickly separated from the first tray **320** by the ice separation heater **290**.

[0637] In this embodiment, when the ice separation heater **290** is not provided, the transparent ice heater **430** may be disposed at a position adjacent to the second tray **380** described above or be disposed at a position adjacent to the first tray **320**.

[0638] The refrigerator may further include a first temperature sensor **33** (or an internal temperature sensor) that senses a temperature of the freezing compartment **32**.

[0639] The controller **800** may control the cold air supply part **900** based on the temperature sensed by the first temperature sensor **33**. The controller **800** may determine whether ice making is completed based on the temperature sensed by the second temperature sensor **700**.

[0640] FIG. **51** is a flowchart for explaining a process of making ice in the ice maker according to an embodiment.

[0641] FIG. **52** is a view for explaining a height reference depending on a relative position of the transparent heater with respect to the ice making cell, and FIG. **53** is a view for explaining an output of the transparent heater per unit height of water within the ice making cell.

[0642] FIG. **54** is a cross-sectional view illustrating a position relationship between a first tray assembly and a second tray assembly at a water supply position. FIG. **55** is a view illustrating a state in which supply of water is complete in FIG. **54**.

[0643] FIG. **56** is a cross-sectional view illustrating a position relationship between a first tray assembly and a second tray assembly at an ice making position, and FIG. **57**

is a view illustrating a state in which a pressing part of the second tray is deformed in a state in which ice making is complete.

[0644] FIG. **58** is a cross-sectional view illustrating a position relationship between a first tray assembly and a second tray assembly in an ice separation process, and FIG. **59** is a cross-sectional view illustrating the position relationship between the first tray assembly and the second tray assembly at the ice separation position.

[0645] Referring to FIGS. **51** to **59**, to make ice in the ice maker **200**, the controller **800** moves the second tray assembly **211** to a water supply position (S**1**).

[0646] In this specification, a direction in which the second tray assembly **211** moves from the ice making position of FIG. **56** to the ice separation position of FIG. **59** may be referred to as forward movement (or forward rotation). On the other hand, the direction from the ice separation position of FIG. **56** to the water supply position of FIG. **54** may be referred to as reverse movement (or reverse rotation).

[0647] The movement to the water supply position of the second tray assembly **211** is detected by a sensor, and when it is detected that the second tray assembly **211** moves to the water supply position, the controller **800** stops the driver **480**.

[0648] At least a portion of the second tray **380** may be spaced apart from the first tray **320** at the water supply position of the second tray assembly **211**.

[0649] At the water supply position of the second tray assembly **211**, the first tray assembly **201** and the second tray assembly **211** define a first angle θ1 with respect to the rotation center C**4**. That is, the first contact surface **322***c* of the first tray **320** and the second contact surface **382***c* of the second tray **380** define a first angle therebetween.

[0650] The water supply starts when the second tray **380** moves to the water supply position (S**2**). For the water supply, the controller **800** turns on the water supply valve **242**, and when it is determined that a predetermined amount of water is supplied, the controller **800** may turn off the water supply valve **242**. For example, in the process of supplying water, when a pulse is outputted from a flow sensor (not shown), and the outputted pulse reaches a reference pulse, it may be determined that a predetermined amount of water is supplied.

[0651] In the water supply position, the second portion **383** of the second tray **380** may surround the first tray **320**. For example, the second portion **383** of the second tray **380** may surround the second portion **323** of the first tray **320**. Accordingly, leakage of the water, which supplied to the ice making cell **320***a*, between the first tray assembly **201** and the second tray assembly **211** while the second tray **380** moves from the water supply position to the ice making position may be reduced. Also, it is possible to reduce a phenomenon in which water expanded in the ice making process leaks between the first tray assembly **201** and the second tray assembly **211** and is frozen.

[0652] After the water supply is completed, the controller **800** controls the driver **480** to allow the second tray assembly **211** to move to the ice making position (S**3**). For example, the controller **800** may control the driver **480** to allow the second tray assembly **211** to move from the water supply position in the reverse direction.

[0653] When the second tray assembly **211** move in the reverse direction, the second contact surface **382***c* of the second tray **380** comes close to the first contact surface **322***c* of the first tray **320**. Then, water between the second contact surface **382***c* of the second tray **380** and the first contact surface **322***c* of the first tray **320** is divided into each of the plurality of second cells **381***a* and then is distributed. When the second contact surface **382***c* of the second tray **380** and the first contact surface **322***c* of the first tray **320** contact each other, water is filled in the first cell **321***a*. As described above, when the second contact surface **382***c* of the second tray **380** contacts the first contact surface **322***c* of the first tray **320**, the leakage of water in the ice making cell **320***a* may be reduced.

[0654] The movement to the ice making position of the second tray assembly **211** is detected by a sensor, and when it is detected that the second tray assembly **211** moves to the ice making position, the controller **800** stops the driver **480**.

[0655] In the state in which the second tray assembly **211** moves to the ice making position, ice making is started (S**4**).

[0656] At the ice making position of the second tray assembly **211**, the second portion **383** of the second tray **380** may face the second portion **323** of the first tray **320**. At least a portion of each of the second portion **383** of the second tray **380** and the second portion **323** of the first tray **320** may extend in a horizontal direction passing through the center of the ice making cell **320***a*. At least a portion of each of the second portion **383** of the second tray **380** and the second portion **323** of the first tray **320** is disposed at the same height or higher than the uppermost end of the ice making cell **320***a*.

[0657] At least a portion of each of the second portion **383** of the second tray **380** and the second portion **323** of the first tray **320** may be lower than the uppermost end of the auxiliary storage chamber **325**.

[0658] At the ice making position of the second tray assembly **211**, the second portion **383** of the second tray **380** may be spaced apart from the second portion **323** of the first tray **320**. The space may extend to a portion having a height equal to or greater than the uppermost end of the ice making cell **320***a* defined by the first portion **322** of the first tray **320**. The space may extend to a point lower than the uppermost end of the auxiliary storage chamber **325**.

[0659] The ice separation heater **290** provides heat to reduce freezing of water in the space between the second portion **383** of the second tray **380** and the second portion **323** of the first tray **320**. As described above, the second portion **383** of the second tray **380** serves as a leakage prevention part. It is advantageous that a length of the leakage prevention part is provided as long as possible. This is because as the length of the leak prevention part increases, an amount of water leaking between the first and second tray assemblies is reduced. A length of the leakage prevention part defined by the second portion **383** may be greater than a distance from the center of the ice making cell **320***a* to the outer circumferential surface of the ice making cell **320***a*.

[0660] A second surface facing the first portion **322** of the first tray **320** at the first portion of the second tray **380** may have a surface area greater than that of the first surface facing the first portion **382** of the second tray **380** at the first portion **322** of the first tray **320**. Due to a difference in surface area, coupling force between the first tray assembly **201** and the second tray assembly **211** may increase.

[0661] The ice making may be started when the second tray **380** reaches the ice making position. Alternatively, when the second tray **380** reaches the ice making position, and the water supply time elapses, the ice making may be started.

[0662] When ice making is started, the controller **800** may control the cold air supply part **900** to supply cool air to the ice making cell **320***a*.

[0663] After the ice making is started, the controller **800** may control the transparent ice heater **430** to be turned on in at least partial sections of the cold air supply part **900** supplying the cold air to the ice making cell **320***a*.

[0664] When the transparent ice heater **430** is turned on, since the heat of the transparent ice heater **430** is transferred to the ice making cell **320***a*, the ice making rate of the ice making cell **320***a* may be delayed.

[0665] According to this embodiment, the ice making rate may be delayed so that the bubbles dissolved in the water inside the ice making cell **320***a* move from the portion at which ice is made toward the liquid water by the heat of the transparent ice heater **430** to make the transparent ice in the ice maker **200**.

[0666] In the ice making process, the controller **800** may determine whether the turn-on condition of the transparent ice heater **430** is satisfied (S**5**).

[0667] In this embodiment, the transparent ice heater **430** is not turned on immediately after the ice making is started, and the transparent ice heater **430** may be turned on only when the turn-on condition of the transparent ice heater **430** is satisfied (S**6**).

[0668] Generally, the water supplied to the ice making cell **320***a* may be water having normal temperature or water having a temperature lower than the normal temperature. The temperature of the water supplied is higher than a freezing point of water. Thus, after the water supply, the temperature of the water is lowered by the cold air, and when the temperature of the water reaches the freezing point of the water, the water is changed into ice. In this embodiment, the transparent ice heater **430** may not be turned on until the water is phase-changed into ice.

[0669] If the transparent ice heater **430** is turned on before the temperature of the water supplied to the ice making cell **320***a* reaches the freezing point, the speed at which the temperature of the water reaches the freezing point by the heat of the transparent ice heater **430** is slow. As a result, the starting of the ice making may be delayed.

[0670] The transparency of the ice may vary depending on the presence of the air bubbles in the portion at which ice is made after the ice making is started. If heat is supplied to the ice making cell **320***a* before the ice is made, the transparent ice heater **430** may operate regardless of the transparency of the ice.

[0671] Thus, according to this embodiment, after the turn-on condition of the transparent ice heater **430** is satisfied, when the transparent ice heater **430** is turned on, power consumption due to the unnecessary operation of the transparent ice heater **430** may be prevented.

[0672] Alternatively, even if the transparent ice heater **430** is turned on immediately after the start of ice making, since the transparency is not affected, it is also possible to turn on the transparent ice heater **430** after the start of the ice making.

[0673] In this embodiment, the controller **800** may determine that the turn-on condition of the transparent ice heater **430** is satisfied when a predetermined time elapses from the set specific time point. The specific time point may be set to at least one of the time points before the transparent ice heater **430** is turned on.

[0674] For example, the specific time point may be set to a time point at which the cold air supply part **900** starts to supply cooling power for the ice making, a time point at which the second tray assembly **211** reaches the ice making position, a time point at which the water supply is completed, and the like.

[0675] In this embodiment, the controller **800** determines that the turn-on condition of the transparent ice heater **430** is satisfied when a temperature sensed by the second temperature sensor **700** reaches a turn-on reference temperature. For example, the turn-on reference temperature may be a temperature for determining that water starts to freeze at the uppermost side (side of the opening **324**) of the ice making cell **320***a*.

[0676] When a portion of the water is frozen in the ice making cell **320***a*, the temperature of the ice in the ice making cell **320***a* is below zero. The temperature of the first tray **320** may be higher than the temperature of the ice in the ice making cell **320***a*.

[0677] Alternatively, although water is present in the ice making cell **320***a*, after the ice starts to be made in the ice making cell **320***a*, the temperature sensed by the second temperature sensor **700** may be below zero.

[0678] Thus, to determine that making of ice is started in the ice making cell **320***a* on the basis of the temperature detected by the second temperature sensor **700**, the turn-on reference temperature may be set to the below-zero temperature.

[0679] That is, when the temperature sensed by the second temperature sensor **700** reaches the turn-on reference temperature, since the turn-on reference temperature is below zero, the ice temperature of the ice making cell **320***a* is below zero, i.e., lower than the below reference temperature. Therefore, it may be indirectly determined that ice is made in the ice making cell **320***a*.

[0680] As described above, when the transparent ice heater **430** is not used, the heat of the transparent ice heater **430** is transferred into the ice making cell **320***a*.

[0681] In this embodiment, when the second tray **380** is disposed below the first tray **320**, the transparent ice heater **430** is disposed to supply the heat to the second tray **380**, the ice may be made from an upper side of the ice making cell **320***a*.

[0682] In this embodiment, since ice is made from the upper side in the ice making cell **320***a*, the bubbles move downward from the portion at which the ice is made in the ice making cell **320***a* toward the liquid water.

[0683] Since density of water is greater than that of ice, water or bubbles may convex in the ice making cell **320***a*, and the bubbles may move to the transparent ice heater **430**.

[0684] In this embodiment, the mass (or volume) per unit height of water in the ice making cell **320***a* may be the same or different according to the shape of the ice making cell **320***a*. For example, when the ice making cell **320***a* is a rectangular parallelepiped, the mass (or volume) per unit height of water in the ice making cell **320***a* is the same. On the other hand, when the ice making cell **320***a* has a shape such as a sphere, an inverted triangle, a crescent moon, etc., the mass (or volume) per unit height of water is different.

[0685] When the cooling power of the cold air supply part **900** is constant, if the heating amount of the transparent ice heater **430** is the same, since the mass per unit height of water in the ice making cell **320***a* is different, an ice making rate per unit height may be different.

[0686] For example, if the mass per unit height of water is small, the ice making rate is high, whereas if the mass per unit height of water is high, the ice making rate is slow.

[0687] As a result, the ice making rate per unit height of water is not constant, and thus, the transparency of the ice may vary according to the unit height. In particular, when ice is made at a high rate, the bubbles may not move from the ice to the water, and the ice may contain the bubbles to lower the transparency. That is, the more the variation in ice making rate per unit height of water decreases, the more the variation in transparency per unit height of made ice may decrease.

[0688] Therefore, in this embodiment, the control part **800** may control the cooling power and/or the heating amount so that the cooling power of the cold air supply part **900** and/or the heating amount of the transparent ice heater **430** is variable according to the mass per unit height of the water of the ice making cell **320***a*.

[0689] In this specification, the variable of the cooling power of the cold air supply part **900** may include one or more of a variable output of the compressor, a variable output of the fan, and a variable opening degree of the refrigerant valve.

[0690] Also, in this specification, the variation in the heating amount of the transparent ice heater **430** may represent varying the output of the transparent ice heater **430** or varying the duty of the transparent ice heater **430**.

[0691] In this case, the duty of the transparent ice heater **430** represents a ratio of the turn-on time and a sum of the turn-on time and the turn-off time of the transparent ice heater **430** in one cycle, or a ratio of the turn-ff time and a sum of the turn-on time and the turn-off time of the transparent ice heater **430** in one cycle.

[0692] In this specification, a reference of the unit height of water in the ice making cell **320***a* may vary according to a relative position of the ice making cell **320***a* and the transparent ice heater **430**.

[0693] For example, as shown in FIG. **52**, view (a), the transparent ice heater **430** at the bottom surface of the ice making cell **320***a* may be disposed to have the same height. In this case, a line connecting the transparent ice heater **430** is a horizontal line, and a line extending in a direction perpendicular to the horizontal line serves as a reference for the unit height of the water of the ice making cell **320***a*. In the case of FIG. **52**, view (a), ice is made from the uppermost side of the ice making cell **320***a* and then is grown.

[0694] On the other hand, as shown in FIG. **52**, view (b), the transparent ice heater **430** at the bottom surface of the ice making cell **320***a* may be disposed to have different heights. In this case, since heat is supplied to the ice making cell **320***a* at different heights of the ice making cell **320***a*, ice is made with a pattern different from that of FIG. **52**, view (a). For example, in FIG. **52**, view (b), ice may be made at a position spaced apart from the uppermost end to the left side of the ice making cell **320***a*, and the ice may be grown to a right lower side at which the transparent ice heater **430** is disposed. Accordingly, in FIG. **52**, view (b), a line (reference line) perpendicular to the line connecting two points of the transparent ice heater **430** serves as a reference for the unit height of water of the ice making cell **320***a*. The reference line of FIG. **52**, view (b) is inclined at a predetermined angle from the vertical line.

[0695] FIG. **53** illustrates a unit height division of water and an output amount of transparent ice heater per unit height when the transparent ice heater is disposed as shown in FIG. **52**, view (a).

[0696] Hereinafter, an example of controlling an output of the transparent ice heater so that the ice making rate is constant for each unit height of water will be described.

[0697] Referring to FIG. **53**, when the ice making cell **320***a* is formed, for example, in a spherical shape, the mass per unit height of water in the ice making cell **320***a* increases from the upper side to the lower side to reach the maximum and then decreases again.

[0698] For example, the water (or the ice making cell itself) in the spherical ice making cell **320***a* having a diameter of about 50 mm is divided into nine sections (section A to section I) by 6 mm height (unit height). Here, it is noted that there is no limitation on the size of the unit height and the number of divided sections.

[0699] When the water in the ice making cell **320***a* is divided into unit heights, the height of each section to be divided is equal to the section A to the section H, and the section I is lower than the remaining sections. Alternatively, the unit heights of all divided sections may be the same depending on the diameter of the ice making cell **320***a* and the number of divided sections,

[0700] Among the many sections, the section E is a section in which the mass of unit height of water is maximum. For example, in the section in which the mass per unit height of water is maximum, when the ice making cell **320***a* has spherical shape, a diameter of the ice making cell **320***a*, a horizontal cross-sectional area of the ice making cell **320***a*, or a circumference of the ice may be maximum.

[0701] As described above, when assuming that the cooling power of the cold air supply part **900** is constant, and the output of the transparent ice heater **430** is constant, the ice making rate in section E is the lowest, the ice making rate in the sections A and I is the fastest.

[0702] In this case, since the ice making rate varies for the height, the transparency of the ice may vary for the height. In a specific section, the ice making rate may be too fast and contain bubbles, thereby lowering the transparency.

[0703] Therefore, in this embodiment, the output of the transparent ice heater **430** may be controlled so that the ice making rate for each unit height is the same or similar while the bubbles move from the portion at which ice is made to the water in the ice making process.

[0704] Specifically, since the mass of the section E is the largest, the output W**5** of the transparent ice heater **430** in the section E may be set to a minimum value. Since the volume of the section D is less than that of the section E, the volume of the ice may be reduced as the volume decreases, and thus it is necessary to delay the ice making rate. Thus, an output W**6** of the transparent ice heater **430** in the section D may be set to a value greater than an output W**5** of the transparent ice heater **430** in the section E.

[0705] Since the volume in the section C is less than that in the section D by the same reason, an output W**3** of the transparent ice heater **430** in the section C may be set to a value greater than the output W**4** of the transparent ice heater **430** in the section D. Since the volume in the section B is less than that in the section C, an output W**2** of the transparent ice heater **430** in the section B may be set to a value greater than the output W**3** of the transparent ice heater **430** in the section C. Since the volume in the section A is less than that in the section B, an output W**1** of the transparent ice heater **430** in the section A may be set to a value greater than the output W**2** of the transparent ice heater **430** in the section B.

[0706] For the same reason, since the mass per unit height decreases toward the lower side in the section E, the output of the transparent ice heater **430** may increase as the lower side in the section E (see W**6**, W**7**, W**8**, and W**9**).

[0707] Thus, according to an output variation pattern of the transparent ice heater **430**, the output of the transparent ice heater **430** is gradually reduced from the first section to the intermediate section after the transparent ice heater **430** is initially turned on.

[0708] The output of the transparent ice heater **430** may be minimum in the intermediate section in which the mass of unit height of water is maximum. The output of the transparent ice heater **430** may again increase step by step from the next section of the intermediate section.

[0709] The output of the transparent ice heater **430** in two adjacent sections may be set to be the same according to the type or mass of the made ice. For example, the output of section C and section D may be the same. That is, the output of the transparent ice heater **430** may be the same in at least two sections.

[0710] Alternatively, the output of the transparent ice heater **430** may be set to the minimum in sections other than the section in which the mass per unit height is the smallest.

[0711] For example, the output of the transparent ice heater **430** in the section D or the section F may be minimum. The output of the transparent ice heater **430** in the section E may be equal to or greater than the minimum output.

[0712] In summary, in this embodiment, the output of the transparent ice heater **430** may have a maximum initial output. In the ice making process, the output of the transparent ice heater **430** may be reduced to the minimum output of the transparent ice heater **430**.

[0713] The output of the transparent ice heater **430** may be gradually reduced in each section, or the output may be maintained in at least two sections.

[0714] The output of the transparent ice heater **430** may increase from the minimum output to the end output. The end output may be the same as or different from the initial output.

[0715] In addition, the output of the transparent ice heater **430** may incrementally increase in each section from the minimum output to the end output, or the output may be maintained in at least two sections.

[0716] Alternatively, the output of the transparent ice heater **430** may be an end output in a section before the last section among a plurality of sections. In this case, the output of the transparent ice heater **430** may be maintained as an end output in the last section. That is, after the output of the transparent ice heater **430** becomes the end output, the end output may be maintained until the last section.

[0717] As the ice making is performed, an amount of ice existing in the ice making cell **320***a* may increase. Thus, when the transparent ice heater **430** continues to increase until the output reaches the last section, the heat supplied to the ice making cell **320***a* may be reduced. As a result, excessive water may exist in the ice making cell **320***a* even after the end of the last section. Therefore, the output of the transparent ice heater **430** may be maintained as the end output in at least two sections including the last section.

[0718] The transparency of the ice may be uniform for each unit height, and the bubbles may be collected in the lowermost section by the output control of the transparent ice heater **430**. Thus, when viewed on the ice as a whole, the bubbles may be collected in the localized portion, and the remaining portion may become totally transparent.

[0719] As described above, even if the ice making cell **320***a* does not have the spherical shape, the transparent ice may be made when the output of the transparent ice heater **430** varies according to the mass for each unit height of water in the ice making cell **320***a*.

[0720] The heating amount of the transparent ice heater **430** when the mass per unit height of water is large may be less than that of the transparent ice heater **430** when the mass per unit height of water is small.

[0721] For example, while maintaining the same cooling power of the cold air supply part **900**, the heating amount of the transparent ice heater **430** may vary so as to be inversely proportional to the mass per unit height of water.

[0722] Also, it is possible to make the transparent ice by varying the cooling power of the cold air supply part **900** according to the mass per unit height of water. For example, when the mass per unit height of water is large, the cold force of the cold air supply part **900** may increase, and when the mass per unit height is small, the cold force of the cold air supply part **900** may decrease.

[0723] For example, while maintaining a constant heating amount of the transparent ice heater **430**, the cooling power of the cold air supply part **900** may vary to be proportional to the mass per unit height of water.

[0724] Referring to the variable cooling power pattern of the cold air supply part **900** in the case of making the spherical ice, the cooling power of the cold air supply part **900** from the initial section to the intermediate section during the ice making process may increase.

[0725] The cooling power of the cold air supply part **900** may be maximum in the intermediate section in which the mass per unit height of water is maximum. The cooling power of the cold air supply part **900** may be reduced again from the next section of the intermediate section.

[0726] Alternatively, the transparent ice may be made by varying the cooling power of the cold air supply part **900** and the heating amount of the transparent ice heater **430** according to the mass per unit height of water.

[0727] For example, the heating power of the transparent ice heater **430** may vary, and the cooling power of the cold air supply part **900** may be proportional to the mass per unit height of water. The heating power of the transparent ice heater **430** may be inversely proportional to the mass per unit height of water.

[0728] According to this embodiment, when one or more of the cooling power of the cold air supply part **900** and the heating amount of the transparent ice heater **430** are controlled according to the mass per unit height of water, the ice making rate per unit height of water may be substantially the same or may be maintained within a predetermined range.

[0729] As illustrated in FIG. **57**, a convex portion **382***f* may be deformed in a direction away from the center of the ice making cell **320***a* by being pressed by the ice. The lower portion of the ice may have the spherical shape by the deformation of the convex portion **382***f*.

[0730] The controller **800** may determine whether the ice making is completed based on the temperature sensed by the second temperature sensor **700** (S**8**).

[0731] When it is determined that the ice making is completed, the controller **800** may turn off the transparent ice heater **430** (S**9**).

[0732] For example, when the temperature sensed by the second temperature sensor **700** reaches a first reference temperature, the controller **800** may determine that the ice making is completed to turn off the transparent ice heater **430**.

[0733] In this case, since a distance between the second temperature sensor **700** and each ice making cell **320***a* is different, in order to determine that the ice making is completed in all the ice making cells **320***a*, the controller **800** may perform the ice separation after a certain amount of time, at which it is determined that ice making is completed, has passed or when the temperature sensed by the second temperature sensor **700** reaches a second reference temperature lower than the first reference temperature.

[0734] When the ice making is completed, the controller **800** operates one or more of the ice maker heater **290** and the transparent ice heater **430** (S**10**).

[0735] When at least one of the ice heater **290** or the transparent ice heater **430** is turned on, heat of the heater is transferred to at least one of the first tray **320** or the second tray **380** so that the ice may be separated from the surfaces (inner surfaces) of one or more of the first tray **320** and the second tray **380**.

[0736] Also, the heat of the heaters **290** and **430** is transferred to the contact surface of the first tray **320** and the second tray **380**, and thus, the first contact surface **322***c* of the first tray **320** and the second contact surface **382***c* of the second tray **380** may be in a state capable of being separated from each other.

[0737] When at least one of the ice separation heater **290** and the transparent ice heater **430** operate for a predetermined time, or when the temperature sensed by the second temperature sensor **700** is equal to or higher than an off reference temperature, the controller **800** is turned off the heaters **290** and **430**, which are turned on (S**10**). Although not limited, the turn-off reference temperature may be set to below zero temperature.

[0738] The controller **800** operates the driver **480** to allow the second tray assembly **211** to move in the forward direction (S**11**). As illustrated in FIG. **58**, when the second tray **380** move in the forward direction, the second tray **380** is spaced apart from the first tray **320**.

[0739] The moving force of the second tray **380** is transmitted to the first pusher **260** by the pusher link **500**. Then, the first pusher **260** descends along the guide slot **302**, and the extension part **264** passes through the opening **324** to press the ice in the ice making cell **320***a*.

[0740] In this embodiment, ice may be separated from the first tray **320** before the extension part **264** presses the ice in the ice making process. That is, ice may be separated from the surface of the first tray **320** by the heater that is turned on.

[0741] In this case, the ice may move together with the second tray **380** while the ice is supported by the second tray **380**.

[0742] For another example, even when the heat of the heater is applied to the first tray **320**, the ice may not be separated from the surface of the first tray **320**.

[0743] Therefore, when the second tray assembly **211** moves in the forward direction, there is possibility that the ice is separated from the second tray **380** in a state in which the ice contacts the first tray **320**.

[0744] In this state, in the process of moving the second tray **380**, the extension part **264** passing through the opening **324** may press the ice contacting the first tray **320**, and thus, the ice may be separated from the tray **320**. The ice separated from the first tray **320** may be supported by the second tray **380** again.

[0745] When the ice moves together with the second tray **380** while the ice is supported by the second tray **380**, the ice may be separated from the tray **250** by its own weight even if no external force is applied to the second tray **380**.

[0746] While the second tray **380** moves, even if the ice does not fall from the second tray **380** by its own weight, when the second pusher **540** contacts the second tray **540** as illustrated in FIGS. **58** and **59** to press the second tray **380**, the ice may be separated from the second tray **380** to fall downward.

[0747] For example, as illustrated in FIG. **58**, while the second tray assembly **311** moves in the forward direction, the second tray **380** may contact the extension part **544** of the second pusher **540**.

[0748] As illustrated in FIG. **58**, when the second tray **380** contacts the second pusher **540**, the first tray assembly **201** and the second tray assembly **211** form a second angle θ**2** therebetween with respect to the rotation center C**4**. That is, the first contact surface **322***c* of the first tray **320** and the second contact surface **382***c* of the second tray **380** form a second angle therebetween. The second angle may be greater than the first angle and may be close to about 90 degrees.

[0749] When the second tray assembly **211** continuously moves in the forward direction, the extension part **544** may press the second tray **380** to deform the second tray **380** and the extension part **544**. Thus, the pressing force of the extension part **544** may be transferred to the ice so that the ice is separated from the surface of the second tray **380**. The ice separated from the surface of the second tray **380** may drop downward and be stored in the ice bin **600**.

[0750] In this embodiment, as shown in FIG. **59**, the position at which the second tray **380** is pressed by the second pusher **540** and deformed may be referred to as an ice separation position.

[0751] As illustrated in FIG. **59**, at the ice separation position of the second tray assembly **211**, the first tray assembly **201** and the second tray assembly **211** may form a third angle θ**3** based on the rotation center C**4**. That is, the first contact surface **322***c* of the first tray **320** and the second contact surface **382***c* of the second tray **380** form the third angle θ**3**. The third angle θ**3** is greater than the second angle θ**2**. For example, the third angle θ**3** is greater than about 90 degrees and less than about 180 degrees.

[0752] At the ice separation position, a distance between a first edge **544***a* of the second pusher **540** and a second contact surface **382***c* of the second tray **380** may be less than that between the first edge **544***a* of the second pusher **540** and the lower opening **406***b* of the second tray supporter **400** so that the pressing force of the second pusher **540** increases.

[0753] An attachment degree between the first tray **320** and the ice is greater than that between the second tray **380** and the ice. Thus, a minimum distance between the first edge **264***a* of the first pusher **260** and the first contact surface **322***c* of the first tray **320** at the ice separation position may be greater than a minimum distance between the second edge **544***a* of the second pusher **540** and the second contact surface **382***c* of the second tray **380**.

[0754] At the ice separation position, a distance between the first edge **264***a* of the first pusher **260** and the line passing through the first contact surface **322***c* of the first tray **320** may be greater than 0 and may be less than about ½ of a radius of the ice making cell **320***a*. Accordingly, since the first edge **264***a* of the first pusher **260** moves to a position close to the first contact surface **322***c* of the first tray **320**, the ice is easily separated from the first tray **320**.

[0755] Whether the ice bin **600** is full may be detected while the second tray assembly **211** moves from the ice making position to the ice separation position.

[0756] For example, the full ice detection lever **520** rotates together with the second tray assembly **211**, and the rotation of the full ice detection lever **520** is interrupted by ice while the full ice detection lever **520** rotates. In this case, it may be determined that the ice bin **600** is in a full ice state. On the other hand, if the rotation of the full ice detection lever **520** is not interfered with the ice while the full ice detection lever **520** rotates, it may be determined that the ice bin **600** is not in the ice state.

[0757] After the ice is separated from the second tray **380**, the controller **800** controls the driver **480** to allow the second tray assembly **211** to move in the reverse direction (S**11**). Then, the second tray assembly **211** moves from the ice separation position to the water supply position.

[0758] When the second tray assembly **211** moves to the water supply position of FIG. **54**, the controller **800** stops the driver **480** (S**1**).

[0759] When the second tray **380** is spaced apart from the extension part **544** while the second tray assembly **211** moves in the reverse direction, the deformed second tray **380** may be restored to its original shape.

[0760] In the reverse movement of the second tray assembly **211**, the moving force of the second tray **380** is transmitted to the first pusher **260** by the pusher link **500**, and thus, the first pusher **260** ascends, and the extension part **264** is removed from the ice making cell **320***a*.

[0761] FIG. **60** is a view illustrating an operation of the pusher link when the second tray assembly moves from the ice making position to the ice separation position.

[0762] FIG. **60**(*a*) illustrates the ice making position, FIG. **60**(*b*) illustrates the water supply position, FIG. **60**(*c*) illustrates the position at which the second tray contacts the second pusher, and FIG. **60**(*d*) illustrates the ice separation position.

[0763] FIG. **61** is a view illustrating a position of the first pusher at the water supply position at which the ice maker is installed in the refrigerator, FIG. **62** is a cross-sectional view illustrating the position of the first pusher at the water supply position at which the ice maker is installed in the refrigerator, and FIG. **63** is a cross-sectional view illustrating a position of the first pusher at the ice separation position at which the ice maker is installed in the refrigerator.

[0764] Referring to FIGS. **60** to **63**, the pushing bar **264** of the first pusher **260** may include the first edge **264***a* and the second edge **264***b* as described above.

[0765] The first pusher **260** may move by receiving power from the driver **480**.

[0766] The control unit **800** may control the first edge **264***a* so as to be disposed at a different position from the ice making position so that a phenomenon in which water supplied into the ice making cell **320***a* at the water supply position is attached to the first pusher **260** and then frozen in the ice making process.

[0767] In this specification, the control of the position by the controller **800** may be understood as controlling the position by controlling the driver **480**.

[0768] The controller **800** may control the position so that the first edge **264***a* is disposed at different positions at the water supply position, the ice making position, and the ice separation position.

[0769] The controller **800** control the first edge **264***a* to allow the first edge **264***a* to move in the first direction in the process of moving from the ice separation position to the water supply position and to allow the first edge **264***a* to additionally move in the first direction in the process of moving from the water supply position to the ice making position.

[0770] Alternatively, the controller **800** controls the first edge **264***a* to allow the first edge **264***a* to move in the first direction in the process of moving from the ice separation position to the water supply position and allow the first edge to move in a second direction different from the first direction in the process of moving from the water supply position to the ice making position.

[0771] For example, the first edge **264***a* may move in the first direction by the first slot **302***a* of the guide slot **302**, and the second edge **264***a* may rotate in a second direction or move in a second direction inclined with the first direction by the second slot **302***b*.

[0772] The first edge **264***a* may be disposed at a first point outside the ice making cell **320***a* at the ice making position and may be controlled to be disposed at a second point of the ice making cell **320***a* during the ice separation process.

[0773] The refrigerator further includes a cover member **100** including a first portion **101** defining a support surface supporting the bracket **220** and a third portion **103** defining the accommodation space **104**. A wall **32***a* defining the freezing compartment **32** may be supported on a top surface of the first portion **101**. The first portion **101** and the third portion **103** may be spaced a predetermined distance from each other and may be connected by the second portion **102**. The second portion **102** and the third portion **103** may define the accommodation space **104** accommodating at least a portion of the ice maker **200**. At least a portion of the guide slot **302** may be defined in the accommodation space **104**.

[0774] For example, the upper end **302***c* of the guide slot **302** may be disposed in the accommodation space **104**. The lower end **302***d* of the guide slot **302** may be disposed outside the accommodation space **104**.

[0775] The lower end **302***d* of the guide slot **302** may be higher than the support wall **221***d* of the bracket **220** and be lower than the upper surface **303***b* of the circumferential wall **303** of the first tray cover **300**. Accordingly, a length of the guide slot **302** may increase without increasing the height of the ice maker **200**.

[0776] The water supply part **240** may be coupled to the bracket **220**. The water supply part **240** may include a first portion **241**, a second portion **242** disposed to be inclined with respect to the first portion **241**, and a third portion extending from both sides of the first portion **241**. The through-hole **244** may be defined in the first portion **241**. Alternatively, the through-hole **244** may be defined between the first portion **241** and the second portion **242**.

[0777] The water supplied to the water supply part **240** may flow downward along the second portion **242** and then be discharged from the water supply part **240** through the through-hole **244**. The water discharged from the water supply part **244** may be supplied to the ice making cell **320***a* through the auxiliary storage chamber **325** and the opening **324** of the first tray **320**.

[0778] The through-hole **244** may be defined in a direction in which the water supply part **240** faces the ice making cell **320***a*.

[0779] The lowermost end **240***a* of the water supply part **240** may be disposed lower than an upper end of the auxiliary storage chamber **325**. The lowermost end **240***a* of the water supply part **240** may be disposed in the auxiliary storage chamber **325**.

[0780] The control unit **800** may control a position of the first edge **264***a* so that the first edge moves in the direction away from the through-hole **244** of the water supply unit **240** in the process of allowing the second tray assembly **211** to move from the ice separation position to the water supply position. For example, the first edge **264***a* may rotate in a direction away from the through-hole **244**.

[0781] When the first edge **264***a* moves away from the through-hole **244**, the contact of the water with the first edge **264***a* in the water supply process may be reduced, and thus, the freezing of the water at the first edge **264***a* is reduced.

[0782] In the process of allowing the second tray assembly **211** to move from the water supply position to the ice making position, the second edge **264***b* may further move in the second direction.

[0783] At the water supply position, the first edge **264***a* may be disposed outside the ice making cell **320***a*. At the water supply position, the first edge **264***a* may be disposed outside the auxiliary storage chamber **325**.

[0784] At the water supply position, the first edge **264***a* may be disposed higher than the lower end of the through-hole **224**.

[0785] At the water supply position, a maximum value of a distance between the center line C**1** of the ice making cell **320***a* and the first edge **264***a* may be greater than that of a distance between the center line C**1** of the ice making cell **320***a* and the storage wall **325***a*.

[0786] At the water supply position, the first edge **264***a* may be disposed higher than the upper end **325***c* of the auxiliary storage chamber **325** and be disposed lower than the upper end **325***b* of the circumferential wall **303** of the first tray cover **300**. In this case, the first edge **264***a* may be disposed close to the ice making cell **320***a* to allow the first edge **264***a* to press the ice at the initial ice separation process, thereby improving the ice separation performance.

[0787] At the ice separation position, a length of the first pusher **260** inserted into the ice making cell **320***a* may be longer than that of the second pusher **541** inserted into the second tray supporter **400**.

[0788] At the ice separation position, the first edge **264***a* may be disposed on an area (the area between the two dotted lines in FIG. **63**) between parallel lines extending in the direction of the first contact surface **322***c* by passing through the highest and lowest points of the shaft **440**.

[0789] Alternatively, at the ice separation position, the first edge **264***a* may be disposed on an extension line extending from the first contact surface **322***c*.

[0790] At the water supply position, the second edge **264***b* may be disposed lower than the third portion **103** of the cover member **100**.

[0791] At the water supply position, the second edge **264***b* may be disposed higher than an upper end **241***b* of the first portion **241** of the water supply **240**.

[0792] At the water supply position, the second edge **264***b* may be higher than a top surface **221***b***1** of the first fixing wall **221***b* of the bracket **220**.

[0793] The controller **800** may control a position of the second edge **264***b* to be closer to the water supply **240** than the first edge **264***a* at the water supply position.

[0794] At the water supply position, the second edge **264***b* may be disposed between the first portion **101** of the cover member **100** and the third portion **103** of the cover member **100**.

[0795] For example, the second edge **264***b* at the water supply position may be disposed in the accommodation space **104**. Accordingly, since a portion of the ice maker **200** is disposed in the accommodation space **104**, the space accommodating food in the freezing compartment **32** may be reduced by the ice maker **200**, and the first pusher **260** may increase in moving length. When the moving length of the first pusher **260** increase, the pressing force pressing the ice by the first pusher **260** may increase during the ice making process.

[0796] At the ice separation position, the second edge **264***b* may be disposed outside the accommodation space **104**.

[0797] At the ice separation position, the second edge **264***b* may be disposed between the support surface **221***d***1** supporting the first tray assembly **201** in the bracket **220** and the first portion of the cover member **100**.

[0798] At the ice separation position, the second edge **264***b* may be lower than the top surface **221***b***1** of the first fixing wall **221***b* of the bracket **220**.

[0799] At the ice separation position, the second edge **264***b* may be disposed outside the ice making cell **320***a*. At the ice separation position, the second edge **264***b* may be disposed outside the auxiliary storage chamber **325**.

[0800] At the ice separation position, the second edge **264***b* may be disposed higher than the support surface **221***d***1** of the support wall **221***d*. At the ice separation position, the second edge **264***b* may be higher than the through hole **241** of the water supply **240**.

[0801] At the iced position, the second edge **264***b* may be disposed higher than the lower end **241***a* of the first portion **241** of the water supply **240**.

[0802] The first portion **241** of the water supply part **240** may extend in the vertical direction as a whole or may partially extend in the vertical direction, and the other portion of the first portion **241** may extend in a direction away from the first pusher **260**. Alternatively, the first portion **241** of the water supply unit **240** may be provided to be farther from the first pusher **260** from the lower end **241***a* to the upper end **241***a*.

[0803] A distance between the second edge **264***b* and the first portion **241** of the water supply **240** at the water supply position may be greater than that between the second edge **264***b* and the first portion **241** of the water supply part **240** at the ice making position.

[0804] A distance between the second edge **264***b* and the portion at which the first portion **241** of the water supply **240** faces the first pusher **260** at the water supply position may be greater than that between the second edge **264***b* and the portion at which the first portion **241** of the water supply part **240** faces the first pusher **260** at the ice separation position.

[0805] FIG. **64** is a view illustrating a position relationship between the through-hole of the bracket and a cold air duct.

[0806] Referring to FIG. **64**, the refrigerator may further include a cold air duct **120** guiding cold air of the cold air supply unit **900**.

[0807] An outlet **121** of the cold air duct **120** may be aligned with the through-hole **222***a* of the bracket **220**.

[0808] The outlet **121** of the cold air duct **120** may be disposed so as not to face at least the guide slot **302**. When the cold air flows directly into the guide slot **302**, freezing may occur in the guide slot **302** so that the first pusher **260** does not move smoothly.

[0809] At least a portion of the outlet **121** of the cold air duct **120** may be disposed higher than an upper end of the circumferential wall **303** of the first tray cover **300**.

[0810] For example, the outlet **121** of the cold air duct **120** may be disposed higher than the opening **324** of the first tray **320**. Therefore, the cold air may flow toward the opening **324** from the upper side of the ice making cell **320***a*. An area of the outlet **121** of the cold air duct **120**, which does not overlap the first tray cover **300**, is larger than that that overlaps the first tray cover **300**. Therefore, the cold air may flow to the upper side of the ice making cell **320***a* without interfering with the first tray cover **300** to cool water or ice of the ice making cell **320***a*.

[0811] That is, the cold air supply part **900** (or cooler) is disposed so that an amount of cold air (or cold) supplied to the first tray assembly is greater than that of cold air supplied to the second tray assembly in which the transparent ice heater **430** is disposed.

[0812] Also, the cold air supply part **900** (or cooler) may be disposed so that more amount of cold air (or cold) may be supplied to the area of the first cell **321***a*, which is farther from the transparent ice heater, than the area of the first cell **321***a*, which is close to the transparent ice heater **430**.

[0813] For example, a distance between the cooler and the area of the first cell **321***a*, which is close to the transparent ice heater **430** is greater than that between the cooler and the area of the first cell **321***a*, which is far from the transparent ice heater **430**.

[0814] A distance between the cooler and the second cell **381***a* may be greater than that between the cooler and the first cell **321***a*.

[0815] FIG. **65** is a view for explaining a method for controlling the refrigerator when a heat transfer amount between cold air and water vary in the ice making process.

[0816] Referring to FIGS. **50** and **65**, cooling power of the cold air supply part **900** may be determined corresponding to the target temperature of the freezing compartment **32**. The cold air generated by the cold air supply part **900** may be supplied to the freezing chamber **32**.

[0817] The water of the ice making cell **320***a* may be phase-changed into ice by heat transfer between the cold water supplied to the freezing chamber **32** and the water of the ice making cell **320***a*.

[0818] In this embodiment, a heating amount of the transparent ice heater **430** for each unit height of water may be determined in consideration of predetermined cooling power of the cold air supply part **900**.

[0819] In this embodiment, the heating amount of the transparent ice heater **430** determined in consideration of the predetermined cooling power of the cold air supply part **900** is referred to as a reference heating amount. The magnitude of the reference heating amount per unit height of water is different.

[0820] However, when the amount of heat transfer between the cold of the freezing compartment **32** and the water in the ice making cell **320***a* is variable, if the heating amount of the transparent ice heater **430** is not adjusted to reflect this, the transparency of ice for each unit height varies.

[0821] In this embodiment, the case in which the heat transfer amount between the cold and the water increase may be a case in which the cooling power of the cold air supply part **900** increases or a case in which the air having a temperature lower than the temperature of the cold air in the freezing compartment **32** is supplied to the freezing compartment **32**.

[0822] On the other hand, the case in which the heat transfer amount between the cold and the water decrease may be a case in which the cooling power of the cold air supply part **900** decreases or a case in which the air having a temperature higher than the temperature of the cold air in the freezing compartment **32** is supplied to the freezing compartment **32**.

[0823] For example, a target temperature of the freezing compartment **32** is lowered, an operation mode of the freezing compartment **32** is changed from a normal mode to a rapid cooling mode, an output of at least one of the compressor or the fan increases, or an opening degree increases, the cooling power of the cold air supply part **900** may increase.

[0824] On the other hand, the target temperature of the freezer compartment **32** increases, the operation mode of the freezing compartment **32** is changed from the rapid cooling mode to the normal mode, the output of at least one of the compressor or the fan decreases, or the opening degree of the refrigerant valve decreases, the cooling power of the cold air supply part **900** may decrease.

[0825] When the cooling power of the cold air supply part **900** increases, the temperature of the cold air around the ice maker **200** is lowered to increase in ice making rate.

[0826] On the other hand, if the cooling power of the cold air supply part **900** decreases, the temperature of the cold air around the ice maker **200** increases, the ice making rate decreases, and also, the ice making time increases.

[0827] Therefore, in this embodiment, when the amount of heat transfer of cold and water increases so that the ice making rate is maintained within a predetermined range lower than the ice making rate when the ice making is performed with the transparent ice heater **430** that is turned off, the heating amount of transparent ice heater **430** may be controlled to increase.

[0828] On the other hand, when the amount of heat transfer between the cold and the water decreases, the heating amount of transparent ice heater **430** may be controlled to decrease.

[0829] In this embodiment, when the ice making rate is maintained within the predetermined range, the ice making rate is less than the rate at which the bubbles move in the portion at which the ice is made, and no bubbles exist in the portion at which the ice is made.

[0830] When the cooling power of the cold air supply part **900** increases, the heating amount of transparent ice heater **430** may increase. On the other hand, when the cooling power of the cold air supply part **900** decreases, the heating amount of transparent ice heater **430** may decrease.

[0831] Hereinafter, the case in which the target temperature of the freezing compartment **32** varies will be described with an example.

[0832] The controller **800** may control the output of the transparent ice heater **430** so that the ice making rate may be maintained within the predetermined range regardless of the target temperature of the freezing compartment **32**.

[0833] For example, the ice making may be started (S**4**), and a change in heat transfer amount of cold and water may be detected (S**31**).

[0834] For example, it may be sensed that the target temperature of the freezing compartment **32** is changed through an input part (not shown).

[0835] The controller **800** may determine whether the heat transfer amount of cold and water increases (S**32**). For example, the controller **800** may determine whether the target temperature increases.

[0836] As the result of the determination in the process (S**32**), when the target temperature increases, the controller **800** may decrease the reference heating amount of transparent ice heater **430** that is predetermined in each of the current section and the remaining sections.

[0837] The variable control of the heating amount of the transparent ice heater **430** may be normally performed until the ice making is completed (S**35**).

[0838] On the other hand, if the target temperature decreases, the controller **800** may increase the reference heating amount of transparent ice heater **430** that is predetermined in each of the current section and the remaining sections. The variable control of the heating amount of the transparent ice heater **430** may be normally performed until the ice making is completed (S**35**).

[0839] In this embodiment, the reference heating mount that increases or decreases may be predetermined and then stored in a memory.

[0840] According to this embodiment, the reference heating amount for each section of the transparent ice heater increases or decreases in response to the change in the heat transfer amount of cold and water, and thus, the ice making rate may be maintained within the predetermined range, thereby realizing the uniform transparency for each unit height of the ice.

What is claimed is:

1. An ice maker comprising:

a first tray assembly configured to define a first portion of a cell that is a space in which liquid is phase-changed into ice;

a second tray assembly configured to define a second portion of the cell, the second tray assembly being in contact the first tray assembly during an ice making process and spaced apart from the first tray assembly during an ice separation process; and

a heater disposed adjacent to the second tray assembly,

wherein the heater is turned on in at least a partial section while a cooler supplies cold so that bubbles dissolved in the liquid within the cell moves from a portion, at which the ice is made, toward the liquid that is in a fluid state to increase a transparency of the ice.

2. The ice maker of claim **1**, further comprising a driver configured to move the second tray assembly.

3. The ice maker of claim **1**, wherein the second tray assembly comprise:

a tray configured to define the second portion of the cell, and a tray case configured to support at least a portion of the tray.

4. The ice maker of claim **3**, wherein a tray case comprises a tray cover and a tray supporter.

5. The ice maker of claim **4**, further comprising a heater case in which the heater is installed.

6. The ice maker of claim **5**, wherein the heater case is integrally formed with the tray supporter.

7. The ice maker of claim **5**, wherein the heater case is separately provided to be coupled to the tray supporter.

8. The ice maker of claim **5**, wherein the heater case includes a heater accommodation portion to accommodate the heater.

9. The ice maker of claim **5**, wherein the heater case includes an opening.

10. An ice maker comprising:

a first tray assembly configured to define a first portion of a cell that is a space in which liquid is phase-changed into ice;

a second tray assembly configured to define a second portion of the cell, the second tray assembly being in contact the first tray assembly during an ice making process and spaced apart from the first tray assembly during an ice separation process; and

a heater disposed adjacent to the first tray assembly,

wherein the heater is turned on in at least a partial section while a cooler supplies cold so that bubbles dissolved in the liquid within the cell moves from a portion, at which the ice is made, toward the liquid that is in a fluid state to increase a transparency of the ice.

11. The ice maker of claim **10**, wherein the first tray assembly is a region in which the heater is disposed.

12. The ice maker of claim **10**, wherein the first tray assembly includes a first tray case contacting the first tray to support at least a portion of the first tray.

13. The maker of claim **12**, wherein the first tray case includes a first tray supporter and a first tray cover.

14. The refrigerator of claim **13**, wherein the first tray supporter and the first tray cover are integrally provided.

15. The refrigerator of claim **13**, wherein the first tray supporter and the first tray cover are separately manufactured configurations coupled to each other.

16. The ice maker of claim **13**, further comprising a bracket configured to define at least a portion of a space that accommodates the first tray assembly and the second tray assembly.

17. The ice maker of claim **16**, wherein the first tray cover is manufactured as a separate part from the bracket and is coupled to the bracket.

18. The ice maker of claim **16**, wherein the first tray cover is integrally formed with the bracket.

19. A refrigerator comprising:

a storage chamber configured to store foods;

a door that opens and closes the storage chamber; and

an ice maker of claim **1**.

20. A refrigerator comprising:

a storage chamber configured to store foods;

a door that opens and closes the storage chamber; and

an ice maker of claim **10**.

* * * * *